Attribute-based Quantum Broadcast Encryption with Composite Policies via Symmetric Unitary t-Designs

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Abstract

In an emerging era of quantum technologies, securing quantum communications is paramount. In this paper, we introduce two frameworks for attribute-based quantum broadcast encryption (AB-QBE), enabling fine-grained access control over sensitive quantum data. The first scheme, Multi-Policy Quantum Private Broadcast Encryption (MP-QPBE), leverages symmetric unitary $\boldsymbol{t}$-designs to construct a protocol where decryption is possible upon satisfying a composite set of access policies simultaneously. This approach ensures that a user can only access the broadcast quantum state if their attributes fulfill multiple predefined criteria. In our MP-QPBE, we first perform symmetrization of the initial quantum message for the encryption of the tensor product of distinct pure states, a scenario not directly addressed by previous quantum private broadcasting schemes. We demonstrate that this method is information-theoretically secure. The second scheme, Component-wise Independent Quantum Broadcast Encryption (CI-QBE), offers an alternative, lossless approach where each quantum message is encrypted independently using a unitary $\mathbf{1}$-design. It provides greater flexibility and is applicable to arbitrary quantum states, including mixed states, without the information loss inherent in symmetrization. We provide a comprehensive security analysis for both constructions, proving their robustness against unauthorized users and adversaries with quantum side information. A comparative analysis highlights the trade-offs between the two schemes in terms of security guarantees, quantum resource requirements, and practical applicability, offering a nuanced perspective on designing secure multi-user quantum communication systems.

Keywords: Linear Secret-Sharing Schemes(LSSS), Symmetric Unitary t-designs, Access Control, Multi-Policy Quantum Private Broadcast Encryption(MP-QPBE), Component-wise Independent Quantum Broadcast Encryption(CI-QBE)

1 Introduction

Quantum communication offers significant capabilities that complement and, in specific scenarios, exceed classical approaches. As we move towards a future of distributed quantum networks, quantum cloud computing, and secure multiparty quantum computation, the need for sophisticated cryptographic primitives to protect quantum data has become increasingly critical. Classical cryptographic techniques, even those deemed quantum-resistant, are insufficient to manage access to and protect the integrity of quantum information itself, which exists as fragile quantum states. This makes it necessary to develop a new generation of quantum cryptographic protocols designed to operate directly on quantum data.

Broadcast encryption [1] is a fundamental cryptographic primitive that enables a sender to securely transmit information to a selected group of users over an insecure channel, while ensuring that unauthorized parties, even if they collude, cannot access the content. In the quantum realm, where the information being broadcast is a quantum state, the challenge is to design protocols that preserve the delicate nature of this quantum information while enforcing access control.

Attribute-based Encryption (ABE) [2] offers a powerful and flexible paradigm for fine-grained access control. In contrast to traditional encryption methods that rely on user identities, ABE defines access based on descriptive attributes. A sender can encrypt a message under a specific policy, and only users whose attributes satisfy this policy can decrypt it. The extension of this paradigm to the quantum domain, known as Attribute-based Quantum Encryption (ABQE), allows for the creation of sophisticated access control systems for quantum data.

In this paper, we develop and analyze two distinct schemes for Attribute-based Quantum Broadcast Encryption.

1. **A Multi-Policy Quantum Private Broadcast Encryption (MP-QPBE) Scheme:** We introduce a construction that requires users to satisfy multiple access policies simultaneously to decrypt a broadcast quantum message. This is achieved by transforming a composite access tree into a Linear Secret-Sharing Scheme (LSSS) and using the secret to select a unitary operator from a symmetric unitary t-design for encryption. A distinctive approach in our construction is the initial symmetrization of the quantum message, which correctly defines the broadcasting of tensor product states on the symmetric subspace.

2. **A Component-wise Independent Quantum Broadcast Encryption (CI-QBE) Scheme:** As an alternative approach that avoids the inherent information loss from symmetrization, we propose a second scheme where individual quantum messages are encrypted independently. This approach offers greater flexibility, supports arbitrary quantum states, and provides a different set of trade-offs in terms of security and resource requirements.
3. **Rigorous Security Analysis:** We provide comprehensive, information-theoretic security proofs for both schemes. We demonstrate their security against unauthorized users and prove their resilience against adversaries with quantum side information, leveraging the properties of unitary designs.
4. **Comparative Study:** We provide a detailed comparison of the MP-QPBE and CI-QBE schemes, evaluating their respective strengths and weaknesses concerning scalability, quantum resource overhead, classical key management, and the nature of their security guarantees.

1.1 Related Work

The concept of broadcast encryption was first introduced by Fiat and Naor [1], who proposed schemes to allow a central authority to securely broadcast to any chosen subset of users. This area has since seen extensive development, with numerous schemes focusing on minimizing key storage, transmission overhead, and computational cost [3]. The paradigm of Attribute-based Encryption (ABE) was introduced by Sahai and Waters [4], which significantly advanced the expressiveness of access control. This was refined into Ciphertext-Policy ABE (CP-ABE) by Bethencourt, Sahai, and Waters [2], where ciphertexts are associated with access policies, a model that directly inspires our work. The construction of Linear Secret-Sharing Schemes (LSSS) from access trees, as formalized by Waters [5] and Lewko and Waters [6], provides the classical underpinning for our attribute-based key distribution. The optimization of these schemes for threshold gates, as seen in the work of Liu, Cao, and Wong [7], is also relevant to our LSSS implementation. The foundation of secure quantum communication is Quantum Key Distribution (QKD), with the seminal BB84 protocol [8] and the E91 protocol [9]. While QKD solves the problem of distributing a secret key between two parties, it does not directly address the one-to-many communication model of broadcasting. Early work on quantum secret sharing [10, 11] explored how to distribute a secret quantum state among multiple parties such that only authorized subsets can reconstruct it, which is conceptually related to the decryption phase of a broadcast scheme. The direct problem of quantum broadcast encryption, where the broadcast message is itself a quantum state, has been explored more recently. The challenge lies in ensuring that unauthorized users, who may possess parts of the key, cannot disturb or gain any information about the broadcast quantum state.

A crucial precursor to our work is the concept of a quantum private broadcast channel, formally defined by Broadbent, Gutoski, and Stebila [12]. They established the foundational definitions of security and correctness for broadcasting quantum states and, most importantly, showed how to construct such a scheme using unitary t-designs. Their work demonstrated that for a message in the symmetric subspace, averaging the encryption over a unitary t-design results in a state that is indistinguishable from a fixed state, thereby ensuring privacy. Our MP-QPBE scheme is a direct extension of this framework, adapting it to an attribute-based context and introducing the symmetrization step to broaden the class of applicable messages. The use of unitary designs as a tool for derandomizing proofs in quantum information theory was explored by Dankert et al. [13] and has been applied in various contexts, including randomized benchmarking and decoupling [14]. Brandao, Harrow, and Horodecki provided explicit constructions for approximate unitary designs [15], making these tools more practical. The hereditary property of designs, which we leverage, was discussed by Lancien, Postle, and Tomamichel [16].

1.2 Overview of our constructions

The development of sophisticated quantum broadcast encryption schemes is motivated by the evolving landscape of quantum technologies and the unique security challenges they present. Our work is driven by the need to create robust, flexible, and provably secure access control mechanisms for multi-user quantum communication environments.

In this paper, we have comparatively studied two aspects of quantum broadcast encryption. The first construction, Multi-Policy Quantum Private Broadcast Encryption(MP-QPBE), is motivated by the work of Broadbent et al. on quantum private broadcasting [12] based on multiple access policies. In this scenario, we deal with a given message of the form $\bigotimes_{j=1}^{t_i} \rho^{(j)}$. The second construction, Component-wise Independent Quantum Broadcast Encryption(CI-QBE), is based on individual access policies, where each message is independent of the other messages, and policies are independent of each other. In the first construction, we construct an attribute-based quantum private broadcast encryption such that the decryption is possible when a user satisfies multiple policies together.

Given multiple attribute-based policy trees $\mathsf{T}^{j_1}, \ldots, \mathsf{T}^{j_{t_i}}$, we construct a single general access-tree $\mathbb{T}_i$. We transform the general access tree into a (t, n)- threshold access tree. Given m sets of access policies, with each policy set with cardinality t_i, we choose m distinct unitary t_i-designs $(\mathfrak{U}_i, w_i)$ for each i-th set of policies Ω_i. Fixing an ordering on each set of unitary t_i-designs, with uniform distribution, we randomly choose a single index k_i^* of the unitary matrices from the unitary t_i-designs corresponding to the associated i-th set of policies. The index of the unitary matrix we share classically using generalized attribute-based LSSS. To share multiple secrets together, we have presented a general LSSS, based on the foundational works [2, 5, 7] when a user satisfies multiple policies. We choose a prime p with $p > \max\{|\mathfrak{U}_i| : i = 1, \ldots, m\}$ and consider a finite field $\mathbb{F}_p$ and construct the LSSS over the field $\mathbb{F}_p$ to the share the secret k_i^* classically. Broadbent et al. [12] showed that any message of the form $\rho^{\otimes t}$ taken from $\mathcal{D}(\mathcal{H}_M^{\otimes t})$ does not satisfy the quantum private broadcast definition, and if the messages are taken from $\mathrm{Sym}(\mathcal{H}_M^{\otimes t})$ then we encrypt; otherwise, abort.

In our construction, given any message $\bigotimes_{j=1}^{t_i} \rho^{(j)} \in \mathcal{D}(\mathcal{H}_M^{\otimes t})$, we first symmetrize it and map it into a symmetric subspace $\mathrm{Sym}(\mathcal{H}_M^{\otimes t})$. Then we encrypt the message using the chosen unitary matrix U_{i,k_i^*} and send the ciphertext to the users. In this case, our study shows that if our messages are only the pure states, then only the decryption is possible based on positive operator-valued measurements(POVM), and we recover an unordered set of all messages. As our goal is to recover all the messages based on the policy set, even if they are unordered, we can recover all the messages for the given set of policies. Our study also suggests the advantage of using multiple smaller unitary t_i-designs, rather than a single unitary t-design for encrypting all the messages and then putting them into a single block diagonal matrix for broadcasting all the ciphertexts together.

In the second construction of CI-QBE, we broadcast the set of all messages in the form of a block diagonal matrix. Unless we need a private broadcast encryption to broadcast a message of the form $\bigotimes_{j=1}^{t_i} \rho^{(j)}$ as an alternative broadcast encryption to address the inherent information loss of the symmetrization-based approach, we also introduce and analyze a CI-QBE scheme based on a sufficiently large unitary 1-design.

In this construction, each quantum message $\rho^{(j)}$ is treated as an independent component and is individually encrypted using a unique secret key k_j^* with a corresponding unitary $U_{k_j^*}$ drawn from a simple unitary 1-design. The resulting individual ciphertexts $\{\sigma^{(j)}\}$ are then assembled into a single, scalable broadcast state using a block-diagonal structure, $\rho_{\mathsf{Broadcast}} = \frac{1}{m}\sum_{j=1}^{m} |j\rangle\langle j| \otimes \sigma^{(j)}$, where an ancillary register indexes each component. The complete set of classical keys $\{k_j^*\}$ is securely distributed to authorized users via a multi-secret LSSS. This component-wise approach is inherently lossless, preserving message order and applicable to arbitrary quantum states, and offers better scalability in terms of quantum resources, particularly for high-dimensional messages, while the MP-QPBE offers a much smaller number of classical keys, consequently a smaller classical share size, stronger security against adversaries with side information, and more suitable for pure states of the form $\bigotimes_{j=1}^{t_i} \rho^{(j)} \in \mathcal{D}(\mathcal{H}_M^{\otimes t_i})$.

1.2.1 Technical Considerations and Contributions

Our framework integrates quantum private broadcasting [12], unitary t-designs [13], attribute-based encryption [2], and linear secret-sharing schemes [17], into a unified quantum broadcast encryption protocol. This integration poses several non-trivial challenges, which we address in this section.

- **Bridging Classical Access Structures with Quantum Encryption.**

The classical LSSS operates over a finite field $\mathbb{F}_p$, while quantum encryption requires unitary operators from $U(d)$. We establish a bridge through the following construction:

1. **Index-based design enumeration:** We impose an ordering on each unitary t_i-design $(\mathfrak{U}_i, w_i)$, associating each unitary $U_{i,k} \in \mathfrak{U}_i$ with a unique index $k \in \{1, \ldots, |\mathfrak{U}_i|\}$. This enumeration is fixed during system setup and known to all players.

2. **Field choice:** We select a prime p satisfying:

$$p > \max_{i \in [m]} |\mathfrak{U}_i|. \tag{1}$$

This ensures that all unitary indices embed naturally into $\mathbb{F}_p$ without collision, enabling the classical LSSS to share quantum encryption keys.

3. **Security transfer:** The LSSS guarantees that for any unauthorized attribute set $S \not\models \mathbb{T}_i$, the conditional distribution of the secret satisfies:

$$\Pr\left[k_i^* = k \,\middle|\, \{\lambda_{i,j}\}_{j \in I_{S_{user}}}\right] = \frac{1}{|\mathfrak{U}_i|} \quad \forall k \in \{1, \ldots, |\mathfrak{U}_i|\}. \tag{2}$$

This uniform uncertainty over the classical index translates directly to quantum security: the adversary's perceived ciphertext becomes the ensemble average over the entire design, yielding the maximally mixed state.

• **Symmetrization of Heterogeneous Message Blocks.**

Prior quantum private broadcasting schemes [12] address messages of the form $\rho^{\otimes t}$, which inherently reside in the symmetric subspace $\mathrm{Sym}(\mathcal{H}_M^{\otimes t})$. Our multi-policy framework requires simultaneous transmission of *distinct* states $\rho_{\Omega_i} = \bigotimes_{j=1}^{t_i} \rho^{(j)}$, which generally lie outside the symmetric subspace.

We apply explicit **symmetrization projection**:

$$\tilde{\rho}_{\Omega_i} = \frac{\Pi_{\mathrm{Sym}(t_i)} \rho_{\Omega_i} \Pi_{\mathrm{Sym}(t_i)}}{\mathrm{Tr}\left(\Pi_{\mathrm{Sym}(t_i)} \rho_{\Omega_i}\right)}, \tag{3}$$

where $\Pi_{\mathrm{Sym}(t_i)} = \frac{1}{t_i!} \sum_{\pi \in \mathfrak{S}_{t_i}} P_d(\pi)$ is the projector onto the symmetric subspace.

Information-theoretic analysis: We discuss the following recoverability results in 3.1:

- **Orthogonal pure states:** If $\{|\psi^{(j)}\rangle\}_{j=1}^{t_i}$ are mutually orthogonal, then the symmetrized state $|\tilde{\psi}_{\Omega_i}\rangle = \frac{1}{\sqrt{t_i!}} \sum_{\pi \in \mathfrak{S}_{t_i}} |\psi^{(\pi(1))}\rangle \otimes \cdots \otimes |\psi^{(\pi(t_i))}\rangle$ allows recovery of the *unordered set* $\{|\psi^{(j)}\rangle\}$ via an appropriate POVM.
- **General mixed states:** The symmetrization map $\rho \mapsto \tilde{\rho}$ is generally *non-invertible* and therefore, it is difficult to perform perfect message recovery for mixed density matrices. This limitation motivates our secondary construction (CI-QBE).

• **Design Selection—Security vs. Efficiency Trade-off.**

A fundamental property of unitary designs is their **nested structure**: every unitary t-design is also a $(t-1)$-design, $(t-2)$-design, and so on down to a 1-design. That is

$$(\mathfrak{U}, w) \text{ is a } t\text{-design} \implies (\mathfrak{U}, w) \text{ is a } t'\text{-design } \forall t' \leq t. \tag{4}$$

This property suggests three possible design strategies for encrypting m policy blocks with parameters $\{t_1, \ldots, t_m\}$:

1. **Single T_Σ-design** where $T_\Sigma = \sum_{i=1}^{m} t_i$: Use one large design for the entire tensor product.
2. **Single $T_{\max}$-design** where $T_{\max} = \max_{i\in[m]} t_i$: Use one design that serves as a t_i-design for each block via the nested property (4).
3. **Distinct t_i-designs**: Use independent designs $\{(\mathfrak{U}_i, w_i)\}_{i=1}^{m}$, one for each block.

We argue that option (3) offers enhanced security guarantees in most practical scenarios, despite potentially higher storage costs. The detailed comparison is provided in Section 5.1.1.

Security vulnerability of shared designs: Consider an adversary $\mathcal{A}$ who is authorized for block j but unauthorized for block $i \neq j$. Under options (1) or (2):

- $\mathcal{A}$ legitimately learns the single T_Σ-design $(\mathfrak{U}, w)$, though $\mathcal{A}$ cannot learn particular unitary encryption operators used for a particular block.
- $\mathcal{A}$ knows that block i is encrypted using some $U_{k_i^*} \in \mathfrak{U}$.
- Although $\mathcal{A}$ does not know k_i^* (protected by LSSS), $\mathcal{A}$ can enumerate all $|\mathfrak{U}|$ candidate decryptions:

$$\left\{(U_k^\dagger)^{\otimes t_i}\sigma_{\Omega_i}(U_k)^{\otimes t_i} : k \in \{1, \ldots, |\mathfrak{U}|\}\right\}. \tag{5}$$

- For approximate t-designs or in the presence of side information, this enumeration may leak partial information about the plaintext.

Security advantage of distinct designs: Under option (3):

- A user learns only a unique $\mathfrak{U}_j$ specific to the authorized block.
- The user has *no knowledge* of which design $\mathfrak{U}_i$ was used for block $i \neq j$.
- The adversary faces uncertainty over both the key index k_i^* *and* the design $\mathfrak{U}_i$ itself for $i \neq j$.
- Because of this, even if the LSSS is weakened (e.g., partial key leakage), the design uncertainty provides an additional security layer.

- **Scalability via Modular Design.**

Beyond security, the modular approach with distinct t_i-designs offers significant *computational advantages*.

A single T_Σ-design (option (1)) would require:

$$|\mathfrak{U}_{T_\Sigma}| = \Omega(d^{2T_\Sigma}) = \Omega\left(d^{2\sum_{i=1}^{m} t_i}\right), \tag{6}$$

which grows *exponentially* in the total number of messages.

Our modular architecture (option (3))achieves:

$$\sum_{i=1}^{m} |\mathfrak{U}_i| = \sum_{i=1}^{m} O(d^{2t_i}). \tag{7}$$

- **Unified Multi-Secret Access Control.**

Standard LSSS shares a single secret among parties. Our framework requires simultaneous sharing of m independent secrets $\{k_i^*\}_{i=1}^m$ under heterogeneous access structures $\{\Gamma_i\}_{i=1}^m$.

We apply **Multi-Secret LSSS (MS-LSSS)** with block-diagonal structure:

$$\mathcal{M} = \begin{pmatrix} M_1 & 0 & \cdots & 0 \\ 0 & M_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & M_m \end{pmatrix} \in \mathbb{F}_p^{\ell_{\text{total}} \times c_{\text{total}}}, \tag{8}$$

where $M_i \in \mathbb{F}_p^{\ell_i \times c_i}$ is the LSSS matrix for access structure Γ_i, $\ell_{\text{total}} = \sum_i \ell_i$, and $c_{\text{total}} = \sum_i c_i$.

Key properties:

1. **Independent reconstruction:** A user with attributes S can reconstruct k_i^* if and only if $S \models \mathbb{T}_i$, independently for each block.
2. **Cross-block privacy:** Authorization for block j reveals no information about k_i^* for $i \neq j$.
3. **Attribute reuse:** The same attribute can contribute to multiple blocks. If attribute α appears in both $\mathbb{T}_i$ and $\mathbb{T}_j$, the user receives shares $(\lambda_{\alpha,i}, \lambda_{\alpha,j})$ for both blocks.

- **Compositional Security Proof.**

The security of our protocol relies on the composition of three mechanisms:

1. **Classical LSSS privacy:** Unauthorized attribute sets gain no information about secrets.
2. **Quantum design averaging:** Uniform uncertainty over design elements produces the Haar-random channel.
3. **Block-diagonal isolation:** Independent processing of blocks prevents cross-block information leakage.

1.3 Paper organization

The remainder of this paper is organized as follows. Section 2 provides the necessary preliminaries. In Section 3, we present our first main contribution: the detailed construction of the Multi-Policy Quantum Private Broadcast Encryption (MP-QPBE) scheme, which uses a symmetrization approach. In Section 4, we introduce our second construction, the Component-wise Independent Quantum Broadcast Encryption (CI-QBE) scheme, which offers a lossless alternative. Section 5 presents a comprehensive discussion and comparative analysis of the two proposed schemes, highlighting their respective advantages and trade-offs. Finally, Section 6 concludes the paper by summarizing our contributions and outlining potential directions for future research. Section 8 contains supplementary derivations supporting the main results.

2 Preliminaries

2.1 Notations

We denote a d-dimensional Hilbert space by $\mathcal{H}_d$. Throughout the paper, we have used the notation $\mathcal{H}_M$ to denote message space, which is a d-dimensional Hilbert space. The set of density operators on $\mathcal{H}_M$ is $\mathcal{D}(\mathcal{H}_M)$, and defined by

$$\mathcal{D}(\mathcal{H}_M) := \{\rho \in \mathcal{L}(\mathcal{H}_M) \mid \rho \geq 0,\, \mathrm{Tr}(\rho) = 1\}.$$

The symmetric subspace of $\mathcal{H}_M^{\otimes t}$ is denoted by $\mathrm{Sym}(d^t)$ or $\mathrm{Sym}(\mathcal{H}_M^{\otimes t})$. The group of all $d \times d$ unitary matrices is denoted by $U(d)$. The symmetric group on n letters is denoted by $\mathfrak{S}_n$. The notation $[d]$ defines the set $\{1, \dots, d\}$. We denote the set of all bounded linear operators on $\mathcal{H}$ by $\mathcal{B}(\mathcal{H})$. For $X \in \mathcal{B}(\mathcal{H})$, the trace norm of X, is denoted by $\|X\|_1$, is defined as $\|X\|_1 := \mathrm{Tr}\sqrt{X^\dagger X}$. Given a linear map $\Phi : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H}')$. The induced $1 \to 1$ norm is defined as $\|\Phi\|_{1\to 1} := \sup_{X \neq 0} \frac{\|\Phi(X)\|_1}{\|X\|_1} = \sup_{\|X\|_1 = 1} \|\Phi(X)\|_1$. The diamond norm of Φ, denoted $\|\Phi\|_\diamond$, is defined as $\|\Phi\|_\diamond := \sup_{d \in \mathbb{N}} \|\Phi \otimes \mathrm{id}_{\mathbb{C}^d}\|_{1\to 1}$, where $\mathrm{id}_{\mathbb{C}^d}$ is the identity map on $\mathcal{B}(\mathbb{C}^d)$, and $\|\Phi \otimes \mathrm{id}_d\|_{1\to 1} = \sup_{\|X\|_1 = 1} \|(\Phi \otimes \mathrm{id}_d)(X)\|_1$. Note that for a completely positive trace-preserving map Φ, the following inequality always holds:

$$\|\Phi\|_{1\to 1} \leq \|\Phi\|_\diamond. \tag{9}$$

2.2 Symmetric subspace

The following description is from [14]. We refer [14] for a good description of the symmetric subspace. Consider the symmetric group $\mathfrak{S}_n$. For every $\pi \in \mathfrak{S}_n$, define

$$P_d(\pi) = \sum_{i_1,\dots,i_t \in [d]} \left| i_{\pi^{-1}(1)}, \dots, i_{\pi^{-1}(t)} \right\rangle \langle i_1, \dots, i_t| \, .$$

Then $P_d(\pi_1, \pi_2) = P_d(\pi_1) P_d(\pi_2)$. Therefore, P_d is a representation of $\mathfrak{S}_t$ on $\mathcal{H}_M^{\otimes t} = (\mathbb{C}^d)^{\otimes t}$.

The symmetric subspace of $\mathcal{H}_M^{\otimes t}$ is defined by

$$\mathrm{Sym}(\mathcal{H}_M^{\otimes t}) := \{|\psi\rangle \in \mathcal{H}_M^{\otimes t} : \forall \pi \in \mathfrak{S}_t, P_d(\pi)\, |\psi\rangle = |\psi\rangle\}.$$

We sometimes denote $\mathrm{Sym}(\mathcal{H}_M^{\otimes t}) = \mathrm{Sym}(d^t)$, when $\dim(\mathcal{H}_M) = d$. The dimension of the symmetric subspace $\mathrm{Sym}(\mathcal{H}_M^{\otimes t})$ is $d_{\mathrm{Sym}} = \binom{d+t-1}{t}$. The set of density operators on $\mathrm{Sym}(\mathcal{H}_M^{\otimes t})$ is defined as $\mathcal{D}(\mathrm{Sym}(\mathcal{H}_M^{\otimes t}))$. Every density matrix from $\mathcal{D}(\mathrm{Sym}(\mathcal{H}_M^{\otimes t}))$ can be written as a real linear combination of rank 1 density matrices [2, 14], that is, $\mathcal{D}(\mathrm{Sym}(\mathcal{H}_M^{\otimes t})) \subset \mathrm{span}_{\mathbb{R}}\{(|\phi\rangle\langle\phi|)^{\otimes t} : |\phi\rangle \in \mathcal{H}_M\}$.

Define

$$\Pi_{\mathrm{Sym}(t)} := \frac{1}{t!} \sum_{\pi \in \mathfrak{S}_t} P_d(\pi). \tag{10}$$

Proposition 1 ([14]) $\Pi_{\mathrm{Sym}(t)}$ *is the orthogonal projector onto* $\mathrm{Sym}(\mathcal{H}_M^{\otimes t})$.

Proof For each $\pi \in \mathfrak{S}_t$, we have,

$$\begin{aligned} P_d(\pi)\Pi_{\mathrm{Sym}(t)} &= P_d(\pi)\frac{1}{t!}\sum_{\sigma\in\mathfrak{S}_t} P_d(\sigma) \\ &= \frac{1}{t!}\sum_{\sigma\in\mathfrak{S}_t} P_d(\pi\sigma) \\ &= \frac{1}{t!}\sum_{\sigma'\in\mathfrak{S}_t,\sigma'=\pi\sigma} P_d(\sigma') \\ &= \frac{1}{t!}\sum_{\sigma'\in\mathfrak{S}_t} P_d(\sigma') \\ &= \Pi_{\mathrm{Sym}(t)}. \end{aligned}$$

Similarly, it is easy to show that $\Pi_{\mathrm{Sym}(t)}P_d(\pi) = \Pi_{\mathrm{Sym}(t)}$ and $\Pi^\dagger_{\mathrm{Sym}(t)}\Pi_{\mathrm{Sym}(t)} = \Pi_{\mathrm{Sym}(t)}$. As $\Pi^\dagger_{\mathrm{Sym}(t)} = \Pi_{\mathrm{Sym}(t)}$, it implies that $\Pi^2_{\mathrm{Sym}(t)} = \Pi_{\mathrm{Sym}(t)}$.

Since $P^\dagger P = P$ is a necessary and sufficient condition for an operator P to be an orthogonal projector, $\Pi_{\mathrm{Sym}(t)}$ is an orthogonal projector. □

From the Proposition 1, it implies that

$$\forall\,|\phi\rangle \in \mathcal{H}_M^{\otimes t}, \quad P_d(\pi)\Pi_{\mathrm{Sym}(t)}\,|\phi\rangle = \Pi_{\mathrm{Sym}(t)}\,|\phi\rangle\,, \tag{11}$$

and $\Pi_{\mathrm{Sym}(t)}\,|\phi\rangle \in \mathrm{Sym}(\mathcal{H}_M^{\otimes t})$. Therefore, $\mathrm{Img}(\Pi_{\mathrm{Sym}(t)}) \subseteq \mathrm{Sym}(\mathcal{H}_M^{\otimes t})$.

2.3 Unitary *t*-designs

In this section, we review some basic definitions and results on unitary t-designs.

Definition 1 ([18]) Let $\mathfrak{U}$ be a finite subset of $U(d)$. A function $w : \mathfrak{U} \longrightarrow [0,1]$ is said to be a weight function on $\mathfrak{U}$ if for all $U \in \mathfrak{U}$, $w(U) > 0$, and $\sum_{U\in\mathfrak{U}} w(U) = 1$.

For our construction, we consider the tensor product definition of unitary t-designs.

Definition 2 (Weighted tensor product definition of Unitary t-designs [18]) Let $\mathfrak{U} \subseteq U(d)$ be a finite set of unitary operators on a d-dimensional complex Hilbert space $\mathbb{C}^d$, and let $w : \mathfrak{U} \to [0,1]$ be a probability distribution over $\mathfrak{U}$, i.e.,

$$\sum_{U\in\mathfrak{U}} w(U) = 1.$$

Then the tuple $(\mathfrak{U}, w)$ is said to be an *exact unitary* t*-design* if

$$\sum_{U\in\mathfrak{U}} w(U)\, U^{\otimes t} \otimes (U^*)^{\otimes t} \;=\; \int_{U(d)} U^{\otimes t} \otimes (U^*)^{\otimes t}\, d\mu_{\mathrm{Haar}}(U), \tag{12}$$

where the integral is taken with respect to the normalized Haar measure μ_{Haar} on the unitary group $U(d)$. That is, the tuple $(\mathfrak{U}, w)$ reproduces the t-th moment operator of the Haar distribution over $U(d)$.

Proposition 2 ([18]) *Every unitary t-design is also a $(t-1)$-design.*

Theorem 3 (Seymour, Zaslavsky 1984 [19]) *Let X be a path-connected topological space endowed with a measure dX with full support and normalized to unity, and let $f\colon X \longrightarrow \mathbb{C}^{n\times m}$ be a continuous integrable function. Then there exists a finite set $Y \subset X$ such that*

$$\frac{1}{|Y|}\sum_{x\in Y} f(x) = \int_X f(x)dX, \tag{13}$$

where $|Y|$ can be any number with a finite number of exceptions.

Corollary 4 ([18]) A minimum unitary t-design $(\mathfrak{U}, w)$ exists for all $d, t \geq 1$.

Definition 3 (Approximate Unitary t-Design [12, 18]) A tuple $(\mathfrak{U}, w)$ with a finite set $\mathfrak{U} \subset U(d)$ and weight function w on $\mathfrak{U}$ is an ε-approximate unitary t-design if

$$\left\| \sum_{U\in\mathfrak{U}} w(U)\, U^{\otimes t} \otimes (U^*)^{\otimes t} - \int_{U(d)} U^{\otimes t} \otimes (U^*)^{\otimes t}\, d\mu(U) \right\|_{1\to 1} < \varepsilon, \tag{14}$$

Note. Different norms are used in the literature to define ϵ-approximate unitary t-designs. Dankert et al. [13] use the diamond norm ($||\cdot||_\diamond$), which provides a stronger notion of approximation suitable for composable security ($||\cdot||_{1\to1} \leq ||\cdot||_\diamond$). Conversely, Brandão et al. [15] use the operator norm ($||\cdot||_\infty$) of the tensor moment operator. Our choice of the $(1 \to 1)$-norm follows the framework of quantum private broadcasting established by Broadbent et al. [12].

2.4 Symmetric unitary t-designs and quantum private broadcasting

In the paper [12], Broadbent et al. defined symmetric unitary t-designs and used it to define and construct a quantum private broadcast encryption scheme. We will use this definition to construct an attribute-based quantum broadcast encryption when a user satisfies multiple access policies together.

Let $\mathcal{H}_M = \mathcal{H}_d$ denote the message space and $\mathcal{H}_C$ be the ciphertext space. Let $\mathcal{K}$ be the set of all possible keys.

Definition 4 (Indistinguishable Ciphertexts for t-Policy Quantum Broadcast Encryption [12, 16]) A t-policy-based quantum broadcast encryption protocol exhibits **ε-indistinguishable ciphertexts** if the ensemble average of encryption operations over the entire key space approximates a fixed state replacement channel restricted to the symmetric subspace, with bounded deviation in the $1 \to 1$ operator norm. That is, there exists a fixed quantum state $\sigma \in \mathcal{D}(\mathcal{H}_d^{\otimes t})$ such that

$$\left\| \left(\mathbb{E}_{k\in\mathcal{K}}[\mathsf{Enc}_k] - \mathcal{R}_\sigma \right)(\rho) \Big|_{\mathrm{Sym}(d^t)} \right\|_{1\to 1} < \varepsilon, \tag{15}$$

where the state replacement channel is defined as $\mathcal{R}_\sigma(\rho) := \mathrm{Tr}(\rho)\sigma$ for arbitrary $\rho \in \mathcal{D}(\mathcal{H}_d^{\otimes t})$.

An encryption protocol achieves ε-indistinguishable ciphertexts against adversaries possessing side information if

$$\left\| \left(\mathbb{E}_{k\in\mathcal{K}}[\mathsf{Enc}_k] - \mathcal{R}_\sigma \right)(\rho) \Big|_{\mathrm{Sym}(d^t)} \right\|_{\diamond} < \varepsilon. \tag{16}$$

Note that security against adversaries with side information provides a stronger security than standard indistinguishability, as the $1 \to 1$ norm is bounded by diamond norm: $\|\cdot\|_{1\to 1} \le \|\cdot\|_\diamond$ [12].

Lemma 5 (Dimension Reduction Property [12]) *Consider a Quantum Private Broadcasting (QPB) scheme operating on t copies, characterized by encryption and decryption mappings*

$$\mathsf{Enc}_k^{(t)}(\rho) = U_k^{\otimes t}\rho(U_k^{\otimes t})^\dagger, \tag{17}$$

$$\mathsf{Dec}_k(\gamma) = U_k^\dagger \gamma U_k, \tag{18}$$

where U_k represents the unitary operator associated with key $k \in \mathcal{K}$. If this scheme achieves perfect correctness and perfect security on the symmetric subspace $\mathrm{Sym}(d^t)$, then the dimensionally reduced scheme for $(t-1)$ copies, defined via partial trace:

$$\mathsf{Enc}_k^{(t-1)}(\rho') := \mathrm{Tr}_1 \left[\mathsf{Enc}_k^{(t)}(\rho' \otimes |\varphi\rangle\langle\varphi|) \right],$$

maintains both perfect correctness and perfect security properties for $(t-1)$ policies, where $|\varphi\rangle$ is an arbitrary reference state. That is, both correctness and indistinguishability of ciphertexts are preserved under dimensional reduction.

Theorem 6 (QPB Construction from Approximate Unitary Designs [12]) *Let $(\mathfrak{U} = \{U_k\}_{k\in\mathcal{K}}, w)$ constitute an ε-approximate unitary t-design with weight function $w : \mathfrak{U} \longrightarrow [0,1]$. For $k \in \mathcal{K}$, $\rho \in \mathcal{D}(\mathrm{Sym}(d^t))$, and $\tau \in \mathcal{D}(\mathcal{H}_d)$, define the encryption transformation as $\mathsf{Enc}_k(\rho) = U_k^{\otimes t}\rho(U_k^{\otimes t})^\dagger$ and the corresponding decryption operation as $\mathsf{Dec}_k(\Lambda) = U_k^\dagger \Lambda U_k$.*

The resulting t-policy QPB scheme satisfies the following, for all quantum states ρ in the symmetric subspace $\mathrm{Sym}(d^t)$:

- ***Correctness:*** *The composition $(\mathsf{Dec}_k^{\otimes t} \circ \mathsf{Enc}_k)$ approximates the identity mapping on $\mathrm{Sym}(d^t)$ with error bounded by δ, where δ depends on the approximation quality of the unitary design.*
- ***Security:*** *The averaged encryption mapping over all keys is ε-close to a fixed state replacement channel in the $1 \to 1$ norm.*

Furthermore, if $(\mathfrak{U}, w)$ forms an exact unitary t-design, then the constructed scheme achieves perfect correctness and perfect security, including robustness against adversaries possessing quantum side information, i.e., we obtain a perfect t-QPB protocol.

Definition 5 (Approximate Symmetric Unitary t-Design [12]) Let $\mathfrak{U} = \{U_k\}_{k\in\mathcal{K}} \subset U(d)$ be a finite collection of unitary operators with associated weight function $w : \mathfrak{U} \longrightarrow [0, 1]$ satisfying the normalization condition $\sum_{k\in\mathcal{K}} w(U_k) = 1$.
The pair $(\mathfrak{U}, w)$ is called an **ε-approximate symmetric unitary t-design** if

$$\left\| \left(\mathbb{E}_{\mathfrak{U}}[\mathcal{E}^{(t)}_{U_k}] - \mathcal{T}^{(t)}_{\mathrm{Haar}} \right) \Big|_{\mathrm{Sym}(d^t)} \right\|^{\mathrm{Sym}(d^t)}_{1\to1} < \varepsilon,$$

where $\mathcal{E}^{(t)}_{U_k}(\rho) = U_k^{\otimes t}\rho(U_k^{\dagger})^{\otimes t}$ represents the t-fold conjugation channel, $\mathcal{T}^{(t)}_{\mathrm{Haar}}(\rho) = \int_{U(d)} U^{\otimes t}\rho(U^{\dagger})^{\otimes t} d\mu_{\mathrm{Haar}}(U)$ denotes the Haar-averaged twirling channel, and μ_{Haar} is the normalized Haar measure on $U(d)$.

Corollary 7 (Perfect Security for Exact Designs [12]) When $(\mathfrak{U}, w)$ constitutes an exact symmetric unitary t-design, the resulting QPB scheme provides perfect security against adversaries possessing arbitrary quantum side information.

Lemma 8 (Hereditary Property of Symmetric Unitary Designs [12]) *If $(\mathfrak{U}, w)$ constitutes a symmetric unitary t-design, then $\mathfrak{U}$ naturally inherits the structure of a symmetric unitary $(t-1)$-design.*

2.5 Classical secret-sharing scheme

We refer to a survey article [17] by Beimel for classical secret sharing. The description in this section is based on the article [17].

Definition 6 [17] Let $P = \{\mathsf{pl}_1, ..., \mathsf{pl}_n\}$ be a set of players. A collection $\Gamma \subseteq \mathcal{P}(P)$ is *monotone* if $B \in \Gamma$ and $B \subseteq C$ imply that $C \in \Gamma$.

Definition 7 [17] An *access structure* is a *monotone collection* $\Gamma \subseteq \mathcal{P}(P)$ of *non-empty* subsets of P.

If $B \in \Gamma$, then B is called an *authorized* set. Let $\mathcal{B}$ be the collection of *unauthorized* sets.

Let $\mathcal{S}$ be a set of secrets. Let R be a finite set of random strings, $\mu : R \longrightarrow \mathbb{R}$ be a probability distribution function, and $\forall j,\ 1 \le j \le n,\ \mathcal{S}_j$ be the domain of shares of

pl_j.
We define a function $\pi : \mathcal{S} \times R \longrightarrow \prod_{i=1}^{n} \mathcal{S}_i$ by $\pi(k, r) := (s_1, ..., s_n)$; where $k \in \mathcal{S}$ is a secret, $r \in R$ is a random string and s_j is a share privately communicated to the party pl_j by the dealer pl_0.

Definition 8 [17] A distribution scheme $\Sigma :=< \pi, \mu >$ with domain of secrets $\mathcal{S}$ is a pair of π and μ.
If $A \subseteq P$, we denote $\pi(k, r)_A$ as the restriction of $\pi(k, r)$ to its A-entries.

Definition 9 [17] Let $\mathcal{S}$ be a finite set of secrets, where $|\mathcal{S}| \geq 2$. A distribution scheme is a *secret-sharing scheme* realizing an access structure Γ if the following two conditions hold

Correctness: The secret k can be reconstructed by any authorized set of players, i.e. $\forall B \in \Gamma$(where $B = \{\mathsf{pl}_{i_1}, ..., \mathsf{pl}_{i_{|B|}}\}$), $\exists\ RECON_B : \prod_{j=1}^{j=|B|} \mathcal{S}_{i_j} \longrightarrow \mathcal{S}$ such that,

$$\forall k \in \mathcal{S}, \quad \Pr[RECON_B(\pi(k, r)_B) = k] = 1.$$

Privacy: Any unauthorized set cannot learn anything about the secret from their shares. i.e. $\forall T \notin \Gamma$, for every two secrets $a, b \in \mathcal{S}$, and for every possible vector of shares $< s_j >_{\mathsf{pl}_j \in T}$:

$$\Pr\Big[\pi(a, r)_T =< s_j >_{\mathsf{pl}_j \in T}\Big] = \Pr\Big[\pi(b, r)_T =< s_j >_{\mathsf{pl}_j \in T}\Big].$$

Definition 10 ((t, n)–Threshold Access Structure, [17]) For fixed integers t and n with $1 \leq t \leq n$, the *threshold access structure* is defined as

$$\Gamma_{t,n} = \{A \subseteq \mathcal{P} : |A| \geq t\}.$$

The first secret-sharing scheme, proposed by Shamir [20], is based on Lagrange interpolation. The Shamir secret-sharing scheme is a linear scheme for (t, n)-threshold access structures, where a set of at least t participants is authorized.

For more general access structures, Benaloh and Leichter proposed a recursive construction based on representing the policy as a monotone Boolean formula [21]. The scheme is built by recursively applying sharing schemes for AND and OR gates. This laid the groundwork for constructions that convert access trees directly to LSSS matrices. Then Karchmer and Wigderson showed that LSSS are equivalent to Monotone Span Programs (MSPs) [22]. An MSP is defined by a labeled matrix over a field, and it accepts an input if the rows corresponding to the input span a target vector.

2.6 Attribute-based quantum encryption

The basic CP-ABE (Ciphertext Policy-Attribute-based Encryption) model includes four components: cloud service provider (CSP), attribute authority (AA), data owner (DO), and data user (DU).

Let Ω be a finite set of attributes. An access policy is a monotone predicate $\mathsf{Policy}\colon 2^{\Omega} \to \{0, 1\}$. For a set $S \subseteq \Omega$, we write $S \vDash \mathsf{Policy}$ if the policy is satisfied.

2.7 Overview of Attribute-based Access Control

Before presenting the formal definitions, we provide an overview of attribute-based access control and the tree-based policy structures.

An *Attribute-based encryption* (ABE) is a paradigm where access depends on the satisfaction of a policy defined by a subset of attributes. Each user in the system possesses a *set of attributes* $S_{user} \subseteq \Omega$, where Ω is the universe of all possible attributes.

Access Policies as Logical Formulas. An *access policy* specifies the conditions under which a user may decrypt a message. Policies are expressed as logical combinations of attributes using AND ($\wedge$) and OR ($\vee$) gates.

Tree-Based Representation of Policies. Policies are naturally represented as *access trees*, where:

- **Leaf nodes** are labeled with individual attributes from Ω.
- **Internal nodes** are labeled with logical gates (AND or OR), or more generally, with *threshold gates*.
- **The root** represents the overall policy.

A **threshold gate** (t, n) at an internal node with n children is satisfied if at least t of its children are satisfied. This generalizes:

- AND gate: (n, n)-threshold (all children must be satisfied)
- OR gate: $(1, n)$-threshold (at least one child must be satisfied)
- k-of-n gate: (k, n)-threshold (at least k children must be satisfied)

From Policies to Secret Sharing. The *Linear Secret-Sharing Schemes* (LSSS) connects access trees to cryptography. Given an access tree T:

1. The tree is converted into an LSSS matrix (M, π), where M is a matrix over a finite field $\mathbb{F}_p$ and π maps each row to an attribute.
2. A secret s (in our case, the index of the encryption unitary) is shared by computing $\vec{\lambda} = M\vec{v}$, where $\vec{v} = (s, r_2, \ldots, r_c)^T$ contains the secret and random values.
3. Each share λ_j is associated with attribute $\pi(j)$ and distributed accordingly.
4. **Reconstruction:** If a user's attribute set S satisfies the policy (i.e., $S \models \mathsf{T}$), then the rows of M corresponding to S span the target vector, allowing recovery of s.
5. **Privacy:** If $S \not\models \mathsf{T}$, the shares reveal no information about s.

This transformation enables *fine-grained access control*: the same broadcast ciphertext can be decrypted by any user whose attributes satisfy the embedded policy, without requiring separate encryptions for each authorized user.

Multi-Policy Broadcasting. In our MP-QPBE construction, we extend this framework to handle *multiple policies simultaneously*. Each policy block $\Omega_i = \{\mathsf{Policy}_{j_1}, \ldots, \mathsf{Policy}_{j_{t_i}}\}$ governs access to a tensor product of quantum messages. The composite access tree $\mathbb{T}_i = \mathtt{AND}(\mathsf{T}^{j_1}, \ldots, \mathsf{T}^{j_{t_i}})$ requires satisfaction of *all* component policies, ensuring that only users with comprehensive authorization can access the complete message set.

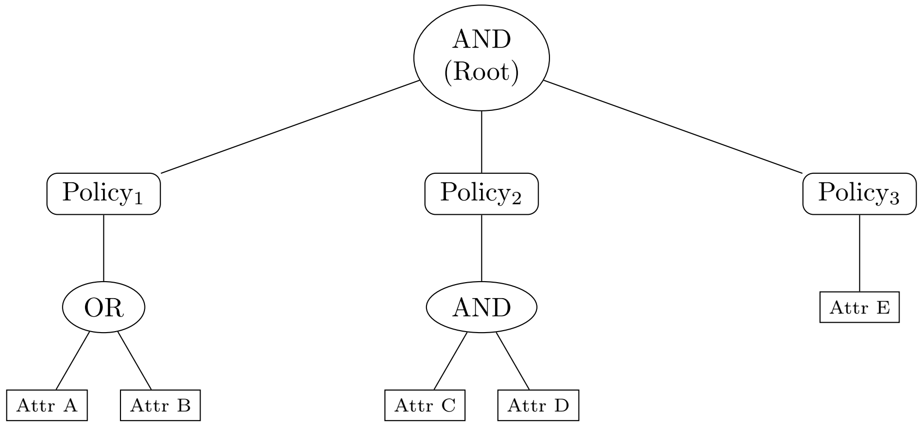

Fig. 1 Composite access tree for a policy set $\Omega_i = \{\mathsf{Policy}_1, \mathsf{Policy}_2, \mathsf{Policy}_3\}$ in MP-QPBE. The root AND gate requires all three policies to be satisfied. Policy_1 requires attribute A or B; Policy_2 requires both C and D; Policy_3 requires E. A user must satisfy all three policies to decrypt the associated quantum message block.

Definition 11 (Attribute-based Quantum Encryption (ABQE)) An *Attribute-based Quantum Encryption* scheme is a quadruple of quantum polynomial-time (QPT) algorithms:

$$\mathsf{ABQE} = (\mathsf{Setup}, \mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec}),$$

defined as follows:

- **Setup:** This is a QPT algorithm that takes a security parameter $\lambda \in \mathbb{N}$, and outputs public parameters PP and a master secret key MSK.

$$(\mathsf{PP}, \mathsf{MSK}) \longleftarrow \mathsf{Setup}(1^\lambda),$$

  where $\lambda \in \mathbb{N}$ is a security parameter, PP denotes public parameters, and MSK is the master secret key.
- **Key Generation:** The key generation algorithm is a QPT algorithm defined as follows:

$$k_S \longleftarrow \mathsf{KeyGen}(\mathsf{MSK}, S),$$

  where $S \subseteq \Omega$ is an attribute set, and k_S is a secret key associated with S.
- **Encryption:**

$$\rho_C \longleftarrow \mathsf{Enc}(\mathsf{PP}, \rho_M, \mathsf{Policy}),$$

  where $\rho_M \in \mathcal{D}(\mathcal{H}_M)$ is a quantum message, $\mathsf{Policy}\colon 2^\Omega \longrightarrow \{0,1\}$, and the ciphertext $\rho_C \in \mathcal{D}(\mathcal{H}_C)$ encodes both the encrypted message and policy.
- **Decryption:**

$$\rho'_M \longleftarrow \mathsf{Dec}(\rho_C, k_S),$$

  where $\rho'_M \in \mathcal{D}(\mathcal{H}_M)$ is the decryption attempt. If $S \vDash \mathsf{Policy}$, then $\Delta(\rho'_M, \rho_M) < \varepsilon(\lambda)$; otherwise ρ'_M is independent of ρ_M.

•**Correctness.** Let $\varepsilon(\lambda)$ be a negligible function. Then for all $\lambda \in \mathbb{N}$, all $\rho_M \in \mathcal{D}(\mathcal{H}_M)$, all policies Policy, and all sets $S \subseteq \Omega$ such that $S \vDash \mathsf{Policy}$, it holds that:

$$\Pr\left[\Delta\big(\mathsf{Dec}(\mathsf{Enc}(\mathsf{PP}, \rho_M, \mathsf{Policy}), k_S),\ \rho_M\big) \le \varepsilon(\lambda)\right] \ge 1 - \varepsilon(\lambda),$$

where $\Delta(\rho, \sigma) := \frac{1}{2}\|\rho - \sigma\|_1$ is the trace distance.

Security.(Quantum Indistinguishability under Chosen Plaintext Attacks (QIND-CPA)). A QABE scheme is *QIND-CPA secure* if for all QPT adversaries $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$, the following experiment yields advantage at most negligible in λ.

QIND-CPA Security Experiment: Let $b \in \{0, 1\}$ be chosen uniformly at random.

1. Challenger runs $(\mathsf{PP}, \mathsf{MSK}) \leftarrow \mathsf{Setup}(1^\lambda)$, and gives PP to $\mathcal{A}_1$.
2. $\mathcal{A}_1$ outputs
$$(\rho_0, \rho_1, \mathsf{Policy}, \Omega),$$
where $\rho_0, \rho_1 \in \mathcal{D}(\mathcal{H}_M)$ are of equal dimension, Policy is an access policy, and 2^Ω is a collection of attribute sets such that for all $S \subseteq \Omega$, we have:
$$S \not\models \mathsf{Policy}.$$
3. For each $S \subseteq \Omega$, the challenger computes $k_S \leftarrow \mathsf{KeyGen}(\mathsf{MSK}, S)$ and gives all $\{k_S\}_{S \subseteq \Omega}$ to $\mathcal{A}_2$.
4. The challenger computes
$$\rho_C^* \leftarrow \mathsf{Enc}(\mathsf{PP}, \rho_b, \mathsf{Policy})$$
and gives ρ_C^* to $\mathcal{A}_2$.
5. $\mathcal{A}_2$ outputs a guess $b' \in \{0, 1\}$.

Define the adversary's advantage as
$$\mathrm{Adv}_{\mathcal{A}}^{\text{QIND-CPA}}(\lambda) := \left| \Pr\left[b' = b\right] - \frac{1}{2} \right|.$$
The scheme is secure if
$$\forall \text{ QPT } \mathcal{A}, \quad \mathrm{Adv}_{\mathcal{A}}^{\text{QIND-CPA}}(\lambda) < \varepsilon(\lambda).$$

2.8 Adversarial Model and Security Goals

To analyze the security of our proposed schemes, we define the capabilities of the adversary and the security properties the protocols must satisfy.

2.8.1 Adversary's Capabilities

We consider a powerful adversary, denoted by $\mathcal{A}$, with the following capabilities:

1. **Computationally Unbounded:** The adversary $\mathcal{A}$ is not restricted to polynomial-time computations. This allows us to prove security in the strong information-theoretic sense, which is independent of any computational hardness assumptions.
2. **Honest-but-Curious (Semi-Honest):** An honest-but-curious (also called semi-honest or passive) adversary is an internal participant of the system [23], who:
    - **Follows the protocol exactly:** The adversary correctly executes all steps of the protocol as specified, using the prescribed algorithms for key generation, encryption, and decryption.

- **Does not deviate or inject faults:** The adversary does not modify messages, send malformed ciphertexts, abort prematurely, or otherwise deviate from the protocol specification.
- **Attempts to learn additional information:** Despite following the protocol, the adversary analyzes all data available to it, including the public parameters, broadcast ciphertext, its own private key shares, and any legitimately received messages, to gain unauthorized information about messages for which it lacks decryption rights.

3. **Collusion:** The adversary $\mathcal{A}$ can represent a coalition of any number of unauthorized users. Let $S_{\mathcal{A}} = \bigcup_{u \in \text{Coalition}} S_u$ be the total set of attributes possessed by the colluding parties. They can pool all their private keys and classical information.
4. **Quantum Side Information:** In the strongest adversarial setting, we grant $\mathcal{A}$ the ability to possess a quantum system, $\mathcal{H}_E$, which may be entangled with the original message state ρ_M. The joint state of the message and the adversary's system is a density operator $\rho_{ME} \in \mathcal{D}(\mathcal{H}_M \otimes \mathcal{H}_E)$. The adversary's goal is to use their side information to distinguish ciphertexts.

Realistic Scenarios. [23] We discuss some realistic examples where the semi-honest model is appropriate:

- Insider threats in regulated environments (employees deterred from overt violations by audit trails).
- Curious service providers (cloud providers with economic incentives to maintain compliance).
- Compromised-but-undetected nodes (systems maintaining normal operation to avoid detection).
- Quantum network infrastructure (intermediate nodes with incentives to preserve state fidelity).
- Multi-party computation with rational parties (long-term relationships where cheating costs outweigh gains).

Note. We refer [Page 619, Section 7.2.2, [23]] for a detailed description of the semi-honest model. Their security proofs, which establish the privacy properties of our construction of unitary t-design-based encryption. We assume that adversaries observe the ciphertext and their key shares but do not actively interfere with the protocol. Since our MP-QPBE and CI-QBE schemes build directly upon their framework, we inherit this assumption. Similarly, the classical LSSS-based secret sharing [17, 20] that we employ for key distribution provides *information-theoretic security* against passive adversaries: unauthorized coalitions learn nothing about the secret from their shares. This guarantee does not extend to active adversaries who might, for example, submit maliciously constructed shares during reconstruction. We acknowledge that the semi-honest assumption does not encompass all practical adversarial scenarios, and we also discuss these limitations in Section 6.

2.8.2 Security Goals

A secure attribute-based quantum broadcast encryption scheme must satisfy two primary goals: correctness for authorized users and privacy against unauthorized adversaries.

1. **Correctness:** For any set of user attributes S that satisfies a given access policy Policy (i.e., $S \vDash \mathsf{Policy}$), the user must be able to correctly recover the original quantum message ρ_M from the broadcast ciphertext ρ_C. That is, if $\rho'_M = \mathsf{Dec}(\rho_C, k_S)$, then the trace distance between the original and decrypted states must be negligible:
$$\frac{1}{2}\left\|\rho_M - \rho'_M\right\|_1 < \varepsilon,$$
for some negligible function $\varepsilon(\lambda)$. For our schemes, in which we use exact designs, we achieve perfect correctness, i.e., $\varepsilon = 0$.
2. **Privacy (Security):** For any adversary $\mathcal{A}$ whose attribute set $S_{\mathcal{A}}$ does not satisfy the access policy ($S_{\mathcal{A}} \nvDash \mathsf{Policy}$), no information about the message ρ_M should be revealed.
   - **Information-Theoretic Security:** The adversary's view of the ciphertext, averaged over all possible choices of the secret key $k \in \mathcal{K}$, must be a fixed state that is independent of the plaintext ρ_M. Let $\sigma_k = \mathsf{Enc}_k(\rho_M)$ be the ciphertext for key k. The adversary's perceived state is:
$$\rho_{\mathcal{A}} = \mathbb{E}_{k \in \mathcal{K}}[\sigma_k] = \sum_{k \in \mathcal{K}} w(k)\,\mathsf{Enc}_k(\rho_M)$$
     The scheme is secure if $\rho_{\mathcal{A}}$ is a fixed state, specifically the maximally mixed state on the relevant Hilbert space (or subspace), e.g., $\rho_{\mathcal{A}} = \frac{\mathbb{I}}{\dim(\mathcal{H}_C)}$.
   - **Perfect Security against Quantum Side Information:** This is the strongest security guarantee. The average encryption channel, $\bar{\mathcal{E}} = \mathbb{E}_{k \in \mathcal{K}}[\mathsf{Enc}_k]$, when acting on any joint state $\rho_{ME} \in \mathcal{D}(\mathcal{H}_M \otimes \mathcal{H}_E)$, must be indistinguishable from a fixed state-replacement channel $\mathcal{R}_{\sigma_E}$. The state-replacement channel is defined as $\mathcal{R}_{\sigma_E}(\rho_{ME}) = \sigma \otimes \mathrm{Tr}_{\mathcal{H}_M}(\rho_{ME})$, where σ is a fixed state on $\mathcal{H}_M$. Perfect security means the diamond norm distance between these two channels is zero:
$$\left\|\bar{\mathcal{E}} - \mathcal{R}_{\sigma_E}\right\|_\diamond = 0$$
     This guarantees that even an adversary with pre-existing entanglement with the message cannot learn anything about it.

2.9 Broadcast encryption

The classical broadcast encryption was proposed by Fiat and Noar in [1]. We refer to [3] for a good survey article on broadcast encryption.

Let $\mathcal{U} = \{u_1, u_2, \ldots, u_N\}$ be the set of N users, and let $R \subseteq \mathcal{U}$, be the set of *revoked* users with $|R| = r$. The revoked users will not be able to decrypt the messages in the next transmissions. Let $\mathcal{K}$ be the key space. Let $\mathsf{Enc} : \mathcal{K} \times \mathcal{M} \longrightarrow \mathcal{C}$ be an encryption algorithm, where $\mathcal{M}$ is the message space, $\mathcal{C}$ is the ciphertext space.

A broadcast encryption scheme enables a sender to deliver a fresh session key $k \in \mathcal{K}$ securely to all non-revoked users $\overline{R} = \mathcal{U} \setminus R$ while preventing any coalition of users in R from learning k. The most basic form of broadcast encryption encrypts the message separately for each authorized user. Each user $u \in \mathcal{U}$ is assigned a unique key $K_u \in \mathcal{K}$. To send a message $M \in \mathcal{M}$, the sender constructs a ciphertext:

$$\mathsf{B} := (u_1, \mathsf{Enc}(K_{u_1}, M)) \| \cdots \| (u_{N-r}, \mathsf{Enc}(K_{u_{N-r}}, M)),$$

where $\{u_1, \ldots, u_{N-r}\} = \overline{R}$, the set of recipients. Each $u \in \overline{R}$ locates its tuple and recovers $M = \mathsf{Enc}^{-1}(K_u, C)$.

Definition 12 (Broadcast Encryption Scheme) [3] A broadcast encryption scheme is a tuple of algorithms $\Pi_{\mathsf{B}} = (\mathsf{Setup}, \mathsf{Broadcast}, \mathsf{Decrypt})$ defined as follows:

1. Setup : For each user $u \in \mathcal{U}$, this algorithm outputs the user's private information $P_u \in \mathcal{P}$, for some $\mathcal{P}$ associated with the scheme.
2. $\mathsf{Broadcast}$: Given a revoked set $R \subseteq \mathcal{U}$ and session key $K \in \mathcal{K}$, output a broadcast message B.
3. $\mathsf{Decrypt}$: A user u runs $\mathsf{Decrypt}(\mathsf{B}, P_u, u)$, which returns K if $u \in \overline{R}$; otherwise, it returns $\perp$.

3 Technical details (Attribute-based quantum private broadcast encryption satisfying multiple access policies)

Definition 13 (Threshold-Gate Access Tree, cf. [7]) A *Threshold-Gate Access Tree* is a rooted tree structure $\mathsf{T} = (V, E)$, where each internal node $v \in V$ possesses a set of children denoted by $\mathsf{child}(v) = \{v_1, v_2, \ldots, v_{n_v}\}$, with $n_v \geq 2$, and is further assigned a threshold parameter $t_v \in \{1, 2, \ldots, n_v\}$.

Each terminal (leaf) node $v \in L \subseteq V$ is uniquely labeled by an attribute $\alpha \in \Omega$, where Ω is the universal set of attributes.

An attribute subset $S \subseteq \Omega$ is said to *satisfy* the tree T if the root node of T is satisfied under the following inductive rule:

- A leaf node labeled by $\alpha \in \Omega$ is satisfied by S if $\alpha \in S$;
- An internal node v is satisfied by S if at least t_v of its child nodes are satisfied by S.

Definition 14 (AND-OR Access Tree, cf. [7]) Let Ω denote a finite set of attributes. An *AND-OR Access Tree* is defined as a tuple $\mathsf{T} = (N, r, \mathsf{child}, \ell)$, where:

- N is a finite set of nodes;
- $r \in N$ is the designated root node;

- $\mathsf{child} : N \to 2^N$ assigns to each node its (possibly empty) set of children, such that:

$$\forall v \in N, \quad \mathsf{child}(v) = \emptyset \iff v \text{ is a leaf;}$$

- $\ell : N \to \Omega \cup \{\text{AND}, \text{OR}\}$ is a labeling function satisfying:

$$\ell(v) = \begin{cases} \alpha \in \Omega, & \text{if } v \text{ is a leaf,} \\ \text{AND or OR}, & \text{if } v \text{ is internal.} \end{cases}$$

The satisfaction relation $\mathsf{Sat}_S(v) \in \{0,1\}$, for $S \subseteq \Omega$, is defined recursively as:

$$\mathsf{Sat}_S(v) := \begin{cases} 1, & \text{if } v \text{ is a leaf and } \ell(v) \in S, \\ 1, & \text{if } \ell(v) = \text{AND and } \forall w \in \mathsf{child}(v),\ \mathsf{Sat}_S(w) = 1, \\ 1, & \text{if } \ell(v) = \text{OR and } \exists w \in \mathsf{child}(v),\ \mathsf{Sat}_S(w) = 1, \\ 0, & \text{otherwise.} \end{cases}$$

We declare that S satisfies the access tree T if and only if $\mathsf{Sat}_S(r) = 1$.

In the context of constructing a broadcast encryption system governed by attribute-based policies, we now define a unified structure to represent multiple policies through a combined logical tree.

Suppose there exist $N_L \in \mathbb{N}$ distinct messages $\rho^{(1)}, \rho^{(2)}, \ldots, \rho^{(N_L)}$, each associated with a separate attribute-based access policy Policy_k, for $k = 1, \ldots, N_L$. Each Policy_k is modeled as an AND-OR access tree $\mathsf{T}^{(k)}$ with logical structure defined over Ω.

We now define a single composite policy tree $\mathsf{T}_{\mathsf{Policy}}$, which integrates all N_L individual policies such that there exists a distinct root-to-leaf path P_k in $\mathsf{T}_{\mathsf{Policy}}$ corresponding to each individual policy Policy_k.

Definition 15 (AND-OR Policy Access Tree, cf. [2]) Let $\{\mathsf{Policy}_k\}_{k=1}^{N_L}$ be a collection of attribute-based policies, each represented by a tree $\mathsf{T}^{(k)}$. Define the *AND-OR Policy Access Tree* as a rooted tree $\mathsf{T}_{\mathsf{Policy}} = (N', r', \mathsf{child}, \ell)$, where:

- N' is a finite set of nodes;
- $r' \in N'$ is the root node;
- $\mathsf{child} : N' \to 2^{N'}$ is the child assignment function;
- $\ell : N' \to \{\text{AND}, \text{OR}\} \cup \{\mathsf{Policy}_1, \ldots, \mathsf{Policy}_{N_L}\}$ is the labeling function satisfying:

$$\ell(v) = \begin{cases} \mathsf{Policy}_k, & \text{if } \mathsf{child}(v) = \emptyset, \\ \text{AND or OR}, & \text{otherwise.} \end{cases}$$

The evaluation of whether a given attribute set $S \subseteq \Omega$ satisfies the composite tree $\mathsf{T}_{\mathsf{Policy}}$ proceeds as follows.

For any leaf node $v \in N'$ such that $\ell(v) = \mathsf{Policy}_k$, we define:

$$\mathsf{Sat}_S(v) := 1 \iff S \models \mathsf{T}^{(k)},$$

where $S \models \mathsf{T}^{(k)}$ indicates that S satisfies the tree $\mathsf{T}^{(k)}$, according to the earlier recursive rule.

Then, the satisfaction relation extends recursively by:

$$\mathsf{Sat}_S(v) := \begin{cases} 1, & \text{if } \ell(v) = \mathsf{Policy}_k \text{ and } S \models \mathsf{T}^{(k)}, \\ 1, & \ell(v) = \text{AND} \wedge \forall w \in \mathsf{child}(v),\ \mathsf{Sat}_S(w) = 1, \\ 1, & \ell(v) = \text{OR} \wedge \exists w \in \mathsf{child}(v),\ \mathsf{Sat}_S(w) = 1, \\ 0, & \text{otherwise.} \end{cases}$$

The attribute set $S \subseteq \Omega$ is said to satisfy the global access policy $\mathsf{T}_{\mathsf{Policy}}$ if and only if $\mathsf{Sat}_S(r') = 1$.

Therefore, within the framework of attribute-based access control (ABAC), each individual policy Policy_k is expressed as an AND-OR logical tree. Our construction integrates these policies into a unified AND-OR Policy Access Tree for logical enforcement of multiple message-specific access conditions.

3.1 Main construction

In this paper, we propose an attribute-based quantum broadcast encryption based on the classical attribute-based access structure.

For a policy set $\Omega_i = \{\mathsf{Policy}_{j_1}, \ldots, \mathsf{Policy}_{j_{t_i}}\}$ with $|\Omega_i| = t_i$ policies, the authority constructs a single general access tree $\mathbb{T}_i$ representing the logical AND of all policies, that is $\mathbb{T}_i = \mathsf{AND}(\mathsf{T}^{j_1}, \cdots, \mathsf{T}^{j_{t_i}})$ and constructs a linear secret-sharing scheme(LSSS). To construct the LSSS, we first convert the AND-OR access tree into a threshold access tree by the following conversion

- An **AND** gate with n_v children is equivalent to a **(n_v, n_v)-threshold gate**.
- An **OR** gate with n_v children is equivalent to a **(1, n_v)-threshold gate**.

Construction of the LSSS matrix from the threshold access tree.
First we translate the logical policy $\mathbb{T}_i$ into the LSSS matrix (M_i, π_i). We implement the standard tree-to-LSSS conversion algorithm based on the foundational work of Bethencourt, Sahai, and Waters [2], using the LSSS methodology of Waters [5] and the threshold gate optimization of Liu, Cao, and Wong [7]. The polynomial evaluation approach for threshold gates is based on Shamir's secret sharing [20] as applied to access trees in CP-ABE constructions [2, 5]. The LSSS conversion methodology follows the works of Lewko and Waters [6] and the works of Liu et. al. [7].

Algorithm 1 Conversion of Threshold Access Tree to LSSS Matrix

Input: An access tree $\mathbb{T}_i$ with root r.
Output: An LSSS matrix M over $\mathbb{F}_p$ and mapping π from rows to attributes.
Data structures: L (a list of pairs $(\vec{u}, \alpha)$ with $\vec{u} \in \mathbb{F}_p^d$), Q (a queue of pairs $(v, \vec{u}_v)$).
Initialize $L \leftarrow [\,]$.
Label the root node r with the vector $\vec{u}_r \leftarrow (1) \in \mathbb{F}_p^1$.
Initialize $Q \leftarrow [(r, \vec{u}_r)]$.
while Q is not empty **do**
  Dequeue $(v, \vec{u}_v)$.
  if v is a leaf node with attribute α **then**
    Append $(\vec{u}_v, \alpha)$ to L.
  else
    v is an internal node with threshold (t_v, n_v) and children $v_1, \ldots, v_{n_v}$
    Let $q_v(x)$ be a random polynomial over $\mathbb{F}_p$ of degree $t_v - 1$.
    Set constant term $q_v(0) \leftarrow$ last element of $\vec{u}_v$.
    Choose remaining coefficients of $q_v(x)$ uniformly at random from $\mathbb{F}_p$.
    for $j = 1$ **to** n_v **do**
      Compute $s_j \leftarrow q_v(j)$ in $\mathbb{F}_p$.
      Set $\vec{u}_{v_j} \leftarrow \vec{u}_v \parallel s_j$.
      Enqueue $(v_j, \vec{u}_{v_j})$ into Q.
    end for
  end if
end while
Let $c \leftarrow \max_{(\vec{u},\alpha)\in L} |\vec{u}|$.
Initialize M as a zero matrix in $\mathbb{F}_p^{|L|\times c}$.
for $j = 1$ **to** $|L|$ **do**
  Let $(\vec{u}, \alpha)$ be the j-th element of L.
  Set $(M)_j \leftarrow \vec{u}$ right-padded with $c - |\vec{u}|$ zeros.
  Set $\pi(j) \leftarrow \alpha$.
end for
return (M, π).

We apply the classical, information-theoretically secure LSSS secret-sharing scheme to share a classical secret—the index k_i^* of a unitary operator U_{i,k_i^*} from a symmetric unitary t_i-design, which we use to encrypt our messages associated with the set of t_i-policies.

Classical secret-sharing scheme construction.

• **Share generation:**

1. **Secret selection:** For each block i, the authority must choose a prime p_i for the LSSS matrix M_i such that $p_i > |\mathfrak{U}_i|$. The LSSS operations for block i are then performed in the field $\mathbb{F}_{p_i}$. For simplicity, we may choose a single large prime p such that $p > \max\{|\mathfrak{U}_i| : i \in [m]\}$, and consider the field $\mathbb{F}_p$ for all blocks. Fix an

ordering of the unitary t_i-designs. For each policy block i, the authority chooses the secret index $s \equiv k_i^* \in \mathbb{F}_p$ at uniformly random so that U_{i,k_i^*} is contained in unitary t_i-design $(\mathfrak{U}_i, w_i)$.

2. **Secret vector construction:** The authority constructs a random column vector $\vec{v} \in \mathbb{F}_p^c$, where c is the number of columns in M_i. The first component is the secret s, and the remaining $c-1$ components are chosen uniformly at random from $\mathbb{F}_p$.

$$\vec{v} = (s, r_2, r_3, \ldots, r_c)^T$$

3. **Share:** The authority computes the vector of shares $\vec{\lambda} = M_i\vec{v}$. The j-th share $\lambda_j = (M_i)_j \cdot \vec{v}$ belongs to the attribute $\pi_i(j)$.

- **Reconstruction:**

1. **Identify shares and coefficients:** An authorized user with an attribute set S_{user} gathers their shares $\{\lambda_{i,j}\}$ and computes the reconstruction coefficients $\{\omega_j\}$ by the LSSS correctness theorem.
2. **Combine shares:** The user applies these coefficients to their shares to recover the secret s:

$$\begin{aligned} s' &= \sum_{j \in I_{S_{user}}} \omega_j \lambda_{i,j} = \sum_{j \in I_{S_{user}}} \omega_j \left((M_i)_j \cdot \vec{v}\right) \\ &= \left(\sum_{j \in I_{S_{user}}} \omega_j (M_i)_j \right) \cdot \vec{v} = \vec{e_1} \cdot \vec{v} = s \end{aligned}$$

The user successfully reconstructs the secret index $s = k_i^*$.

Theorem 9 (Correctness of LSSS [2, 5, 7]) *Let (M_i, π_i) be the LSSS matrix generated from an access policy $\mathbb{T}_i$. For any set of attributes S that satisfies $\mathbb{T}_i$, let $I_S = \{j \mid \pi_i(j) \in S\}$. There exists a set of constants $\{\omega_j \in \mathbb{F}_p\}_{j \in I_S}$, which can be computed in polynomial time, such that $\sum_{j \in I_S} \omega_j (M_i)_j = \vec{e_1}^T = (1, 0, \ldots, 0)$.*

Theorem 10 (Security of LSSS [2, 5, 7]) *Let (M_i, π_i) be as above. For any set of attributes S that does not satisfy $\mathbb{T}_i$, let $I_S = \{j \mid \pi_i(j) \in S\}$. The vector $\vec{e_1}$ is not in the linear span of the row vectors $\{(M_i)_j\}_{j \in I_S}$. Consequently, for a secret vector $\vec{v} = (s, r_2, \ldots, r_c)^T$ where s and r_k, $k = 2, \ldots, c$ are chosen uniformly at random from $\mathbb{F}_p$, the shares $\{\lambda_j = (M_i)_j \cdot \vec{v}\}_{j \in I_S}$ reveal no information about the secret s.*

Classical multi-secret sharing. Now we generalize the previous construction of Liu et. al. for multi-secret sharing to share all the classical keys together associated with each set of policies.

Given m access structures $\Gamma_1, \ldots, \Gamma_m$ with corresponding LSSS matrices (M_k, π_k) where $M_k \in \mathbb{F}_p^{\ell_k \times c_k}$, we construct an MS-LSSS as follows:

• **Composite Matrix Construction:** Define $\mathcal{M} \in \mathbb{F}_p^{\ell \times c}$ where $\ell = \sum_{k=1}^{m} \ell_k$ and $c = \sum_{k=1}^{m} c_k$:

$$\mathcal{M} = \begin{pmatrix} M_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & M_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & M_m \end{pmatrix}.$$

• **Extended Labeling Function:** Define $\varrho : [\ell] \to \Omega \times [m]$ by

$$\varrho(i) = \begin{cases} (\pi_1(i), 1) & \text{if } i \in [1, \ell_1] \\ (\pi_2(i - \ell_1), 2) & \text{if } i \in [\ell_1 + 1, \ell_1 + \ell_2] \\ \vdots & \vdots \\ (\pi_m(i - \sum_{j=1}^{m-1} \ell_j), m) & \text{if } i \in [\sum_{j=1}^{m-1} \ell_j + 1, \ell] \end{cases},$$

where Ω is the set of all attributes.

• **Share Generation:** For secrets $\mathbf{s} = (s_1, \ldots, s_m)^T$, construct the following vector

$$\mathbf{R} = \begin{pmatrix} s_1 \\ \mathbf{r}^{(1)} \\ s_2 \\ \mathbf{r}^{(2)} \\ \vdots \\ s_m \\ \mathbf{r}^{(m)} \end{pmatrix} \in \mathbb{F}_p^c,$$

where $\mathbf{r}^{(k)} \in \mathbb{F}_p^{c_k - 1}$ with each component chosen uniformly at random.

• **Share Distribution:** Compute $\mathbf{\Lambda} = \mathcal{M}\mathbf{R}$. A user with attribute set $S_{user} \subseteq \Omega$ satisfying policy set Ω_k receives

$$\Lambda_{S_{user}} = \{\Lambda_i : \exists k \in [m], \varrho(i) = (A, k) \text{ for some } A \in S_{user}\}.$$

Note: The set of attributes S_{user} is a multiset here.

The reconstruction algorithm is described below.

Algorithm 2 Multi-Policy MS-LSSS Secret Reconstruction

Input: User attribute set $S_{user} \subseteq \Omega$, target policy set index $k \in [m]$, shares $\Lambda_{S_{user}}$

Output: Secret s_k if S_{user} satisfies policy set Ω_k, $\perp$ otherwise

Define $I_{S_{user},k} = \{i \in [\ell] : \varrho(i) = (A, k) \text{ for some } A \in S_{user}\}$

Extract relevant shares $\{\Lambda_i\}_{i \in I_{S_{user},k}}$ for k-th policy set Ω_k

Extract submatrix $M_{S_{user},k} = \{M_i : i \in I_{S_{user},k}\}$

if $\mathbf{e}_1 \in \text{Span}(M_{S_{user},k})$ **and** S_{user} satisfies $\mathbb{T}_k = \mathsf{AND}(\mathsf{T}^{j_1}, \ldots, \mathsf{T}^{j_{t_k}})$ **then**

  Find $\{\omega_i\}_{i \in I_{S_{user},k}}$ such that $\sum_{i \in I_{S_{user},k}} \omega_i M_i = \mathbf{e}_1^T$

  Compute $s_k = \sum_{i \in I_{S_{user},k}} \omega_i \Lambda_i$

  return s_k

else

  return $\perp$

end if

Let $\mathcal{H}_M^{\otimes t_i} = \bigotimes_{j=1}^{t_i} \mathcal{H}_M$ be the message space, and a message is associated with i-th policy set Ω_i. Let $\mathcal{H}_C^{\otimes t_i} = \bigotimes_{j=1}^{t_i} \mathcal{H}_C$ be the ciphertext space. We define the encryption map $\mathsf{Enc}_{\Omega_i} : \mathcal{D}(\mathcal{H}_M^{\otimes t_i}) \longrightarrow \mathcal{D}(\mathcal{H}_C^{\otimes t_i})$. Let $\rho_{\Omega_i} = \rho^{(1)} \otimes \rho^{(2)} \otimes \cdots \otimes \rho^{(t_i)}$ be the quantum message, which is a tensor product of mutually orthogonal ordered pure states, associated with the policy set Ω_i.

Inspired by the work [12] of Broadbent et. al., we propose the following definition.

Definition 16 (Multi-Policy-based Quantum Private Broadcast Encryption (MP-QPBE)) Given a set Ω_i of t_i policies, an Ω_i-policy set-based quantum private broadcast encryption scheme consists of

- An encryption map
$$\mathsf{Enc}_{\Omega_i} : \mathcal{D}(\mathcal{H}_M^{\otimes t_i}) \longrightarrow \mathcal{D}(\mathcal{H}_C^{\otimes t_i})$$
- A decryption map
$$\mathsf{Dec}_{\Omega_i} : \mathcal{D}(\mathcal{H}_C^{\otimes t_i}) \longrightarrow \mathcal{D}(\mathcal{H}_M^{\otimes t_i})$$

satisfying the correctness condition:

$$\left\| (\mathsf{Dec}_{\Omega_i} \circ \mathsf{Enc}_{\Omega_i}) \,|_{\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i})} - \mathbb{I}_{\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i})} \right\|_{\diamond} < \delta \tag{19}$$

for some constant $\delta > 0$, where $\| \cdot \|_{\diamond}$ denotes the diamond norm.

Given a policy set $\Omega_i = \{\mathsf{Policy}_{j_1}, \mathsf{Policy}_{j_2}, \ldots, \mathsf{Policy}_{j_{t_i}}\}$, each policy Policy_{j_k} corresponds to message $\rho^{(k)}$. We assume that the messages are orthogonal pure states $\rho^{(i,k)} = |\psi_{i,k}\rangle\langle\psi_{i,k}|$ with $\langle\psi_{i,j}|\psi_{i,k}\rangle = \delta_{jk}$. The state associated with the policy set Ω_i is

$$\rho_{\Omega_i} = \bigotimes_{k=1}^{t_i} \rho^{(k)} \in \mathcal{D}(\mathcal{H}_M^{\otimes t_i}).$$

The total number of messages in the system is $N_L = \sum_{i=1}^{m} t_i$, as each policy is associated with exactly one message.

In [Remark 5.2., Page 12,[12]], it is mentioned that QPB with unitary t-designs cannot be used to broadcast states of the form $\rho^{\otimes t} \notin \mathcal{D}(\mathrm{Sym}(\mathcal{H}_M^{\otimes t}))$. To address this, in the pre-broadcasting stage, the protocol suggests performing a projective measurement $\{\tau_{\mathrm{Sym}}, \mathbb{I} - \tau_{\mathrm{Sym}}\}$ to determine whether the state is in the symmetric subspace.

In our construction, we take any message from $\mathcal{D}(\mathcal{H}_M^{\otimes t})$, then we perform symmetrization first and map it into $\mathcal{D}(\mathrm{Sym}(\mathcal{H}_M^{\otimes t}))$.
For $\rho_{\Omega_i} \in \mathcal{D}(\mathcal{H}_M^{\otimes t_i})$, the symmetrized state is

$$\tilde{\rho}_{\Omega_i} = \frac{\Pi_{\mathrm{Sym}(t_i)} \rho_{\Omega_i} \Pi_{\mathrm{Sym}(t_i)}}{\mathrm{Tr}\left[\Pi_{\mathrm{Sym}(t_i)} \rho_{\Omega_i} \Pi_{\mathrm{Sym}(t_i)}\right]},$$

where the symmetric projector is

$$\Pi_{\mathrm{Sym}(t_i)} = \frac{1}{t_i!} \sum_{\pi \in \mathfrak{S}_{t_i}} P_d(\pi),$$

and $P_d(\pi)$ is the permutation operator, and

$$P_d(\pi)|i_1\rangle \otimes |i_2\rangle \otimes \cdots \otimes |i_{t_i}\rangle = |i_{\pi^{-1}(1)}\rangle \otimes |i_{\pi^{-1}(2)}\rangle \otimes \cdots \otimes |i_{\pi^{-1}(t_i)}\rangle.$$

The operator $\Pi_{\mathrm{Sym}(t_i)}$ is an orthogonal projector, satisfying $\Pi_{\mathrm{Sym}(t_i)}^2 = \Pi_{\mathrm{Sym}(t_i)}$ and $\Pi_{\mathrm{Sym}(t_i)}^{\dagger} = \Pi_{\mathrm{Sym}(t_i)}$.

We consider both cases when ρ_{Ω_i} is a tensor product of pure states and a tensor product of mixed density matrices. Let the initial state be a tensor product of t_i pure, mutually orthogonal states

$$\rho_{\Omega_i} = \bigotimes_{k=1}^{t_i} \rho^{(i,k)} = \left(\bigotimes_{k=1}^{t_i} |\psi_{i,k}\rangle\langle\psi_{i,k}| \right),$$

where $\langle\psi_{i,j}|\psi_{i,k}\rangle = \delta_{jk}$. Then the symmetrized state $\tilde{\rho}_{\Omega_i}$ is given by

$$\tilde{\rho}_{\Omega_i} = \frac{\frac{1}{(t_i!)^2} \sum_{\pi,\sigma \in \mathfrak{S}_{t_i}} P_d(\pi) \left(\bigotimes_{k=1}^{t_i} |\psi_{i,k}\rangle\langle\psi_{i,k}| \right) P_d(\sigma)^{\dagger}}{\mathrm{Tr}\left[\Pi_{\mathrm{Sym}(t_i)} \rho_{\Omega_i} \Pi_{\mathrm{Sym}(t_i)}\right]}.$$

We compute $\tilde{\rho}_{\Omega_i}$ as follows

$$\frac{\Pi_{\mathrm{Sym}(t_i)} \rho_{\Omega_i} \Pi_{\mathrm{Sym}(t_i)}}{\mathrm{Tr}\left[\Pi_{\mathrm{Sym}(t_i)} \rho_{\Omega_i} \Pi_{\mathrm{Sym}(t_i)}\right]}$$

$$
\begin{aligned}
&= \frac{\left(\frac{1}{t_i!}\sum_{\pi\in\mathfrak{S}_{t_i}} P_d(\pi)\right)\left(\bigotimes_{k=1}^{t_i}\rho^{(i,k)}\right)\left(\frac{1}{t_i!}\sum_{\sigma\in\mathfrak{S}_{t_i}} P_d(\sigma)\right)}{\mathrm{Tr}\left[\Pi_{\mathrm{Sym}(t_i)}\rho_{\Omega_i}\Pi_{\mathrm{Sym}(t_i)}\right]} \\
&= \frac{\frac{1}{(t_i!)^2}\sum_{\pi\in\mathfrak{S}_{t_i}}\sum_{\sigma\in\mathfrak{S}_{t_i}} P_d(\pi)\left(\bigotimes_{k=1}^{t_i}\rho^{(i,k)}\right)P_d(\sigma)}{\mathrm{Tr}\left[\Pi_{\mathrm{Sym}(t_i)}\rho_{S\Omega_i}\Pi_{\mathrm{Sym}(t_i)}\right]}.
\end{aligned}
$$

Since the permutation operators $P_d(\sigma)$ are unitary, we have $P_d(\sigma)^{-1} = P_d(\sigma)^\dagger$. As summing over all $\sigma \in \mathfrak{S}_{t_i}$ is the same as summing over all $\sigma^{-1} \in \mathfrak{S}_{t_i}$, we get

$$
\tilde{\rho}_{\Omega_i} = \frac{\frac{1}{(t_i!)^2}\sum_{\pi,\sigma\in\mathfrak{S}_{t_i}} P_d(\pi)\left(\bigotimes_{k=1}^{t_i}|\psi_{i,k}\rangle\langle\psi_{i,k}|\right)P_d(\sigma)^\dagger}{\mathrm{Tr}\left[\Pi_{\mathrm{Sym}(t_i)}\rho_{\Omega_i}\Pi_{\mathrm{Sym}(t_i)}\right]}.
$$

Let

$$
\mathcal{N} = \mathrm{Tr}\left[\Pi_{\mathrm{Sym}(t_i)}\rho_{\Omega_i}\Pi_{\mathrm{Sym}(t_i)}\right].
$$

We know that, $\mathrm{Tr}(ABC) = \mathrm{Tr}(CAB)$ and using the idempotent property of projectors $\Pi^2 = \Pi$, we get,

$$
\begin{aligned}
\mathcal{N} &= \mathrm{Tr}\left[\rho_{\Omega_i}\Pi^2_{\mathrm{Sym}(t_i)}\right] \\
&= \mathrm{Tr}\left[\rho_{\Omega_i}\Pi_{\mathrm{Sym}(t_i)}\right] \\
&= \mathrm{Tr}\left[\left(\bigotimes_{k=1}^{t_i}|\psi_{i,k}\rangle\langle\psi_{i,k}|\right)\left(\frac{1}{t_i!}\sum_{\pi\in\mathfrak{S}_{t_i}} P_d(\pi)\right)\right] \\
&= \frac{1}{t_i!}\sum_{\pi\in\mathfrak{S}_{t_i}}\mathrm{Tr}\left[\left(\bigotimes_{k=1}^{t_i}|\psi_{i,k}\rangle\langle\psi_{i,k}|\right)P_d(\pi)\right].
\end{aligned}
$$

Let $|\Psi\rangle = |\psi_{i,1}\rangle \otimes \cdots \otimes |\psi_{i,t_i}\rangle$. Then

$$
\begin{aligned}
\mathrm{Tr}[(|\Psi\rangle\langle\Psi|)P_d(\pi)] &= \langle\Psi|\, P_d(\pi)\,|\Psi\rangle \\
&= \left\langle \psi_{i,1},\ldots,\psi_{i,t_i} \middle| \psi_{i,\pi^{-1}(1)},\ldots,\psi_{i,\pi^{-1}(t_i)}\right\rangle \\
&= \prod_{k=1}^{t_i}\left\langle \psi_{i,k} \middle| \psi_{i,\pi^{-1}(k)}\right\rangle.
\end{aligned}
$$

Due to the mutual orthogonality of the states, $\langle\psi_{i,k}|\psi_{i,j}\rangle = \delta_{kj}$. The product on the right-hand side is non-zero only if every term is non-zero. This requires $k = \pi^{-1}(k)$ for all $k \in \{1,\ldots,t_i\}$. This condition holds if and only if π is the identity permutation $\pi = \mathrm{id}$.

- If $\pi = \mathrm{id}$, then $\prod_{k=1}^{t_i}\langle\psi_{i,k}|\psi_{i,k}\rangle = \prod_{k=1}^{t_i} 1 = 1$.
- If $\pi \neq \mathrm{id}$, there exists at least one k such that $\pi(k) \neq k$. The product will contain a term $\left\langle\psi_{i,j}\middle|\psi_{i,\pi^{-1}(j)}\right\rangle$ where $j \neq \pi^{-1}(j)$, which is zero. Thus, the entire product is zero.

Therefore, the summation over all permutations collapses to a single non-zero term corresponding to $\pi = \mathrm{id}$.

$$\mathcal{N} = \frac{1}{t_i!}\left(1 + \sum_{\pi\in\mathfrak{S}_{t_i},\pi\neq\mathrm{id}} 0\right) = \frac{1}{t_i!}.$$

Finally, we get,

$$\begin{aligned}
\tilde{\rho}_{\Omega_i} &= \frac{\frac{1}{(t_i!)^2}\sum_{\pi,\sigma\in\mathfrak{S}_{t_i}} P_d(\pi)\left(\bigotimes_{k=1}^{t_i}\rho^{(i,k)}\right)P_d(\sigma)^\dagger}{\frac{1}{t_i!}} \\
&= t_i!\cdot\left(\frac{1}{(t_i!)^2}\sum_{\pi,\sigma\in\mathfrak{S}_{t_i}} P_d(\pi)\left(\bigotimes_{k=1}^{t_i}\rho^{(i,k)}\right)P_d(\sigma)^\dagger\right) \\
&= \frac{1}{t_i!}\sum_{\pi,\sigma\in\mathfrak{S}_{t_i}} P_d(\pi)\left(\bigotimes_{k=1}^{t_i}|\psi_{i,k}\rangle\langle\psi_{i,k}|\right)P_d(\sigma)^\dagger.
\end{aligned}$$

This completes the proof.

Choice of symmetric unitary t-design.

• **Choice I.** Consider a symmetric unitary t-design $(\mathfrak{U}, w)$, where $t = \sum_{i=1}^m t_i$. For each, Ω_i we select $U_{k_i^*} \in (\mathfrak{U}, w)$ from $(\mathfrak{U}, w)$ such that $U_{k_i}^{\otimes t_i}\tilde{\rho}_{\Omega_i}(U_{k_i}^\dagger)^{\otimes t_i} \in \mathcal{D}(\mathrm{Sym}^{p_i}(\mathcal{H}_M))$.

Using Lemma 8, it is straightforward to see that $(\mathfrak{U}, w)$ is also unitary p_i designs for every $i \in [m]$. We define the encryption map Enc_{Ω_i} by

$$\begin{aligned}
\mathsf{Enc}_{\Omega_i} : \mathcal{D}(\mathcal{H}_M^{\otimes t_i}) &\longrightarrow \mathcal{D}(\mathcal{H}_C^{\otimes t_i}) \\
\rho_{\Omega_i} &\longmapsto U_{k_i}^{\otimes t_i}\tilde{\rho}_{\Omega_i}(U_{k_i}^\dagger)^{\otimes t_i},
\end{aligned}$$

where

1. $\dim(\mathcal{H}_M^{\otimes t_i}) = (2^n)^{t_i} = 2^{n\cdot t_i}$
2. $\dim(\mathrm{Sym}((\mathcal{H}_M)^{\otimes t_i})) = \binom{2^n+t_i-1}{t_i}$
3. The map is trace-preserving and completely positive.

Finally, we define the broadcast ciphertext as

$$\rho_{\mathsf{Broadcast}} := \frac{1}{m}\sum_{i=1}^m\left[|i\rangle\langle i|_{\mathrm{index}} \otimes \sigma_{\Omega_i}\right],$$

where $\{|i\rangle\}_{i=1}^m$ is the orthonormal basis of $\mathcal{H}_{\mathrm{index}}$, and $\sigma_{\Omega_i} = U_{k_i}^{\otimes t_i}\tilde{\rho}_{\Omega_i}(U_{k_i}^\dagger)^{\otimes t_i}$.

• **Choice II.** Choose m different unitary t_i-designs $\{(\mathfrak{U}_i, w_i)\}_{i=1}^m$ for each block. Fix an ordering for each unitary t_i-designs and choose $U_{i,k_i^*} \in (\mathfrak{U}_i, w_i)$ at uniformly random.

For every block, We define the encryption map Enc_{Ω_i}, similarly by

$$\begin{aligned}\mathsf{Enc}_{\Omega_i} : \mathcal{D}(\mathcal{H}_M^{\otimes t_i}) &\longrightarrow \mathcal{D}(\mathcal{H}_C^{\otimes t_i}) \\ \rho_{\Omega_i} &\longmapsto U_{i,k_i}^{\otimes t_i} \tilde{\rho}_{\Omega_i} (U_{i,k_i}^{\dagger})^{\otimes t_i}.\end{aligned}$$

Similarly, we define the broadcast ciphertext as

$$\rho_{\mathsf{Broadcast}} := \frac{1}{m} \sum_{i=1}^{m} \Big[|i\rangle\langle i|_{\text{index}} \otimes \sigma_{\Omega_i} \Big],$$

where $\{|i\rangle\}_{i=1}^m$ is the orthonormal basis of $\mathcal{H}_{\text{index}}$ and $\sigma_{\Omega_i} = U_{i,k_i}^{\otimes t_i} \tilde{\rho}_{\Omega_i} (U_{i,k_i}^{\dagger})^{\otimes t_i}$.

For our construction, **Choice II** is more efficient, and we use it for our quantum broadcast encryption.

A schematic diagram of the MP-QPBE construction is presented in Figure 2.

Comparison between Choice I and Choice II. From the explicit construction of ϵ-approximate unitary t-design, by Brandao, Harrow, and Horodecki [15], we get

$$|\mathfrak{U}| = O\left(t \cdot d^{2t} \cdot \text{poly}(\log(t/d), 1/\varepsilon)\right),$$

where $d = 2^n$.

• **Space complexity.**

• *Choice I.* We have $t = \sum_{i=1}^m t_i$, and we consider a single unitary t-design. In this case

$$\begin{aligned}|\mathfrak{U}| &= O\left(d^{2t}\right) \\ &= O\left(d^{2\sum_{i=1}^m t_i}\right) \\ &= O\left((2^n)^{2\sum_{i=1}^m t_i}\right) = O\left(2^{2n\sum_{i=1}^m t_i}\right)\end{aligned}$$

• *Choice II.* In this case, we have m independent designs. The size of the design for block i is $|\mathfrak{U}_i|$, with order t_i:

$$|\mathfrak{U}_i| = O\left(d^{2t_i}\right)$$

The total storage required is the sum of the sizes of these individual designs:

$$\sum_{i=1}^{m} |\mathfrak{U}_i| = \sum_{i=1}^{m} O\left(d^{2t_i}\right) = O\left(\sum_{i=1}^{m} d^{2t_i}\right)$$

To compare, consider the simple case where $t_i = \tau$ for all i.

– Choice I size: $O(d^{2m\tau})$
– Choice II size: $O(m \cdot d^{2\tau})$

Since $d^{2m\tau} = (d^{2\tau})^m$, the size for Choice I is exponentially larger than for Choice II.

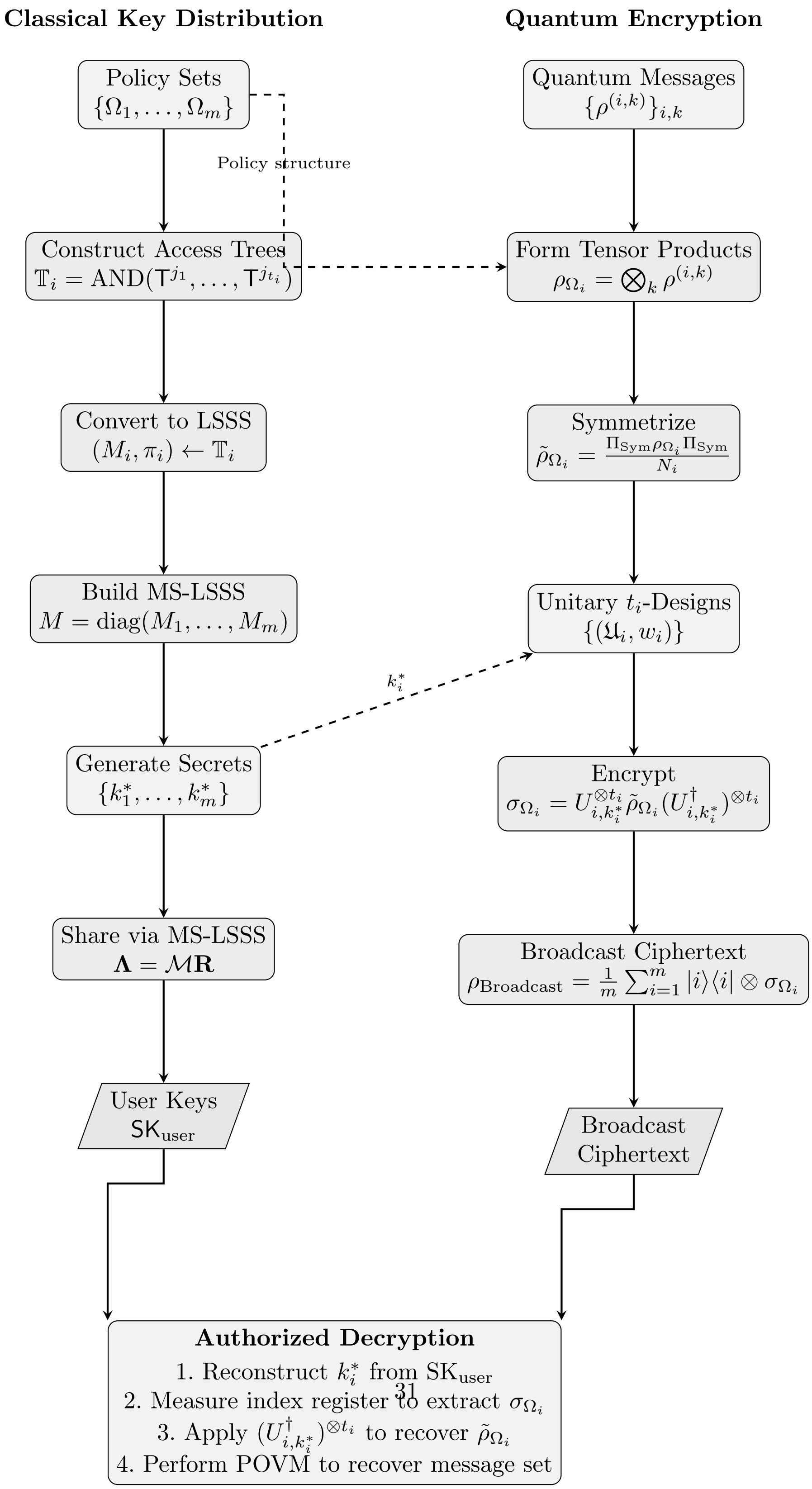

Fig. 2 Schematic overview of the MP-QPBE construction. **Left:** Classical key distribution via Multi-Secret LSSS. Policy sets are converted to access trees, then to LSSS matrices, which are combined into a block-diagonal MS-LSSS structure. Secret unitary indices $\{k_i^*\}$ are shared, producing user-specific key shares. **Right:** Quantum encryption pipeline. Messages are grouped into tensor products, symmetrized onto $\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i})$, encrypted using the corresponding unitary from the t_i-design, and assembled into a block-diagonal broadcast state. **Bottom:** Authorized users combine their classical key shares with the quantum ciphertext to recover the original messages.

- **Time complexity.**

- **Choice I:** $T_{\text{Choice I}} = O(|\mathfrak{U}|) = O\left(d^{2\sum t_i}\right)$.
- **Choice II:** $T_{\text{Choice II}} = O\left(\sum_{i=1}^{m} |\mathfrak{U}_i|\right) = O\left(\sum_{i=1}^{m} d^{2t_i}\right)$. This is exponentially more efficient.

Encryption Time. For each block i we compute $U^{\otimes t_i}\tilde{\rho}_{\Omega_i}(U^\dagger)^{\otimes t_i}$. If the circuit for a single U has gate complexity T_U, then the complexity of applying it t_i times in parallel is simply $t_i \cdot T_U$.

$$T_{\text{Enc}}(i) = O(t_i \cdot T_U)$$

This cost is dependent upon the number of policies t_i for a specific block being encrypted. Therefore, it is identical for both Choice I and Choice II. In both cases, for block i, we select a unitary from either a single large design or the smaller designs and apply its t_i-th tensor power.
The encryption algorithm is described in Algorithm 4.

Table 1 Comparison of Design Choices

Aspect	**Choice I (Single large design)**	**Choice II (Multiple smaller designs)**
Design Order	Requires a single design of high order $t = \sum_{i=1}^{m} t_i$.	Requires m designs of smaller orders $t_1, t_2, \ldots, t_m$.
Construction Feasibility	Comparatively difficult. The existence of unitary t-designs for $t > 3$ is known, but explicit constructions are rare, and the size is very large.	Much more feasible. Low-order designs (e.g., $t_i = 1, 2, 3$) are well understood and have known efficient constructions (e.g., the Clifford group for a 3-design).
Design Size $\lvert\mathfrak{U}\rvert$	The size grows polynomially with d and t. A lower bound is $\lvert\mathfrak{U}\rvert \geq \binom{d^2+t-1}{t}$. For $t = \sum t_i$, this becomes exceptionally large. e.g., $\lvert\mathfrak{U}\rvert \approx O(d^{2t}) = O(d^{2\sum t_i})$.	The size of each design $\lvert\mathfrak{U}_i\rvert$ is much smaller, with $\lvert\mathfrak{U}_i\rvert \approx O(d^{2t_i})$. The total storage for all designs is $\sum \lvert\mathfrak{U}_i\rvert$, which is vastly smaller than the size of the single large design.
Classical Key Size	The secret key k_i^* is an index to a single, massive list of unitaries. The field size p must be larger than $\lvert\mathfrak{U}\rvert$, i.e., $p > O(d^{2\sum t_i})$.	The secret key k_i^* is an index to a smaller list $\mathfrak{U}_i$. The prime p must satisfy $p > \max\{\lvert\mathfrak{U}_i\rvert : i = 1, \ldots, m\}$, which is a much smaller and more manageable constraint.
Quantum Complexity	The quantum operation is $U_{k_i^*}^{\otimes t_i}$. The complexity of implementing $U_{k_i^*}$ itself might be high, but this part is comparable to Choice II.	The quantum operation is $U_{i,k_i^*}^{\otimes t_i}$. No significant difference in the encryption step itself, but the overall practical implementability is based on the feasibility of the designs.
Conclusion	**Theoretically possible but practically infeasible** for any non-trivial number of policies or blocks.	**More practical and efficient.** This is the only viable approach for a real-world implementation.

Remark 1 Because of the Lemma 8, every block of $\rho_{\mathsf{Broadcast}}$ is information theoretically secure.

Recoverability and Decryption. A user, possessing an attribute set S_{user} and the corresponding private key $\mathbf{\Lambda}_{S_{user}}$, targets the k-th message block. The user first uses the public parameters to identify the LSSS sub-scheme (M_k, π_k) governing access to this block. The core of this phase is to determine if the user's attributes satisfy the access policy $\mathbb{T}_k$. This is established by checking if the target vector $\vec{e}_1 = (1, 0, \ldots, 0)^T \in \mathbb{F}_p^{c_k}$ lies within the linear span of the row vectors of M_k corresponding to the user's attributes. Let $I_k := \{j \mid \pi_k(j) \in S_{user}\}$ be the set of indices of these authorized rows. If $\vec{e}_1 \notin \operatorname{span}(\{(M_k)_j\}_{j \in I_k})$, the user is unauthorized, and the decryption process terminates.

If authorized, the linear algebraic properties of the LSSS guarantee the existence of a set of reconstruction coefficients $\{\omega_j \in \mathbb{F}_p\}_{j \in I_k}$ that can be efficiently computed, such that $\sum_{j \in I_k} \omega_j (M_k)_j = \vec{e}_1^T$. The user then retrieves the relevant shares $\{\Lambda_i\}_{i \in I_{S_{user},k}}$ from their private key and computes a linear combination using these coefficients to reconstruct the secret index $s_k = k_k^*$ of the unitary operator

$$k_k^* = \sum_{i \in I_{S_{user},k}} \omega_i \Lambda_i \pmod{p}.$$

Using the recovered classical key k_k^*, the user identifies the specific unitary operator U_{k,k_k^*} from the publicly known unitary t_k-design $(\mathfrak{U}_k, w_k)$. The decryption operation consists of applying the adjoint of the t_k-fold tensor power of this unitary to the extracted ciphertext

$$\tilde{\rho}_{\Omega_k} = (U_{k,k_k^*}^\dagger)^{\otimes t_k} \sigma_{\Omega_k} U_{k,k_k^*}^{\otimes t_k}$$

Now the user can decrypt the broadcast quantum ciphertext. The full broadcast state for the policy set Ω_i is $\rho_{\mathsf{Broadcast}} = \frac{1}{m} \sum_{i=1}^m |i\rangle\langle i|_{\text{index}} \otimes \sigma_{\Omega_i}$, which is a block-diagonal state. To isolate the ciphertext for the k-th block, the user performs a projective measurement on the ancillary index register $\mathcal{H}_{\text{index}}$. The measurement is defined by the projector $P_k = |k\rangle\langle k|_{\text{index}} \otimes \mathbb{I}_{\mathcal{H}_C}$. Upon obtaining the outcome k, the post-measurement state collapses to the desired ciphertext, σ_{Ω_k}.

If the sender used a tensor product of t_k mutually orthogonal pure states, $\rho_{\Omega_k} = \bigotimes_{j=1}^{t_k} |\psi_j\rangle\langle\psi_j|$, then recovery of the set of messages is possible. The state $\tilde{\rho}_{\Omega_k}$ is the pure state $\left|\tilde{\Psi}\right\rangle\left\langle\tilde{\Psi}\right|$, where $\left|\tilde{\Psi}\right\rangle = \frac{1}{\sqrt{t_k!}} \sum_{\pi \in \mathfrak{S}_{t_k}} P(\pi)(\bigotimes_j |\psi_j\rangle)$. The user performs a discriminating POVM measurement $\{E_J\}$ on $\tilde{\rho}_{\Omega_k}$. Each POVM element $E_J = \left|\tilde{\Psi}_J\right\rangle\left\langle\tilde{\Psi}_J\right|$ is a projector onto a fully symmetric state constructed from a unique, unordered subset J of t_k orthogonal vectors from a pre-established basis. The measurement outcome J^* reveals the unordered set of pure states $\{|\psi\rangle\langle\psi| \mid |\psi\rangle \in J^*\}$ that were originally sent. The Algorithm 5 describes the decryption algorithm.

If the original messages were general mixed states, the symmetrization map $\rho_{\Omega_k} \longmapsto \tilde{\rho}_{\Omega_k}$ is non-invertible as it is not a unitary operator and also not a one-to-one map. The entanglement and classical correlations present in the symmetrized state prevent the recovery of the individual tensor factors. For example if we consider two different initial states $\rho_1 = |0\rangle\langle 0| \otimes |1\rangle\langle 1|$ and $\rho_2 = |1\rangle\langle 1| \otimes |0\rangle\langle 0|$, after symmetrization, we get,

$$\tilde{\rho}_1 = \tilde{\rho}_2 = \frac{1}{2}(|01\rangle\langle 01| + |01\rangle\langle 10| + |10\rangle\langle 01| + |10\rangle\langle 10|)$$

This state cannot be uniquely desymmetrized back to either ρ_1 or ρ_2.

Remark 2 A special case is when all messages associated with a policy set Ω_i are identical, i.e., $\rho_{\Omega_i} = \rho^{\otimes t_i}$. This is the case originally considered by the scheme of Broadbent et al. [12]. A state of the form $\rho^{\otimes t_i}$ is always symmetric. For any permutation $\pi \in \mathfrak{S}_{t_i}$, the corresponding permutation operator $P(\pi)$ commutes with $\rho^{\otimes t_i}$

$$P(\pi)(\rho^{\otimes t_i})P(\pi)^\dagger = (P(\pi)\rho^{\otimes t_i})P(\pi)^\dagger = (\rho^{\otimes t_i}P(\pi))P(\pi)^\dagger = \rho^{\otimes t_i}$$

This means the state already lies within the symmetric subspace. The action of the symmetric projector $\Pi_{\mathrm{Sym}(t_i)}$ on $\rho^{\otimes t_i}$ is simply the identity

$$\Pi_{\mathrm{Sym}(t_i)}\rho^{\otimes t_i} = \rho^{\otimes t_i} \quad \text{and} \quad \mathrm{Tr}\Big[\Pi_{\mathrm{Sym}(t_i)}\rho^{\otimes t_i}\Big] = \mathrm{Tr}\Big[\rho^{\otimes t_i}\Big] = 1$$

Therefore, the symmetrization step in your protocol becomes trivial for such states

$$\tilde{\rho}_{\Omega_i} = \frac{\Pi_{\mathrm{Sym}(t_i)}\rho^{\otimes t_i}\Pi_{\mathrm{Sym}(t_i)}}{\mathrm{Tr}\Big[\Pi_{\mathrm{Sym}(t_i)}\rho^{\otimes t_i}\Big]} = \frac{\rho^{\otimes t_i}}{1} = \rho^{\otimes t_i}$$

In this case, no information is lost. The encryption and decryption are well-defined and can be done as follows

- **Encryption:** $\sigma_{\Omega_i} = U_{i,k_i^*}^{\otimes t_i}(\rho^{\otimes t_i})(U_{i,k_i^*}^\dagger)^{\otimes t_i} = (U_{i,k_i^*}\rho U_{i,k_i^*}^\dagger)^{\otimes t_i}$.
- **Decryption:** The authorized user recovers k_i^* and applies $(U_{i,k_i^*}^\dagger)^{\otimes t_i}$, perfectly recovering the original state $\rho^{\otimes t_i}$.

Algorithm 3 Multi-Policy Quantum Private Broadcast Encryption (MP-QPBE) Part-1

Require: Policy sets $\{\Omega_1, \dots, \Omega_m\}$ with $|\Omega_i| = t_i$, messages $\{\rho^{(i,k)}\}_{i\in[m],k\in[t_i]}$ where $\rho^{(i,k)} \in \mathcal{D}(\mathcal{H}_M)$ are orthogonal for fixed i; a collection of unitary t_i-designs $\{(\mathfrak{U}_i, w_i)\}_{i=1}^m$; the set of secret unitary indices $\{k_i^* \in \mathbb{F}_p\}_{i=1}^m$ from the unitary t_i-designs $\{(\mathfrak{U}_i, w_i)\}_{i=1}^m$, where p is a prime such that $\max\{|\mathfrak{U}_i| \mid i = 1, \dots, m\} < p$; Public parameters $\mathsf{PP} = \{(M_i, \pi_i), (\mathfrak{U}_i, w_i)\}_{i=1}^m$.

Ensure: Broadcast ciphertext $\rho_{\mathsf{Broadcast}}$

Phase 1: Policy-Message Association

for $i = 1$ to m **do**

  Define $S_i = \{\mathsf{Policy}_{j_1}^{(i)}, \dots, \mathsf{Policy}_{j_{t_i}}^{(i)}\}$

  Map: $\mathsf{Policy}_{j_k}^{(i)} \mapsto \rho^{(i,k)} \in \mathcal{D}(\mathcal{H}_M)$

end for

Phase 2: Tensor Product Construction

for $i = 1$ to m **do**

  Form the composite state for the current block:

$$\rho_{\Omega_i} = \bigotimes_{k=1}^{t_i} \rho^{(i,k)} \in \mathcal{D}(\mathcal{H}_M^{\otimes t_i})$$

end for

Phase 3: Symmetrization

for $i = 1$ to m **do**

  Construct the symmetric projector:

$$\Pi_{\mathrm{Sym}(t_i)} = \frac{1}{t_i!} \sum_{\pi \in \mathfrak{S}_{t_i}} P(\pi)$$

  where $P(\pi)$ is the unitary representation of permutation π

  Compute the normalization constant $\mathcal{N}_i = \mathrm{Tr}\big[\Pi_{\mathrm{Sym}(t_i)} \rho_{\Omega_i}\big]$. If messages are orthogonal, $\mathcal{N}_i = 1/t_i!$, and messages are general density matrices, then $\mathcal{N} = \sum_{j=1}^D \frac{\lambda_j}{t_i!} \sum_{\pi \in \mathfrak{S}_{t_i}} \langle \phi_j | P(\pi) | \phi_j \rangle$

  Project the state onto the symmetric subspace and normalize:

$$\tilde{\rho}_{\Omega_i} = \frac{\Pi_{\mathrm{Sym}(t_i)} \rho_{\Omega_i} \Pi_{\mathrm{Sym}(t_i)}}{\mathcal{N}_i} \in \mathcal{D}(\mathrm{Sym}(d^{t_i}))$$

end for

Phase 4: Quantum Encryption

Choose a unitary t_i-design $(\mathfrak{U}_i, w_i)$ from the public parameters PP, for encrypting the message ρ_{Ω_i}.

Fix an ordering on the unitary matrices in unitary t_i-design $(\mathfrak{U}_i, w_i)$.

Choose a secret index k_i^* from $\{1, \dots, |\mathfrak{U}_i|\}$, for block i with uniform distribution.

Select the corresponding unitary $U_{i,k_i^*} \in (\mathfrak{U}_i, w_i)$ from the unitary t-design based on the public parameters.

for $i = 1$ to m **do**

  Apply the tensor power t_i (chosen for each set of policies) of the unitary U_{i,k_i^*} to encrypt the symmetrized state:

$$\sigma_{\Omega_i} = U_{i,k_i^*}^{\otimes t_i} \tilde{\rho}_{\Omega_i} (U_{i,k_i^*}^\dagger)^{\otimes t_i} \in \mathcal{D}(\mathrm{Sym}^{t_i}(\mathcal{H}_C))$$

end for

Algorithm 4 Multi-Policy Quantum Private Broadcast Encryption (MP-QPBE) Part-2, Continued

Phase 6: Broadcast State Assembly

Construct the broadcast ciphertext:

$$\rho_{\mathsf{Broadcast}} = \frac{1}{m}\sum_{i=1}^{m}\Big[|i\rangle\langle i|_{\text{index}} \otimes \sigma_{\Omega_i}\Big]$$

where $\{|i\rangle\}_{i=1}^{m}$ is the orthonormal basis of $\mathcal{H}_{\text{index}}$

Convert the AND-OR policy access tree into a threshold gate access tree.

Convert the threshold-gate access tree into an LSSS matrix $\mathcal{M}$ over $\mathbb{F}_p$ along with the associated attribute map ϱ.

Share the secrets $\{k_1^*, \ldots, k_m^*\}$ using $(\mathcal{M}, \varrho)$.

return $\rho_{\mathsf{Broadcast}} \in \mathcal{D}(\mathcal{H}_{\mathsf{B}})$

Algorithm 5 Multi-Policy Quantum Private Broadcast Decryption (MP-QPBD)

Require: User's attribute set S_{user}; user's private key $\mathbf{\Lambda}_{S_{user}}$ (a set of shares from the MS-LSSS); the broadcast ciphertext $\rho_{\mathsf{Broadcast}}$; the index $k \in [m]$ of the desired block; Global Public Parameters $\mathsf{PP} = \{(\mathcal{M}, \varrho), \{(\mathfrak{U}_i, w_i)\}_{i=1}^m\}$.

Ensure: The set of decrypted pure states $\{\rho^{(k,j)}\}_{j=1}^{t_k}$ or the recovered symmetric state $\tilde{\rho}_{\Omega_k}$ or $\perp$.

Phase 1: Authorization Check and Classical Key Reconstruction

Let (M_k, π_k) be the LSSS matrix for the target policy set Ω_k, which is a sub-block of the global matrix $\mathcal{M}$.

Identify the user's relevant row indices for policy k: $I_{S_{user},k} := \{j \mid \varrho(j) = (A, k) \text{ for some } A \in S_{user}\}$.

Let the submatrix of rows be $\mathcal{M}_{S_{user},k} = \{\mathcal{M}_j\}_{j \in I_{S_{user},k}}$. These correspond to rows of M_k.

if $\vec{e}_1 = (1, 0, \ldots, 0) \notin \operatorname{span}(\{(M_k)_j \mid \pi_k(j) \in S_{user}\})$ **then**

  return $\perp$ (User is not authorized for block k).

end if

Compute reconstruction coefficients $\{\omega_j \in \mathbb{F}_p\}_{j \in I_{S_{user},k}}$ such that $\sum_{j \in I_{S_{user},k}} \omega_j (M_k)_{\pi_k^{-1}(\varrho(j))} = \vec{e}_1^{\,T}$.

Collect the corresponding shares $\{\Lambda_j\}_{j \in I_{S_{user},k}}$ from $\mathbf{\Lambda}_{S_{user}}$.

Reconstruct the secret unitary index $s_k = k_k^*$:

$$k_k^* = \sum_{j \in I_{S_{user},k}} \omega_j \Lambda_j \pmod{p}$$

Phase 2: Quantum State Extraction

Perform a projective measurement on the index register $\mathcal{H}_{\text{index}}$ of $\rho_{\mathsf{Broadcast}}$ using the projector $|k\rangle\langle k|_{\text{index}}$.

The post-measurement state is the ciphertext σ_{Ω_k}.

Phase 3: Quantum Unitary Decryption

Retrieve the unitary U_{k,k_k^*} from the public description of $(\mathfrak{U}_k, w_k)$ using the recovered index k_k^*.

Apply the inverse unitary operation to recover the symmetrized state:

$$\tilde{\rho}_{\Omega_k} = (U_{k,k_k^*}^{\dagger})^{\otimes t_k} \sigma_{\Omega_k} U_{k,k_k^*}^{\otimes t_k}$$

Phase 4: Individual Message Recovery

if the original messages were a known set of orthogonal pure states **then**

  The user must know the set of possible basis states $\mathcal{B} = \{|\phi_1\rangle, \ldots, |\phi_N\rangle\}$. The information about the basis states must be communicated via a separate classical channel.

  For every unique subset $J \subset \mathcal{B}$ of size t_k, construct the projector $E_J = \left|\tilde{\Psi}_J\middle\rangle\middle\langle\tilde{\Psi}_J\right|$, where $\left|\tilde{\Psi}_J\right\rangle = \frac{1}{\sqrt{t_k!}} \sum_{\pi \in \mathfrak{S}_{t_k}} P(\pi)(\bigotimes_{|\psi\rangle \in J} |\psi\rangle)$.

  The set $\{E_J\}$ forms a POVM. Perform this measurement on $\tilde{\rho}_{\Omega_k}$.

  Let the outcome be J^*. The recovered unordered set of messages is $\{|\psi\rangle\langle\psi| \mid |\psi\rangle \in J^*\}$.

  return Unordered set $\{\rho^{(k,j)}\}_{j=1}^{t_k}$.

else if the original messages were general mixed states **then**

  Individual message recovery is impossible. The process terminates here.

  return $\tilde{\rho}_{\Omega_k}$.

end if

3.2 Security analysis for MP-QPBE

Theorem 11 (Information-Theoretic Security of MP-QBE) *Let S_{user} be the attribute set of a user such that S_{user} does not satisfy the access policy $\mathbb{T}_i$ for a given block i. Then, the user's view of the ciphertext block σ_{Ω_i} is information-theoretically secure. Specifically, the state perceived by the unauthorized user, averaged over all possible secret keys consistent with their information, is the maximally mixed state on the symmetric subspace:*

$$\rho_{unauth}^{(i)} = \frac{\mathbb{I}_{\mathrm{Sym}(t_i)}}{\dim(\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i}))} = \frac{\mathbb{I}_{\mathrm{Sym}(t_i)}}{\binom{d+t_i-1}{t_i}}, \tag{20}$$

where $\mathbb{I}_{\mathrm{Sym}(t_i)}$ denotes the identity operator on the symmetric subspace $\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i})$ and $d = \dim(\mathcal{H}_M)$.

Proof Let the secret key index for block i be denoted by $k_i^* \in \{1, 2, \ldots, |\mathfrak{U}_i|\}$, where $(\mathfrak{U}_i, w_i)$ is the unitary t_i-design used for encryption. The LSSS matrix (M_i, π_i) is constructed from the access tree $\mathbb{T}_i$, and the secret $s = k_i^*$ is shared using a random vector

$$\vec{v}_i = (k_i^*, r_2, r_3, \ldots, r_{c_i})^T \in \mathbb{F}_p^{c_i}, \tag{21}$$

where $r_2, \ldots, r_{c_i}$ are chosen independently and uniformly at random from $\mathbb{F}_p$, and c_i is the number of columns in M_i.

The shares distributed to users are computed as $\vec{\lambda}_i = M_i \vec{v}_i$. An unauthorized user with attribute set S_{user} that does not satisfy T_i possesses only the shares $\{\lambda_{i,j}\}_{j \in I_{S_{user}}}$, where $I_{S_{user}} = \{j : \pi_i(j) \in S_{user}\}$.

By the security property of LSSS (Theorem 10), since $S_{user} \not\models \mathbb{T}_i$, the target vector $\vec{e}_1 = (1, 0, \ldots, 0)^T$ does not lie in the row span of the submatrix $\{(M_i)_j\}_{j \in I_{S_{user}}}$. This implies that for any two distinct values $s, s' \in \mathbb{F}_p$ of the secret, the conditional distributions of the unauthorized user's shares are identical:

$$\Pr\left[\{(M_i)_j \cdot \vec{v}_i\}_{j \in I_{S_{user}}} = \{\lambda_j\}_{j \in I_{S_{user}}} \,\middle|\, k_i^* = s\right] = \Pr\left[\{(M_i)_j \cdot \vec{v}_i\}_{j \in I_{S_{user}}} = \{\lambda_j\}_{j \in I_{S_{user}}} \,\middle|\, k_i^* = s'\right]. \tag{22}$$

Therefore, from the perspective of the unauthorized user, the secret key index k_i^* remains uniformly distributed over $\{1, 2, \ldots, |\mathfrak{U}_i|\}$, regardless of the shares they possess. For any $k \in \{1, \ldots, |\mathfrak{U}_i|\}$:

$$\begin{aligned}\Pr\left[k_i^* = k \,\middle|\, \{\lambda_{i,j}\}_{j \in I_{S_{user}}}\right] &= \Pr\left[k_i^* = k\right] \\ &= \frac{1}{|\mathfrak{U}_i|},\end{aligned} \tag{23}$$

as both events are independent. The encryption of the symmetrized message $\tilde{\rho}_{\Omega_i} \in \mathcal{D}(\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i}))$ using the secret key k_i^* produces the ciphertext:

$$\sigma_{\Omega_i} = U_{i,k_i^*}^{\otimes t_i} \, \tilde{\rho}_{\Omega_i} \, (U_{i,k_i^*}^{\dagger})^{\otimes t_i}, \tag{24}$$

where $U_{i,k_i^*} \in \mathfrak{U}_i$ is the unitary operator indexed by k_i^*.

Since the unauthorized user cannot determine k_i^* from their available information, they must consider all possible values of the key. By the Equation (23), the quantum state perceived by the unauthorized user is the ensemble average over all keys:

$$\rho_{\mathrm{unauth}}^{(i)} = \mathbb{E}_{k_i^*}\left[\sigma_{\Omega_i}\right] = \sum_{k=1}^{|\mathfrak{U}_i|} \Pr\left[k_i^* = k\right] \cdot U_{i,k}^{\otimes t_i} \, \tilde{\rho}_{\Omega_i} \, (U_{i,k}^{\dagger})^{\otimes t_i}. \tag{25}$$

For an unweighted unitary t_i-design (i.e., $w_i(U) = 1/|\mathfrak{U}_i|$ for all $U \in \mathfrak{U}_i$), this becomes:

$$\rho_{\text{unauth}}^{(i)} = \frac{1}{|\mathfrak{U}_i|} \sum_{k=1}^{|\mathfrak{U}_i|} U_{i,k}^{\otimes t_i} \, \tilde{\rho}_{\Omega_i} \, (U_{i,k}^{\dagger})^{\otimes t_i}. \tag{26}$$

More generally, for a weighted unitary t_i-design with weight function $w_i : \mathfrak{U}_i \to [0, 1]$ satisfying $\sum_{U \in \mathfrak{U}_i} w_i(U) = 1$, and assuming the LSSS uniformly distributes over the design elements, we have:

$$\rho_{\text{unauth}}^{(i)} = \sum_{U_k \in \mathfrak{U}_i} w_i(U_k) \, U_k^{\otimes t_i} \, \tilde{\rho}_{\Omega_i} \, (U_k^{\dagger})^{\otimes t_i}. \tag{27}$$

By the defining property of a unitary t-design (Definition 2), for any operator X acting on $\mathcal{H}_M^{\otimes t_i}$:

$$\sum_{U_k \in \mathfrak{U}_i} w_i(U_k) \, U_k^{\otimes t_i} \, X \, (U_k^{\dagger})^{\otimes t_i} = \int_{U(d)} U^{\otimes t_i} \, X \, (U^{\dagger})^{\otimes t_i} \, d\mu_{\text{Haar}}(U), \tag{28}$$

where μ_{Haar} denotes the normalized Haar measure on the unitary group $U(d)$.

The right-hand side of Equation (28) is known as the *Haar twirling channel* or *t_i-fold twirl*:

$$\mathcal{T}_{\text{Haar}}^{(t_i)}(X) := \int_{U(d)} U^{\otimes t_i} \, X \, (U^{\dagger})^{\otimes t_i} \, d\mu_{\text{Haar}}(U). \tag{29}$$

Since $\tilde{\rho}_{\Omega_i} \in \mathcal{D}(\text{Sym}(\mathcal{H}_M^{\otimes t_i}))$, we apply Theorem 16, which states that for any operator X supported on the symmetric subspace:

$$\mathcal{T}_{\text{Haar}}^{(t_i)}(X) = \frac{\text{Tr}(X)}{\dim(\text{Sym}(\mathcal{H}_M^{\otimes t_i}))} \cdot \mathbb{I}_{\text{Sym}(t_i)}. \tag{30}$$

Combining Equations (26), (28), and (30), we obtain:

$$\begin{aligned}
\rho_{\text{unauth}}^{(i)} &= \sum_{U_k \in \mathfrak{U}_i} w_i(U_k) \, U_k^{\otimes t_i} \, \tilde{\rho}_{\Omega_i} \, (U_k^{\dagger})^{\otimes t_i} \\
&= \int_{U(d)} U^{\otimes t_i} \, \tilde{\rho}_{\Omega_i} \, (U^{\dagger})^{\otimes t_i} \, d\mu_{\text{Haar}}(U) \\
&= \mathcal{T}_{\text{Haar}}^{(t_i)}(\tilde{\rho}_{\Omega_i}) \\
&= \frac{\text{Tr}(\tilde{\rho}_{\Omega_i})}{\dim(\text{Sym}(\mathcal{H}_M^{\otimes t_i}))} \cdot \mathbb{I}_{\text{Sym}(t_i)}.
\end{aligned} \tag{31}$$

Since $\tilde{\rho}_{\Omega_i}$ is a valid quantum state (density operator), we have $\text{Tr}(\tilde{\rho}_{\Omega_i}) = 1$. The dimension of the symmetric subspace is

$$\dim(\text{Sym}(\mathcal{H}_M^{\otimes t_i})) = \binom{d + t_i - 1}{t_i}. \tag{32}$$

Therefore, the unauthorized user's perceived state is:

$$\rho_{\text{unauth}}^{(i)} = \frac{\mathbb{I}_{\text{Sym}(t_i)}}{\binom{d+t_i-1}{t_i}}, \tag{33}$$

which is the maximally mixed state on the symmetric subspace $\text{Sym}(\mathcal{H}_M^{\otimes t_i})$.

The maximally mixed state $\rho_{\text{unauth}}^{(i)}$ is independent of the original message ρ_{Ω_i} (and hence of $\tilde{\rho}_{\Omega_i}$). This means that:

(a) No measurement performed by the unauthorized user on $\rho^{(i)}_{\text{unauth}}$ can yield any information about the original plaintext.
(b) For any two distinct messages ρ_{Ω_i} and ρ'_{Ω_i}, the corresponding perceived states are identical:

$$\mathbb{E}_{k_i^*}\left[\mathrm{Enc}_{k_i^*}(\tilde{\rho}_{\Omega_i})\right] = \mathbb{E}_{k_i^*}\left[\mathrm{Enc}_{k_i^*}(\tilde{\rho}'_{\Omega_i})\right] = \frac{\mathbb{I}_{\mathrm{Sym}(t_i)}}{\binom{d+t_i-1}{t_i}}. \tag{34}$$

Since the security of each block i is established independently, and the total broadcast state $\rho_{\text{Broadcast}} = \frac{1}{m}\sum_{i=1}^{m} |i\rangle\langle i|_{\text{index}} \otimes \sigma_{\Omega_i}$ is a block-diagonal (classically correlated) state with respect to the index register, the information-theoretic security extends to the entire broadcast. An unauthorized user for block i perceives:

$$\rho^{(i)}_{\text{Broadcast,unauth}} = |i\rangle\langle i|_{\text{index}} \otimes \frac{\mathbb{I}_{\mathrm{Sym}(t_i)}}{\binom{d+t_i-1}{t_i}}, \tag{35}$$

which reveals no information about the message ρ_{Ω_i}.

This completes the proof. □ □

Now we discuss the security against quantum side information for all messages in the symmetric subspace $\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i})$.

Theorem 12 (Perfect Security of MP-QPBE) *Let the MP-QPBE scheme for a policy block i use an exact symmetric unitary t_i-design $(\mathfrak{U}_i, w_i)$ as its key space. The scheme achieves perfect security against adversaries with quantum side information for all messages in the symmetric subspace $\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i})$. Specifically, for any joint state $\rho_{ME} \in \mathcal{D}(\mathcal{H}_M^{\otimes t_i} \otimes \mathcal{H}_E)$ where the message register M is supported on the symmetric subspace, the diamond norm distance between the average encryption channel and the state-replacement channel is zero:*

$$\left\|\bar{\mathcal{E}}_{\text{MP}} - \mathcal{R}_{\sigma_{\text{MP}} \otimes \mathrm{id}_E}\right\|_\diamond = 0, \tag{36}$$

where $\sigma_{\text{MP}} = \frac{\mathbb{I}_{\mathrm{Sym}(t_i)}}{\dim(\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i}))}$ is the maximally mixed state on the symmetric subspace.

Proof Let the adversary possess a quantum system described by Hilbert space $\mathcal{H}_E$. The joint state of the message and the adversary's side information is:

$$\rho_{ME} \in \mathcal{D}(\mathcal{H}_M^{\otimes t_i} \otimes \mathcal{H}_E), \tag{37}$$

where we require that the reduced state on the message register satisfies $\rho_M = \mathrm{Tr}_E(\rho_{ME}) \in \mathcal{D}(\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i}))$.

The encryption channel for a specific key k corresponding to unitary $U_k \in \mathfrak{U}_i$ acts on the message register while leaving the adversary's system unchanged:

$$\mathcal{E}_k(\rho_{ME}) = \left(U_k^{\otimes t_i} \otimes \mathbb{I}_E\right) \rho_{ME} \left((U_k^\dagger)^{\otimes t_i} \otimes \mathbb{I}_E\right). \tag{38}$$

The average encryption channel $\bar{\mathcal{E}}_{\text{MP}}$ is the weighted average over all keys in the design:

$$\begin{aligned}\bar{\mathcal{E}}_{\text{MP}}(\rho_{ME}) &= \sum_{U_k \in \mathfrak{U}_i} w_i(U_k)\, \mathcal{E}_k(\rho_{ME}) \\ &= \sum_{U_k \in \mathfrak{U}_i} w_i(U_k) \left(U_k^{\otimes t_i} \otimes \mathbb{I}_E\right) \rho_{ME} \left((U_k^\dagger)^{\otimes t_i} \otimes \mathbb{I}_E\right)\end{aligned} \tag{39}$$

$$= \left(\sum_{U_k \in \mathfrak{U}_i} w_i(U_k)\, U_k^{\otimes t_i} (\cdot) (U_k^\dagger)^{\otimes t_i} \right) \otimes \mathrm{id}_E\, (\rho_{ME}). \tag{40}$$

The last equality follows from the fact that the encryption acts as the identity on $\mathcal{H}_E$. We can write this as:

$$\bar{\mathcal{E}}_{\mathrm{MP}} = \mathcal{T}^{(t_i)}_{\mathfrak{U}_i} \otimes \mathrm{id}_E, \tag{41}$$

where $\mathcal{T}^{(t_i)}_{\mathfrak{U}_i}$ denotes the twirling channel induced by the unitary t_i-design:

$$\mathcal{T}^{(t_i)}_{\mathfrak{U}_{t_i}}(X) = \sum_{U_k \in \mathfrak{U}_{t_i}} w_i(U_k)\, U_k^{\otimes t_i}\, X\, (U_k^\dagger)^{\otimes t_i}. \tag{42}$$

Since $(\mathfrak{U}_i, w_i)$ is an *exact* symmetric unitary t_i-design, for any operator X supported on the symmetric subspace:

$$\mathcal{T}^{(t_i)}_{\mathfrak{U}_i}(X) = \mathcal{T}^{(t_i)}_{\mathrm{Haar}}(X) = \int_{U(d)} U^{\otimes t_i}\, X\, (U^\dagger)^{\otimes t_i}\, d\mu_{\mathrm{Haar}}(U). \tag{43}$$

For the message part of ρ_{ME}, which is supported on $\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i})$, Theorem 16 gives:

$$\mathcal{T}^{(t_i)}_{\mathrm{Haar}}(\rho_M) = \frac{\mathrm{Tr}(\rho_M) \cdot \mathbb{I}_{\mathrm{Sym}(t_i)}}{\dim(\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i}))} = \frac{\mathbb{I}_{\mathrm{Sym}(t_i)}}{D_{\mathrm{Sym}(t_i)}} =: \sigma_{\mathrm{MP}}, \tag{44}$$

where $D_{\mathrm{Sym}(t_i)} := \dim(\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i})) = \binom{d+t_i-1}{t_i}$.

To evaluate $\bar{\mathcal{E}}_{\mathrm{MP}}(\rho_{ME})$, we use the fact that the Haar twirling channel, when applied to one subsystem of a bipartite state, produces a product state. Specifically, for any bipartite state ρ_{AB}:

$$(\mathcal{T}^{(t)}_{\mathrm{Haar}} \otimes \mathrm{id}_B)(\rho_{AB}) = \mathcal{T}^{(t)}_{\mathrm{Haar}}(\rho_A) \otimes \rho_B, \tag{45}$$

where $\rho_A = \mathrm{Tr}_B(\rho_{AB})$ and $\rho_B = \mathrm{Tr}_A(\rho_{AB})$.

This property follows from the linearity of the twirl and the fact that $\mathcal{T}^{(t)}_{\mathrm{Haar}}$ projects onto the trivial representation (for states on irreducible subspaces like the symmetric subspace), which decouples the systems.

Let $\{|e_j\rangle\}$ be a basis for $\mathcal{H}_E$ and write $\rho_{ME} = \sum_{j,k} A_{jk} \otimes |e_j\rangle\langle e_k|$ where A_{jk} are operators on $\mathcal{H}_M^{\otimes t_i}$. Then:

$$\begin{aligned}
(\mathcal{T}^{(t_i)}_{\mathrm{Haar}} \otimes \mathrm{id}_E)(\rho_{ME}) &= \sum_{j,k} \mathcal{T}^{(t_i)}_{\mathrm{Haar}}(A_{jk}) \otimes |e_j\rangle\langle e_k| \\
&= \sum_{j,k} \frac{\mathrm{Tr}(A_{jk})}{D_{\mathrm{Sym}(t_i)}} \mathbb{I}_{\mathrm{Sym}(t_i)} \otimes |e_j\rangle\langle e_k| \\
&= \frac{\mathbb{I}_{\mathrm{Sym}(t_i)}}{D_{\mathrm{Sym}(t_i)}} \otimes \sum_{j,k} \mathrm{Tr}(A_{jk}) |e_j\rangle\langle e_k| \\
&= \frac{\mathbb{I}_{\mathrm{Sym}(t_i)}}{D_{\mathrm{Sym}(t_i)}} \otimes \mathrm{Tr}_M(\rho_{ME}) \\
&= \sigma_{\mathrm{MP}} \otimes \rho_E.
\end{aligned} \tag{46}$$

Therefore:

$$\bar{\mathcal{E}}_{\mathrm{MP}}(\rho_{ME}) = \sigma_{\mathrm{MP}} \otimes \rho_E = \frac{\mathbb{I}_{\mathrm{Sym}(t_i)}}{D_{\mathrm{Sym}(t_i)}} \otimes \mathrm{Tr}_M(\rho_{ME}). \tag{47}$$

The state-replacement channel $\mathcal{R}_\sigma$ is defined for a fixed state σ as:

$$\mathcal{R}_\sigma(X) = \mathrm{Tr}(X) \cdot \sigma. \tag{48}$$

For our purposes, we define the state-replacement channel acting on the joint system while preserving the adversary's marginal:

$$\mathcal{R}_{\sigma_{\mathrm{MP}} \otimes \mathrm{id}_E}(\rho_{ME}) := \sigma_{\mathrm{MP}} \otimes \mathrm{Tr}_M(\rho_{ME}). \tag{49}$$

Comparing with Equation (47), we see that:

$$\bar{\mathcal{E}}_{\mathrm{MP}}(\rho_{ME}) = \mathcal{R}_{\sigma_{\mathrm{MP}} \otimes \mathrm{id}_E}(\rho_{ME}) \tag{50}$$

for *all* input states ρ_{ME} with message register supported on $\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i})$.

Since $\bar{\mathcal{E}}_{\mathrm{MP}}$ and $\mathcal{R}_{\sigma_{\mathrm{MP}} \otimes \mathrm{id}_E}$ agree on all inputs within the domain of interest (states with message register on the symmetric subspace), the difference channel is the zero map on this domain:

$$\left(\bar{\mathcal{E}}_{\mathrm{MP}} - \mathcal{R}_{\sigma_{\mathrm{MP}} \otimes \mathrm{id}_E}\right)(\rho_{ME}) = 0 \quad \forall\, \rho_{ME} \text{ with } \rho_M \in \mathcal{D}(\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i})). \tag{51}$$

Since our channels act identically on all relevant inputs and the diamond norm supremum is achieved on states within the relevant domain for CPTP maps, we have:

$$\left\|\bar{\mathcal{E}}_{\mathrm{MP}} - \mathcal{R}_{\sigma_{\mathrm{MP}} \otimes \mathrm{id}_E}\right\|_\diamond = 0. \tag{52}$$

The diamond norm being zero implies *perfect indistinguishability*: no quantum operation, including those utilizing arbitrary entanglement with the ciphertext, can distinguish between:

(a) The actual encryption of a message ρ_M to produce ciphertext $\bar{\mathcal{E}}_{\mathrm{MP}}(\rho_{ME})$, and
(b) The replacement of ρ_M with the fixed state σ_{MP} while preserving all correlations with the environment, i.e., $\mathcal{R}_{\sigma_{\mathrm{MP}} \otimes \mathrm{id}_E}(\rho_{ME})$.

Since the output of the average encryption channel is completely independent of the input message (being always $\sigma_{\mathrm{MP}} \otimes \rho_E$), the adversary's side information ρ_E becomes uncorrelated with the ciphertext's message content. Any pre-existing entanglement between the message and the adversary's system is completely destroyed by the encryption process.

This establishes that MP-QPBE with an exact symmetric t_i-design provides *perfect security against adversaries with quantum side information.* □

4 Technical details (Construction of a Component-wise Independent Quantum Broadcast Encryption Scheme (CI-QBE))

If the encryption is not private broadcast encryption for a set Ω_i of t_i policies, then, instead of grouping messages into tensor products that are then symmetrized, we may treat each message as an independent entity to broadcast together for all policies.

Let the total number of distinct policies in the system be N_L. Each policy Policy_j for $j \in \{1, \ldots, N_L\}$ is associated with a single quantum message $\rho^{(j)} \in \mathcal{D}(\mathcal{H}_M)$.

Definition 17 (Component-wise Independent QBE (CI-QBE)) The CIE-QBE scheme is a tuple of four algorithms: $(\mathsf{Setup}, \mathsf{KeyGen}, \mathsf{Encrypt}, \mathsf{Decrypt})$.

1. $\mathsf{Setup}(1^\lambda, \{\mathsf{Policy}_j\}_{j=1}^{N_L})$: The setup algorithm, run by the attribute authority, proceeds as follows:
   - For each policy Policy_j, represented by an access tree $\mathbb{T}_j$, the authority generates a corresponding LSSS matrix (M_j, π_j) over a prime field $\mathbb{F}_p$, where $M_j \in \mathbb{F}_p^{\ell_j \times c_j}$.
   - A single composite MS-LSSS matrix $\mathcal{M}$ is constructed by placing these individual matrices on the block diagonal:

$$\mathcal{M} = \begin{pmatrix} M_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & M_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & M_{N_L} \end{pmatrix} \in \mathbb{F}_p^{\ell \times c},$$

     where $\ell = \sum \ell_j$ and $c = \sum c_j$. A corresponding composite labeling function ϱ is also defined as previous construction of general LSSS.
   - The authority selects a unitary 1-design or 2-design for stronger security, denoted by $(\mathfrak{U}_1, w_1)$, which is a finite set of unitary operators on $\mathcal{H}_M$. The prime p must satisfy $p > |\mathfrak{U}_1|$.
   - The **Public Parameters** PP are $(\mathcal{M}, \varrho, (\mathfrak{U}_1, w_1))$. The master secret key MSK consists of N_L secrets $\{s_j\}_{j=1}^{N_L}$ where each $s_j \equiv k_j^*$ is an index chosen uniformly at random from $\{1, \ldots, |\mathfrak{U}_1|\}$.
2. $\mathsf{KeyGen}(\mathsf{MSK}, S_{user})$: To generate a private key for a user with attribute set S_{user}, the authority performs the following:
   - For each of the N_L secrets $k_j^* \in \mathsf{MSK}$, it creates a random vector $\vec{v}_j = (k_j^*, r_{j,2}, \ldots, r_{j,c_j})^T \in \mathbb{F}_p^{c_j}$.
   - It constructs the full secret vector $\mathbf{R} = (\vec{v}_1^T, \vec{v}_2^T, \ldots, \vec{v}_{N_L}^T)^T \in \mathbb{F}_p^c$.
   - The global vector of shares is computed: $\mathbf{\Lambda} = \mathcal{M}\mathbf{R}$.
   - The user's private key $\mathsf{SK}_{\text{user}}$ consists of the subset of shares corresponding to their attributes: $\{\Lambda_i \mid \varrho(i) \in S_{user} \times \{1, \ldots, N_L\}\}$.
3. $\mathsf{Encrypt}(\{\rho^{(j)}\}_{j=1}^{N_L}, \{k_j^*\}_{j=1}^{N_L})$: To encrypt the set of N_L messages
   - For each message $\rho^{(j)}$, the broadcaster retrieves the corresponding secret key k_j^* and the associated unitary $U_{k_j^*} \in \mathfrak{U}_1$.
   - Each message is encrypted independently

$$\sigma^{(j)} = U_{k_j^*} \rho^{(j)} (U_{k_j^*}^\dagger)$$

   - The final broadcast ciphertext is assembled into a single block-diagonal state, where each block corresponds to one encrypted message. An ancillary index register $\mathcal{H}_{\text{index}}$ of dimension N_L is used

$$\rho_{\mathsf{Broadcast}} = \frac{1}{|N_L|} \sum_{j=1}^{N_L} |j\rangle\langle j|_{\text{index}} \otimes \sigma^{(j)} \in \mathcal{D}(\mathcal{H}_{\text{index}} \otimes \mathcal{H}_M)$$

4. $\mathsf{Decrypt}(\rho_{\mathsf{Broadcast}}, \mathsf{SK}_{\mathbf{user}}, j)$: To decrypt the j-th message:

- The user first reconstructs the classical key k_j^* using their attribute set S_{user} and their shares $\mathsf{SK}_{\mathrm{user}}$, by applying the LSSS reconstruction algorithm for the j-th sub-block of $\mathcal{M}$. If unauthorized, this step fails.
- The user performs a projective measurement on the index register of $\rho_{\mathsf{Broadcast}}$ with the projector $P_j = |j\rangle\langle j|_{\mathrm{index}} \otimes \mathbb{I}_{\mathcal{H}_M}$ to extract the state $\sigma^{(j)}$.
- The user applies the inverse unitary transformation using the recovered key k_j^*

$$\rho'^{(j)} = (U_{k_j^*}^\dagger)\sigma^{(j)}U_{k_j^*}$$

This decryption is perfect, yielding $\rho'^{(j)} = \rho^{(j)}$.

4.1 Security analysis for CI-QBE

We prove the CI-QBE is also perfectly secure against adversaries with quantum side information and is also information-theoretically secure. Both proofs are similar to the security proofs for MP-QPBE. We present only the security against adversaries with quantum side information.

Theorem 13 (Perfect Security of CI-QBE) *Let the CI-QBE scheme for an individual message j use an exact unitary 1-design $(\mathfrak{U}_1, w)$ as its key space. The scheme is perfectly secure against adversaries with quantum side information for this message.*

Proof The proof structure is parallel to that for MP-QPBE, but applied to a single message system. Let the joint state for message j and the adversary's system be $\rho_{M_jE} \in \mathcal{D}(\mathcal{H}_M \otimes \mathcal{H}_E)$. The encryption channel for a key k (corresponding to $U_k \in \mathfrak{U}_1$) is

$$\mathcal{E}_k(\rho_{M_jE}) = (U_k \otimes \mathbb{I}_E)\rho_{M_jE}(U_k^\dagger \otimes \mathbb{I}_E)$$

The average encryption channel, $\bar{\mathcal{E}}_{\mathrm{CIE}}$, is the average over the 1-design

$$\bar{\mathcal{E}}_{\mathrm{CI}}(\rho_{M_jE}) = \sum_{U_k \in \mathfrak{U}_1} w(U_k)(U_k \otimes \mathbb{I}_E)\rho_{M_jE}(U_k^\dagger \otimes \mathbb{I}_E)$$

By definition of an exact unitary 1-design, the averaged channel is the single-system Haar twirling channel $\mathcal{T}_{\mathrm{Haar}}^{(1)}$, which depolarizes any input state to the maximally mixed state on $\mathcal{H}_M$

$$\mathcal{T}_{\mathrm{Haar}}^{(1)}(\rho_{M_j}) = \frac{\mathbb{I}_M}{d},$$

where $d = \dim(\mathcal{H}_M)$. Applying this to the joint system yields

$$\begin{aligned}
\bar{\mathcal{E}}_{\mathrm{CI}}(\rho_{M_jE}) &= (\mathcal{T}_{\mathrm{Haar}}^{(1)} \otimes \mathrm{id}_E)(\rho_{M_jE}) \\
&= \frac{\mathbb{I}_M}{d} \otimes \mathrm{Tr}_{\mathcal{H}_M}[\rho_{M_jE}] \\
&= \frac{\mathbb{I}_M}{d} \otimes \rho_E
\end{aligned}$$

The final state of the ciphertext is fixed to $\frac{\mathbb{I}_M}{d}$, independent of the message ρ_{M_j}. Defining the replacement channel $\mathcal{R}_{\sigma_{\mathrm{CI}}}$ with the fixed state $\sigma_{\mathrm{CI}} = \frac{\mathbb{I}_M}{d} \otimes \rho_E$, we find that $\bar{\mathcal{E}}_{\mathrm{CI}}(\rho_{M_jE}) = \mathcal{R}_{\sigma_{\mathrm{CI}}}(\rho_{M_jE})$. Therefore, the diamond norm distance is zero, and the scheme is perfectly secure. □

5 Comparative Analysis and Resource Estimation

In this section, we provide a comprehensive comparison between the **MP-QPBE** and **CI-QBE** constructions. We analyze their relative merits in terms of ciphertext structure, qubit requirements, classical overhead, security guarantees, and computational complexity. We conclude with a detailed feasibility assessment for near-term and fault-tolerant quantum devices.

Throughout this section, let $d = \dim(\mathcal{H}_M)$ denote the message dimension, m denote the number of policy blocks, and t_i denote the number of messages in block i, so that the total number of messages is $N_L = \sum_{i=1}^{m} t_i$.

5.1 Structural Comparison

5.1.1 Design Selection: Single Shared Design vs. Distinct Designs

We now provide a rigorous comparison between using a single shared $T_{\max}$-design (where $T_{\max} = \max_i t_i$) versus distinct t_i-designs for each block. This comparison encompasses resource requirements, security guarantees, and practical considerations.

Consider m policy blocks with design parameters $\{t_1, \ldots, t_m\}$. Define:

$$T_{\max} := \max_{i \in [m]} t_i, \tag{53}$$

$$T_\Sigma := \sum_{i=1}^{m} t_i, \tag{54}$$

$$\bar{t} := \frac{T_\Sigma}{m} = \frac{1}{m} \sum_{i=1}^{m} t_i \quad \text{(average block size)}. \tag{55}$$

Option A: Single $T_{\max}$-Design (Shared).

Use a single design $\mathfrak{U}$ that is a $T_{\max}$-design. By the nested property (4), $\mathfrak{U}$ serves as a valid t_i-design for all $i \in [m]$ since $t_i \leq T_{\max}$.

Resource analysis:

- **Design size:** $|\mathfrak{U}| = \Omega(d^{2T_{\max}})$.
- **Storage:** $S_A = O(|\mathfrak{U}| \cdot d^2) = O(d^{2T_{\max}+2})$ complex numbers.
- **Field size:** $p > |\mathfrak{U}| = \Omega(d^{2T_{\max}})$.
- **LSSS secret space:** All m secrets drawn from $\{1, \ldots, |\mathfrak{U}|\}$.

Option B: Distinct t_i-Designs.

Use independent designs $\{\mathfrak{U}_i\}_{i=1}^{m}$, where $\mathfrak{U}_i$ is a t_i-design.

Resource analysis:

- **Design sizes:** $|\mathfrak{U}_i| = \Omega(d^{2t_i})$ for each i.
- **Total storage:** $S_B = \sum_{i=1}^{m} O(|\mathfrak{U}_i| \cdot d^2) = O\left(\sum_{i=1}^{m} d^{2t_i+2}\right)$.
- **Field size:** $p > \max_i |\mathfrak{U}_i| = \Omega(d^{2T_{\max}})$ (same as Option A).
- **LSSS secret space:** Secret k_i^* drawn from $\{1, \dots, |\mathfrak{U}_i|\}$ (block-specific).

Resource Comparison.

The storage ratio depends on the distribution of $\{t_i\}$:

$$\frac{S_B}{S_A} = \frac{\sum_{i=1}^{m} d^{2t_i+2}}{d^{2T_{\max}+2}} = \sum_{i=1}^{m} d^{2(t_i - T_{\max})} = \sum_{i=1}^{m} d^{-2(T_{\max} - t_i)}. \tag{56}$$

Case 1: Homogeneous blocks ($t_1 = t_2 = \cdots = t_m = t$).

$$T_{\max} = t, \quad \frac{S_B}{S_A} = \sum_{i=1}^{m} d^0 = m. \tag{57}$$

Distinct designs require m times more storage.

Case 2: Heterogeneous blocks (one large, others small). Let $t_1 = T_{\max}$ and $t_i = 1$ for $i \geq 2$.

$$\frac{S_B}{S_A} = d^0 + (m-1) \cdot d^{-2(T_{\max}-1)} = 1 + \frac{m-1}{d^{2(T_{\max}-1)}}. \tag{58}$$

Storage costs are nearly identical.

Case 3: General heterogeneous blocks.

$$\frac{S_B}{S_A} = \sum_{i=1}^{m} d^{-2(T_{\max} - t_i)} \leq m \cdot d^0 = m. \tag{59}$$

The overhead is at most a factor of m, and typically much smaller when t_i values vary.

Security Comparison.

The critical distinction lies in the adversary's knowledge and attack capabilities.

Threat model: Consider an adversary $\mathcal{A}$ with attribute set S satisfying $S \models \mathbb{T}_j$ (authorized for block j) but $S \not\models \mathbb{T}_i$ (unauthorized for block $i \neq j$).

Option A (Single shared design):

1. $\mathcal{A}$ legitimately obtains $\mathfrak{U}$ to decrypt block j.
2. $\mathcal{A}$ knows the ciphertext $\sigma_{\Omega_i} = U_{k_i^*}^{\otimes t_i} \tilde{\rho}_{\Omega_i} (U_{k_i^*}^{\dagger})^{\otimes t_i}$ for block i.
3. $\mathcal{A}$ can compute the candidate decryption set:

$$\mathcal{C}_{\mathcal{A}}^{(i)} = \left\{ (U_k^{\dagger})^{\otimes t_i} \sigma_{\Omega_i} (U_k)^{\otimes t_i} : U_k \in \mathfrak{U} \right\}. \tag{60}$$

4. $|\mathcal{C}_{\mathcal{A}}^{(i)}| = |\mathfrak{U}| = \Omega(d^{2T_{\max}})$, but crucially, $\mathcal{A}$ knows this is the *complete* set of possibilities.

Table 2 Resource and security comparison: Single $T_{\max}$-design vs. distinct t_i-designs. Here $T_{\max} = \max_i t_i$, $T_\Sigma = \sum_i t_i$, and $\Delta_i = T_{\max} - t_i \geq 0$.

Criterion	Single $T_{\max}$-Design (Shared)	Distinct t_i-Designs (Our Construction)
Design size	$\lvert\mathfrak{U}\rvert = \Omega(d^{2T_{\max}})$ (single design)	$\sum_i \lvert\mathfrak{U}_i\rvert = \sum_i \Omega(d^{2t_i})$
Storage	$O(d^{2T_{\max}+2})$	$O\left(\sum_i d^{2t_i+2}\right) = O\left(d^{2T_{\max}+2} \sum_i d^{-2\Delta_i}\right)$
Field size p	$p > d^{2T_{\max}}$	$p > d^{2T_{\max}}$ (same)
Symmetrization (per block i**)**	$O(t_i^2 \log d)$ (Schur) or $O(t_i! \cdot t_i \log d)$	$O(t_i^2 \log d)$ (Schur) or $O(t_i! {\cdot} t_i \log d)$ (same)
Encryption (per block i**)**	$t_i \cdot C(U)$ where $C(U) = O(\log^2 d)$	$t_i \cdot C(U_i)$ where $C(U_i) = O(\log^2 d)$ (same)
Design knowledge	All users learn $\mathfrak{U}$ (shared across all blocks)	User authorized for block i learns only $\mathfrak{U}_i$
Adversary's candidate set	Adversary can enumerate all $\lvert\mathfrak{U}\rvert = \Omega(d^{2T_{\max}})$ decryptions for *any* block	Adversary cannot enumerate decryptions for unauthorized blocks (design unknown)
Security model	Relies solely on LSSS key secrecy	Defense in depth: LSSS key secrecy *and* design obfuscation
Side-channel resistance	Weaker: design structure known to all authorized users	Stronger: design uncertainty provides additional protection
Key compromise impact	Compromise of k_i^* reveals one decryption; design $\mathfrak{U}$ remains available for attacks on other blocks	Compromise of k_i^* reveals $U_{i,k_i^*} \in \mathfrak{U}_i$; provides no information about other designs $\mathfrak{U}_j$

Option B (Distinct designs):

1. $\mathcal{A}$ obtains only $\mathfrak{U}_j$ (specific to block j).
2. $\mathcal{A}$ knows the ciphertext σ_{Ω_i} but *does not know* which design $\mathfrak{U}_i$ for $i \neq j$ was used.
3. $\mathcal{A}$ cannot construct the candidate set (60) because the design elements are unknown.
4. **Security:** Even if LSSS is partially compromised (e.g., $\mathcal{A}$ learns that k_i^*), $\mathcal{A}$ still cannot identify the corresponding unitaries without knowing $\mathfrak{U}_i$.

5.1.2 Ciphertext Structure and Hilbert Space Dimension

The two constructions differ fundamentally in how they organize the broadcast ciphertext.

MP-QPBE Scheme.

The broadcast state has the form:

$$\rho^{\mathrm{MP}}_{\mathsf{Broadcast}} = \frac{1}{m} \sum_{i=1}^{m} |i\rangle\langle i|_{\mathsf{index}} \otimes \sigma_{\Omega_i}, \tag{61}$$

where $\sigma_{\Omega_i} \in \mathcal{D}(\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i}))$ is the encrypted state for block i, supported on the symmetric subspace.

The dimension of the symmetric subspace $\mathrm{Sym}(\mathcal{H}_M^{\otimes t})$ is given by

$$\dim\bigl(\mathrm{Sym}(\mathcal{H}_M^{\otimes t})\bigr) = \binom{d+t-1}{t} = \frac{(d+t-1)!}{t!\,(d-1)!}. \tag{62}$$

The total Hilbert space dimension for the MP-QPBE ciphertext is:

$$\dim(\mathcal{H}_{\mathrm{MP}}) = \sum_{i=1}^{m} \binom{d+t_i-1}{t_i}. \tag{63}$$

CI-QBE Scheme.

The broadcast state has the form:

$$\rho^{\mathrm{CI}}_{\mathsf{Broadcast}} = \frac{1}{N_L} \sum_{j=1}^{N_L} |j\rangle\langle j|_{\mathsf{index}} \otimes \sigma^{(j)}, \tag{64}$$

where each $\sigma^{(j)} \in \mathcal{D}(\mathcal{H}_M)$ is an independently encrypted single message.
The total Hilbert space dimension is:

$$\dim(\mathcal{H}_{\mathrm{CI}}) = N_L \cdot d = d \sum_{i=1}^{m} t_i. \tag{65}$$

This scales linearly in both d and N_L.

5.1.3 Qubit Requirements

MP-QPBE Scheme.

The ciphertext σ_{Ω_i} is supported on $\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i})$ with dimension $D_i^{\mathrm{MP}} = \binom{d+t_i-1}{t_i}$. The number of qubits required to represent this space is:

$$n_i^{\mathrm{MP}} = \lceil \log_2 D_i^{\mathrm{MP}} \rceil = \left\lceil \log_2 \binom{d+t_i-1}{t_i} \right\rceil. \tag{66}$$

For the full broadcast with m blocks with sequential processing:

$$n^{\mathrm{MP}}_{\mathrm{total}} = \lceil \log_2 m \rceil + \max_{i \in [m]} n_i^{\mathrm{MP}}. \tag{67}$$

CI-QBE Scheme.

Each encrypted message $\sigma^{(j)}$ lives in $\mathcal{H}_M$ with dimension d. The broadcast state requires an index register of dimension N_L and a message register of dimension d:

$$n^{\mathrm{CI}}_{\mathrm{total}} = \lceil \log_2 N_L \rceil + \lceil \log_2 d \rceil. \tag{68}$$

For $d = 2^n$ (i.e., n-qubit messages):

$$n^{\mathrm{MP}}_{\mathrm{total}} \approx \lceil \log_2 m \rceil + t_{\max} \cdot n - \log_2(t_{\max}!), \tag{69}$$

$$n_{\text{total}}^{\text{CI}} = \lceil \log_2 N_L \rceil + n, \tag{70}$$

where $t_{\max} = \max_i t_i$. CI-QBE requires fewer qubits when $t_{\max} > 1 + \frac{\log_2 N_L}{n}$.

5.1.4 Classical Key Requirements

MP-QPBE Scheme.

- **Number of secrets:** m independent secrets $\{k_i^*\}_{i=1}^m$, one per policy block.
- **Secret domain:** Each $k_i^* \in \{1, \ldots, |\mathfrak{U}_i|\}$, where $\mathfrak{U}_i$ is a unitary t_i-design.
- **Field size:** The LSSS operates over $\mathbb{F}_p$ with $p > \max_i |\mathfrak{U}_i|$.
- **LSSS matrix:** Block-diagonal structure $\mathcal{M} = \operatorname{diag}(M_1, \ldots, M_m)$ with total rows $\ell_{\text{total}} = \sum_i \ell_i$.

CI-QBE Scheme.

- **Number of secrets:** $N_L = \sum_i t_i$ independent secrets $\{k_j^*\}_{j=1}^{N_L}$, one per message.
- **Secret domain:** Each $k_j^* \in \{1, \ldots, |\mathfrak{U}_1|\}$, where $\mathfrak{U}_1$ is a unitary 1-design.
- **Field size:** Smaller prime $p > |\mathfrak{U}_1|$ suffices.
- **LSSS matrix:** Larger matrix with N_L secret columns.

MP-QPBE requires fewer secrets but larger t-designs (hence larger field). CI-QBE requires more secrets but simpler 1-designs. The total private key size $|\mathsf{SK}_{\text{user}}|$ is typically larger for CI-QBE due to the increased number of shares.

5.2 Security Comparison

MP-QPBE Scheme.

MP-QPBE provides security for the *joint state* $\tilde{\rho}_{\Omega_i} = \bigotimes_{j=1}^{t_i} \rho^{(j)}$ (after symmetrization). The use of a unitary t_i-design ensures that the encryption channel matches the Haar-random channel up to the t_i-th moment:

$$\frac{1}{|\mathfrak{U}_i|} \sum_{U \in \mathfrak{U}_i} U^{\otimes t_i} X (U^\dagger)^{\otimes t_i} = \int_{U(d)} U^{\otimes t_i} X (U^\dagger)^{\otimes t_i} \, d\mu_{\text{Haar}}(U) \tag{71}$$

for all operators X on $\mathcal{H}_M^{\otimes t_i}$.

In the work of Broadbent et al. [12], they proved that the unitary t-design based QPB gives protection against adversaries who:

1. Possess quantum side information correlated with the plaintext.
2. Attempt joint measurements on the t_i-partite ciphertext.
3. Seek to learn collective properties (correlations, entanglement structure) of the message set.

CI-QBE Scheme.

CI-QBE provides *independent* security for each message $\rho^{(j)}$. The security proof shows that each component is perfectly protected:

$$\rho_{\mathcal{A}}^{(j)} = \frac{1}{|\mathfrak{U}_1|} \sum_{U \in \mathfrak{U}_1} U \rho^{(j)} U^\dagger = \frac{\mathbb{I}_d}{d}. \tag{72}$$

However, CI-QBE does *not* provide security guarantees for the joint state. An adversary performing collective measurements might extract information about correlations between messages, even if individual messages remain hidden.

For applications requiring protection of message correlations (e.g., quantum secret sharing, distributed quantum computation), MP-QPBE is essential. For independent message distribution, CI-QBE suffices with lower overhead.

5.3 Complexity Analysis

We now provide rigorous complexity bounds for all protocol components, with complete derivations.

5.3.1 Symmetrization Circuit Complexity

The symmetrization step in MP-QPBE requires projecting onto the symmetric subspace:

$$\tilde{\rho}_{\Omega_i} = \frac{\Pi_{\mathrm{Sym}(t_i)} \rho_{\Omega_i} \Pi_{\mathrm{Sym}(t_i)}}{\mathrm{Tr}(\Pi_{\mathrm{Sym}(t_i)} \rho_{\Omega_i})}, \tag{73}$$

where $\Pi_{\mathrm{Sym}(t)} = \frac{1}{t!} \sum_{\pi \in \mathfrak{S}_t} P_d(\pi)$ is the projector onto $\mathrm{Sym}(\mathcal{H}_M^{\otimes t})$.

We analyze two implementation approaches.

Approach A: Direct Permutation Sum.

The projector can be implemented as a linear combination of permutation operators:

$$\Pi_{\mathrm{Sym}(t)} = \frac{1}{t!} \sum_{\pi \in \mathfrak{S}_t} P_d(\pi). \tag{74}$$

Step 1: Permutation operator implementation. Each permutation $\pi \in \mathfrak{S}_t$ can be decomposed into transpositions. A transposition $(i\,j)$ on registers i and j is implemented by a SWAP gate:

$$\mathrm{SWAP}_{ij} = \sum_{a,b=0}^{d-1} |a\rangle\langle b|_i \otimes |b\rangle\langle a|_j\,. \tag{75}$$

For $d = 2^n$, each SWAP gate on n-qubit registers decomposes into $3n$ CNOT gates:

$$\mathrm{SWAP} = \mathrm{CNOT}_{12} \cdot \mathrm{CNOT}_{21} \cdot \mathrm{CNOT}_{12}. \tag{76}$$

Any permutation $\pi \in \mathfrak{S}_t$ can be written as a product of at most $t-1$ transpositions. Thus:

$$C(\pi) \le (t-1) \times 3n = 3n(t-1) \text{ CNOT gates.} \tag{77}$$

Step 2: Linear combination of unitaries (LCU). To implement $\Pi_{\mathrm{Sym}(t)} = \frac{1}{t!}\sum_\pi P_d(\pi)$, we use the LCU technique with:

- An ancilla register of $\lceil \log_2(t!) \rceil$ qubits to index permutations.
- A preparation unitary PREP that creates uniform superposition: $\mathsf{PREP}\,|0\rangle = \frac{1}{\sqrt{t!}}\sum_\pi |\pi\rangle$.
- A controlled-permutation oracle $\mathsf{SEL} = \sum_\pi |\pi\rangle\langle\pi| \otimes P_d(\pi)$.

The circuit structure is:

$$\Pi_{\mathrm{Sym}(t)} \propto (\mathsf{PREP}^\dagger \otimes I) \cdot \mathsf{SEL} \cdot (\mathsf{PREP} \otimes I). \tag{78}$$

Step 3: Gate count analysis.

- PREP: Preparing uniform superposition over $t!$ states requires $O(t!)$ gates in the worst case, or $O(t \log t)$ gates using efficient algorithms for specific groups [24].
- SEL: The controlled-permutation oracle requires $O(t!)$ controlled-$P_d(\pi)$ operations. Each controlled permutation requires $O(t \cdot n)$ Toffoli gates.

Total complexity for direct implementation:

$$C_{\mathrm{sym}}^{\mathrm{direct}}(t,d) = O(t! \cdot t \cdot n) = O(t! \cdot t \cdot \log d) \text{ elementary gates.} \tag{79}$$

For small t:

t	$t!$	Gate complexity
2	2	$O(\log d)$
3	6	$O(\log d)$
4	24	$O(\log d)$
5	120	$O(\log d)$

Approach B: Schur Transform.

For larger t, the Schur transform provides an asymptotically more efficient approach.

Step 1: Schur-Weyl duality. The space $(\mathbb{C}^d)^{\otimes t}$ decomposes under the joint action of $U(d)$ and $\mathfrak{S}_t$ as:

$$(\mathbb{C}^d)^{\otimes t} \cong \bigoplus_{\lambda \vdash t,\, \ell(\lambda) \le d} V_\lambda^{U(d)} \otimes S_\lambda^{\mathfrak{S}_t}, \tag{80}$$

where the sum is over partitions λ of t with at most d parts, $V_\lambda^{U(d)}$ is the irreducible representation of $U(d)$ labeled by λ, and $S_\lambda^{\mathfrak{S}_t}$ is the Specht module of $\mathfrak{S}_t$ labeled by λ.

Step 2: Symmetric subspace identification. The symmetric subspace corresponds to the trivial partition $\lambda = (t)$ (one row of length t):

$$\mathrm{Sym}((\mathbb{C}^d)^{\otimes t}) = V_{(t)}^{U(d)} \otimes S_{(t)}^{\mathfrak{S}_t}. \tag{81}$$

Since $S_{(t)}^{\mathfrak{S}_t}$ is one-dimensional (the trivial representation), projection onto the symmetric subspace is equivalent to projecting onto the $\lambda = (t)$ sector in the Schur basis.

Step 3: Schur transform circuit. The Schur transform is a unitary $U_{\text{Sch}} : (\mathbb{C}^d)^{\otimes t} \to \bigoplus_\lambda V_\lambda \otimes S_\lambda$ that block-diagonalizes the representation. In the Schur basis, the state is represented as:

$$U_{\text{Sch}} |\psi\rangle = \sum_{\lambda,p,q} c_{\lambda,p,q} |\lambda\rangle |p\rangle_\lambda |q\rangle_\lambda , \tag{82}$$

where $|\lambda\rangle$ labels the irrep, $|p\rangle_\lambda$ indexes the $U(d)$ basis within V_λ, and $|q\rangle_\lambda$ indexes the $\mathfrak{S}_t$ basis within S_λ.

Step 4: Complexity of Schur transform. The quantum Schur transform can be implemented efficiently [25, 26]:

Theorem 14 (Bacon, Chuang, Harrow [25]) *The Schur transform on $(\mathbb{C}^d)^{\otimes t}$ can be implemented with:*

$$C_{\text{Sch}}(t, d) = O\big(\text{poly}(t, \log d)\big) = O(t^2 \log d) \tag{83}$$

two-qubit gates, using $O(t \log t + t \log d)$ ancilla qubits.

Derivation of the complexity bound: The algorithm proceeds iteratively, adding one tensor factor at a time. At step k, the algorithm updates the Schur basis for $(\mathbb{C}^d)^{\otimes k}$ to $(\mathbb{C}^d)^{\otimes(k+1)}$ using Clebsch-Gordan coefficients.

1. **Iteration count:** $t-1$ iterations (adding registers 2 through t).
2. **Per-iteration cost:** Each iteration involves:
   - Computing which irrep μ results from $\lambda \otimes \Box$ (where $\Box$ is the fundamental representation). This requires $O(t)$ comparisons.
   - Applying Clebsch-Gordan transforms, which are $O(\log d)$-qubit unitaries with $O(\log d)$ gate complexity each.
   - Updating the multiplicity register, requiring $O(\log t)$ gates.
3. **Total:** $(t-1) \times O(t \log d) = O(t^2 \log d)$ gates.

Step 5: Projection in Schur basis. After applying U_{Sch}, projection onto $\text{Sym}((\mathbb{C}^d)^{\otimes t})$ is trivial: measure the λ-register and post-select on $\lambda = (t)$. Alternatively, controlled on $\lambda = (t)$, swap into a clean register.
The measurement has success probability:

$$p_{\text{sym}} = \text{Tr}(\Pi_{\text{Sym}(t)}\rho) = \frac{\binom{d+t-1}{t}}{d^t}, \tag{84}$$

which for $d = 2$, $t = 2$ gives $p_{\text{sym}} = 3/4$.

Step 6: Total symmetrization complexity via Schur transform.

$$C_{\text{sym}}^{\text{Schur}}(t, d) = 2 \cdot C_{\text{Sch}}(t, d) + O(t) = O(t^2 \log d), \tag{85}$$

where the factor of 2 accounts for the Schur transform and its inverse (after projection).

Comparison of Symmetrization Approaches.

Table 3 Comparison of symmetrization circuit implementations.

Approach	Gate Complexity	Ancilla Qubits	Best For
Direct (LCU)	$O(t! \cdot t \cdot \log d)$	$O(\log(t!)) = O(t \log t)$	$t \leq 4$
Schur transform	$O(t^2 \log d)$	$O(t \log t + t \log d)$	$t \geq 5$

The crossover point is approximately $t^* \approx 5$: for $t \leq 4$, direct implementation is simpler; for $t \geq 5$, the Schur transform is asymptotically superior since $t^2 \ll t!$.

5.3.2 Encryption Circuit Complexity

After symmetrization, encryption applies $U_{k_i^*}^{\otimes t_i}$ to the state $\tilde{\rho}_{\Omega_i}$.

Gate Complexity for Tensor Power.

$$C_{\text{enc}}(t, U) = t \cdot C(U), \tag{86}$$

where $C(U)$ is the gate complexity of a single unitary from the design.

Design-Specific Complexities.

(a) Clifford Group (Exact 3-Design). The n-qubit Clifford group Cliff_n forms an exact unitary 3-design. Each Clifford unitary can be decomposed into [27]:

$$C(U_{\text{Cliff}}) = O(n^2/\log n) \text{ gates} \tag{87}$$

using the canonical form decomposition. For practical purposes, $C(U_{\text{Cliff}}) = O(n^2)$.

For $d = 2^n$:

$$C_{\text{enc}}^{\text{Cliff}}(t) = t \cdot O(n^2) = O(t \cdot \log^2 d). \tag{88}$$

(b) Random Quantum Circuits (Approximate t-Design). Random circuits of sufficient depth form approximate t-designs [15, 28]:

Theorem 15 (Brandão, Harrow, Horodecki [15]) *A random quantum circuit on n qubits with depth $D = O(n \cdot t)$ using gates from a universal gate set forms an ε-approximate t-design with $\varepsilon = e^{-\Omega(D/n)}$.*

The gate complexity per unitary is:

$$C(U_{\text{rand}}) = O(n^2 \cdot t) = O(t \cdot \log^2 d). \tag{89}$$

For the tensor power:

$$C_{\mathrm{enc}}^{\mathrm{rand}}(t) = t \cdot O(t \cdot \log^2 d) = O(t^2 \cdot \log^2 d). \tag{90}$$

(c) Exact t-Designs for $t > 3$. Explicit constructions of exact t-designs for $t > 3$ are less common. One approach uses:

- **Unitary t-groups:** Finite subgroups of $U(d)$ forming exact t-designs exist for specific (t, d) pairs.
- **Approximate constructions:** Use random circuits with depth $O(nt)$.

Encryption Complexity.

$$C_{\mathrm{enc}}(t_i) = \begin{cases} O(t_i \cdot \log^2 d) & \text{Clifford group, } t_i \leq 3, \\ O(t_i^2 \cdot \log^2 d) & \text{Random circuits, general } t_i. \end{cases} \tag{91}$$

5.3.3 Total Quantum Gate Complexity

MP-QPBE Scheme.

Combining symmetrization and encryption for all m blocks:

$$C_{\mathrm{quantum}}^{\mathrm{MP}} = \sum_{i=1}^{m} \big[C_{\mathrm{sym}}(t_i, d) + C_{\mathrm{enc}}(t_i)\big]. \tag{92}$$

Using the Schur transform for symmetrization and Clifford encryption:

$$\begin{aligned} C_{\mathrm{quantum}}^{\mathrm{MP}} &= \sum_{i=1}^{m} \big[O(t_i^2 \log d) + O(t_i \log^2 d)\big] \\ &= O\left(\log d \sum_{i=1}^{m} t_i^2\right) + O\left(\log^2 d \sum_{i=1}^{m} t_i\right) \\ &= O\left(\log^2 d \cdot \sum_{i=1}^{m} t_i^2\right). \end{aligned} \tag{93}$$

CI-QBE Scheme.

No symmetrization is required. Each of the $N_L = \sum_i t_i$ messages is encrypted independently:

$$C_{\mathrm{quantum}}^{\mathrm{CI}} = \sum_{j=1}^{N_L} C(U_{k_j^*}) = N_L \cdot O(\log^2 d) = O\left(\log^2 d \cdot \sum_{i=1}^{m} t_i\right). \tag{94}$$

Complexity Ratio.

$$\frac{C_{\mathrm{quantum}}^{\mathrm{MP}}}{C_{\mathrm{quantum}}^{\mathrm{CI}}} = O\left(\frac{\sum_i t_i^2}{\sum_i t_i}\right) = O\left(\frac{\sum_i t_i^2}{N_L}\right). \tag{95}$$

By the Cauchy-Schwarz inequality, $\sum_i t_i^2 \geq (\sum_i t_i)^2/m = N_L^2/m$, so:

$$\frac{C_{\text{quantum}}^{\text{MP}}}{C_{\text{quantum}}^{\text{CI}}} \geq \frac{N_L}{m} = \bar{t}, \tag{96}$$

where $\bar{t} = N_L/m$ is the average block size. MP-QPBE has higher quantum complexity by a factor of at least $\bar{t}$.

5.3.4 Classical Resource Complexity

LSSS Matrix Construction.

Converting an access tree T to an LSSS matrix (M, π) uses the recursive algorithm of [29].

Algorithm complexity: For a tree with ℓ leaves (attributes) and depth h:

1. Assign the root vector $\vec{v}_{\text{root}} = (1, 0, \ldots, 0) \in \mathbb{F}_p^c$.
2. For each internal node with threshold (k, n), expand the parent vector into n child vectors using a (k, n)-threshold LSSS.
3. Each expansion at depth j adds at most j new columns.

The matrix dimensions are $M \in \mathbb{F}_p^{\ell \times c}$ where $c \leq \ell$. The construction requires:

$$C_{\text{tree}\to\text{LSSS}}(\mathsf{T}) = O(\ell \cdot h) \text{ field operations.} \tag{97}$$

For the block-diagonal MS-LSSS with m blocks:

$$C_{\text{LSSS}}^{\text{total}} = \sum_{i=1}^{m} O(\ell_i \cdot h_i) = O\left(\sum_{i=1}^{m} \ell_i \cdot h_i\right). \tag{98}$$

Secret Sharing (Share Generation).

Computing shares $\mathbf{\Lambda} = \mathcal{M}\mathbf{R}$:

- Matrix $\mathcal{M} \in \mathbb{F}_p^{\ell_{\text{total}} \times c_{\text{total}}}$ where $\ell_{\text{total}} = \sum_i \ell_i$ and $c_{\text{total}} = \sum_i c_i$.
- Vector $\mathbf{R} \in \mathbb{F}_p^{c_{\text{total}}}$ containing secrets and random padding.

Complexity:

$$C_{\text{share}} = O(\ell_{\text{total}} \cdot c_{\text{total}}) \text{ multiplications in } \mathbb{F}_p. \tag{99}$$

For block-diagonal structure, this simplifies to:

$$C_{\text{share}} = \sum_{i=1}^{m} O(\ell_i \cdot c_i). \tag{100}$$

Secret Reconstruction.

Given an authorized attribute set S for block i, reconstruction involves:

1. Identify rows $I_S = \{j : \pi_i(j) \in S\}$.
2. Find coefficients $\{\omega_j\}_{j \in I_S}$ such that $\sum_{j \in I_S} \omega_j (M_i)_j = \vec{e_1}^T$.
3. Compute $k_i^* = \sum_{j \in I_S} \omega_j \lambda_j$.

Step 2 requires solving a linear system of size $|I_S| \times c_i$:

$$C_{\text{reconstruct}} = O(c_i^3) \text{ using Gaussian elimination, or } O(c_i^2) \text{ with precomputation.} \tag{101}$$

Design Storage.

MP-QPBE Scheme. This scheme requires storing m designs $\{\mathfrak{U}_i\}_{i=1}^m$, where $\mathfrak{U}_i$ is a t_i-design.

- Explicit storage: $\sum_{i=1}^m |\mathfrak{U}_i| \cdot d^2$ complex numbers.
- For exact t-designs, $|\mathfrak{U}| = \Omega(d^{2t})$ [30].

CI-QBE Scheme. This scheme requires storing a single 1-design $\mathfrak{U}_1$.

- Minimal 1-design: Consider the Weyl-Heisenberg group, which is a unitary 1-design with $|\mathfrak{U}_1| = d^2$.
- Storage: $d^2 \cdot d^2 = d^4$ complex numbers.

Storage ratio:

$$\frac{\text{Storage}^{\text{MP}}}{\text{Storage}^{\text{CI}}} = \frac{\sum_i d^{2t_i+2}}{d^4} = \sum_i d^{2t_i-2} = \sum_i d^{2(t_i-1)}. \tag{102}$$

For $t_i \geq 2$, MP-QPBE requires poly(d) more storage.

5.3.5 Comparison: Single Large Design vs. Multiple Smaller Designs

A key design choice in MP-QPBE is using m independent t_i-designs rather than a single T-design where $T = \sum_i t_i$.

Table 4 Resource comparison: single T-design vs. multiple t_i-designs.

Resource	Single T-Design	Multiple t_i-Designs (Our Construction)
Design size	$\Omega(d^{2T})$	$\sum_i O(d^{2t_i})$
Storage	$O(d^{2T+2})$	$O\left(\sum_i d^{2t_i+2}\right)$
Symmetrization	$O(T^2 \log d)$ or $O(T! \cdot T \log d)$	$\sum_i O(t_i^2 \log d)$
Encryption	$O(T \cdot \log^2 d)$	$\sum_i O(t_i \cdot \log^2 d) = O(T \cdot \log^2 d)$

5.4 Practical Feasibility Assessment

5.4.1 NISQ-Compatible Regime

For near-term implementation on Noisy Intermediate-Scale Quantum (NISQ) devices, we identify parameter regimes where both schemes are feasible.

Constraints.

Current NISQ devices (e.g., IBM Quantum, IonQ, Rigetti) offer:

- 50–200 physical qubits
- Two-qubit gate fidelities $\sim 99\%$–99.9%
- Circuit depths $\lesssim 100$ before decoherence dominates

NISQ-Feasible Parameters.

- Message dimension: $d = 2$ (single-qubit messages), i.e., $n = 1$.
- Block sizes: $t_i \leq 3$ (to use Clifford group as exact 3-design).
- Number of blocks: $m \leq 4$.

Resource Estimates for NISQ Regime.

Both schemes are **implementable on current hardware** for proof-of-concept demonstrations in this regime. Here is a table based on a small example.

Table 5 Resource requirements in the NISQ-compatible regime ($d = 2$, $t_i \leq 3$, $m \leq 4$).

Resource	**MP-QPBE**	**CI-QBE**
Qubits (sequential)	$2 + 3 = 5$	$\lceil \log_2 12 \rceil + 1 = 5$
Qubits (parallel)	$2 + 12 = 14$	$4 + 12 = 16$
Symmetrization gates	≤ 20 per block	0
Encryption gates	≤ 3 Clifford per block	≤ 1 Clifford per message
Total gates	$\lesssim 50$	$\lesssim 20$
Circuit depth	$\lesssim 30$	$\lesssim 10$

5.4.2 Fault-Tolerant Quantum Computing

For fault-tolerant quantum computers with error-corrected logical qubits:

Possible Capabilities.

- Arbitrary message dimension d.
- Large block sizes t_i using Schur transform ($O(t_i^2 \log d)$ complexity).
- Approximate t-designs via random circuits for $t > 3$.

Scalability.
Both schemes scale polynomially in all parameters:

$$C_{\text{total}}^{\text{MP}} = O\left(\log^2 d \cdot \sum_i t_i^2\right) + O\left(\sum_i \ell_i h_i\right), \tag{103}$$

$$C_{\text{total}}^{\text{CI}} = O\left(N_L \cdot \log^2 d\right) + O\left(\sum_j \ell_j h_j\right). \tag{104}$$

Full-scale implementation is feasible once fault-tolerant quantum computers become mature enough.

Here is a comparison table of both constructions based on resource estimation and feasibility.

Table 6 Summary comparison of MP-QPBE and CI-QBE.

Criterion	**MP-QPBE**	**CI-QBE**
Security guarantee	Collective (stronger)	Component-wise
Side-channel resistance	High (t-design)	Moderate (1-design)
Quantum complexity	Higher ($\propto \sum t_i^2$)	Lower ($\propto \sum t_i$)
Classical complexity	Lower (m secrets)	Higher (N_L secrets)
Design requirements	t_i-designs (harder)	1-design (easier)
Ciphertext dimension	$\sum \binom{d+t_i-1}{t_i}$	$N_L \cdot d$
NISQ feasibility	$t_i \leq 3$ (Clifford)	All t_i

We also present an overall comparison of these two quantum broadcast encryption schemes in the following Table 8.

6 Conclusion and Future works

In this paper, we have introduced two complementary frameworks for attribute-based quantum broadcast encryption: Multi-Policy Quantum Private Broadcast Encryption (MP-QPBE) and Component-wise Independent Quantum Broadcast Encryption (CI-QBE). Both constructions leverage Linear Secret-Sharing Schemes (LSSS) for classical key distribution and unitary designs for quantum encryption, achieving information-theoretic security against computationally unbounded adversaries.

The MP-QPBE scheme, built upon symmetric unitary t-designs, enables the simultaneous enforcement of multiple access policies on tensor products of quantum messages. By employing a symmetrization procedure, this construction provides strong security guarantees against adversaries with quantum side information, including protection of collective properties such as correlations and entanglement structure. However, this approach forces the chosen messages to be pure and mutually orthogonal states, as the symmetrization map is non-invertible; for general mixed states, it is not possible to recover the exact message in such cases.

Table 7 Comparative Analysis of Quantum Broadcast Encryption Schemes (Part-1)

Aspect	**MP-QPBE (Symmetrization-Based)**	**CI-QBE (Component-wise Independent)**
Information integrity	The non-invertible symmetrization map erases message order and is not applicable to general mixed states. This is a trade-off for its specialized security features.	Guarantees complete fidelity by preserving the order, structure, and quantum state type (pure, mixed, or entangled) of all broadcast messages.
Security & integrity model	Provides an all-or-nothing decryption property for a message block. Any adversarial tampering is likely to corrupt the entire symmetric state, preventing partial decryption.	Security is handled on a per-message basis. Adversarial tampering is isolated to individual ciphertexts without affecting others.
Cryptographic privacy	Employs a unitary t_i-design to provide information-theoretic security against *joint attacks* across the entire t_i-message block, making the encrypted block indistinguishable from a Haar-randomly encrypted state up to the t_i-th moment. Protects collective properties (correlations, entanglement structure) of the message set against adversaries with quantum side information.	Employs a unitary 1-design to secure each message individually. This protects individual messages but does not secure against adversaries seeking to find correlations between the independently encrypted messages. Component-wise security does not imply collective security.
Qubits requirement	Number of qubits per block is $\lceil \log_2(\binom{d+t_i-1}{t_i}) \rceil$. For sequential processing of m blocks: $Q^{\mathrm{MP}}_{\mathrm{total}} = \lceil \log_2 m \rceil + \max_i \lceil \log_2 \binom{d+t_i-1}{t_i} \rceil$.	Number of qubits for t_i messages: $\lceil \log_2(t_i) \rceil + \lceil \log_2(d) \rceil$. For full broadcast: $Q^{\mathrm{CI}}_{\mathrm{total}} = \lceil \log_2 N_L \rceil + \lceil \log_2 d \rceil$. This is generally more efficient for high-dimensional states and is highly scalable.
Quantum gate complexity	High encryption complexity: $C^{\mathrm{MP}}_{\mathrm{quantum}} = \sum_{i=1}^{m} [C_{\mathrm{sym}}(t_i, d) + t_i \cdot C(U)]$. Symmetrization via Schur transform: $C_{\mathrm{sym}}(t_i, d) = O(t_i^2 \log d)$; via direct LCU: $C_{\mathrm{sym}}(t_i, d) = O(t_i! \cdot t_i \cdot \log d)$. For Clifford designs: total complexity is $O(\log^2 d \cdot \sum_i t_i^2)$.	Low encryption complexity: $C^{\mathrm{CI}}_{\mathrm{quantum}} = \sum_{j=1}^{N_L} C(U_{k_j^*}) = O(N_L \cdot \log^2 d)$. No symmetrization overhead. Complexity ratio: $C^{\mathrm{MP}}/C^{\mathrm{CI}} \geq \bar{t}$ where $\bar{t} = N_L/m$ is the average block size.
Classical key management	**High Efficiency**. Requires only m secrets for m blocks. This minimizes the LSSS construction's size and the classical computation required for key reconstruction. However, requires larger prime field $p > \max_i \lvert \mathfrak{U}_i \rvert$ where $\lvert \mathfrak{U}_i \rvert = \Omega(d^{2t_i})$ for t_i-designs.	**Higher Overhead**. Requires $N_L = \sum t_i$ secrets in total. This increases the size of the LSSS matrix and the classical resources needed for key management. Smaller field suffices: $p > \lvert \mathfrak{U}_1 \rvert$ where $\lvert \mathfrak{U}_1 \rvert = O(d^2)$ for 1-designs.

Table 8 Comparative Analysis of Quantum Broadcast Encryption Schemes (Part-2) Continued

Aspect	MP-QPBE (Symmetrization-Based)	CI-QBE (Component-wise Independent)
Design requirements	Requires m independent t_i-designs $\{\mathfrak{U}_i\}_{i=1}^m$. Storage: $O(\sum_i d^{2t_i+2})$ complex numbers. For $t_i \leq 3$: Clifford group (exact 3-design) with $\lvert\mathrm{Cliff}_1\rvert = 24$ for $d = 2$.	Requires a single 1-design $\mathfrak{U}_1$. Storage: $O(d^4)$ complex numbers. Weyl-Heisenberg group provides minimal 1-design with $\lvert\mathfrak{U}_1\rvert = d^2$.
Ciphertext dimension	$\dim(\mathcal{H}_{\mathrm{MP}}) = \sum_{i=1}^m \binom{d+t_i-1}{t_i}$. Scales as $O(d^{t_i})$ for fixed t_i; grows rapidly with t_i.	$\dim(\mathcal{H}_{\mathrm{CI}}) = N_L \cdot d$. Scales linearly in both N_L and d; highly scalable.
NISQ feasibility	Feasible for $d = 2$, $t_i \leq 3$, $m \leq 4$ using Clifford group. Requires $\lesssim 50$ gates, depth $\lesssim 30$. Symmetrization adds implementation complexity.	Feasible for all t_i values on NISQ devices. Requires $\lesssim 20$ gates, depth $\lesssim 10$. Simpler circuit structure.
Security analysis	**Strongest security**: information-theoretic security against computationally unbounded adversaries with quantum side information performing joint measurements on t_i-partite ciphertext. Unauthorized users perceive maximally mixed state $\mathbb{I}_{\mathrm{Sym}(t_i)}/\binom{d+t_i-1}{t_i}$ independent of plaintext.	**Component-wise security**: each message $\rho^{(j)}$ individually protected; unauthorized users perceive $\mathbb{I}_d/d$. Does not protect correlations between messages. Sufficient for independent message distribution.

The CI-QBE scheme addresses this limitation by treating each quantum message independently, encrypting it with a unitary drawn from a 1-design. This component-wise approach preserves message order, supports arbitrary quantum states, including mixed states, and offers superior scalability in terms of quantum resources. The trade-off is that CI-QBE provides security guarantees only at the individual message level, rather than collective security over message blocks.

Our comparative analysis reveals that the choice between MP-QPBE and CI-QBE depends on the specific application requirements: MP-QPBE is preferable when protecting correlations among messages is essential and messages are pure states, while CI-QBE is better suited for scenarios requiring flexibility in message types and efficient resource utilization.

A limitation of both constructions is the reliance on the semi-honest adversarial model, which assumes that all parties follow the protocol specification. While this model is standard in the literature on classical secret-sharing, it does not account for adversaries who may actively deviate from the protocol.

Extensions to Stronger Adversarial Models. We acknowledge that the semi-honest model does not capture all real-world threats. Extending our schemes to resist **active (malicious) adversaries** would require the following:

- **Verifiable secret sharing:** Replacing the LSSS with a verifiable variant [31] that allows recipients to confirm their shares are consistent without revealing the secret.
- **Authentication of quantum states:** Incorporating quantum authentication codes [32] to detect tampering with the broadcast ciphertext.
- **Robust decryption procedures:** Designing decryption algorithms that detect and reject malformed ciphertexts rather than producing incorrect outputs.

Exploring these extensions constitutes a natural direction for future research. Additionally, investigating efficient constructions of high-order unitary designs and their integration into practical quantum communication systems remains an important open problem.

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8 Appendix

In this section, we derive results that are used in our constructions.

Theorem 16 *Let $\mathcal{H}_M$ be a d-dimensional Hilbert space. Let $\mathcal{H}_M^{\otimes t_i}$ be the t_i-fold tensor product of this space. Let $\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i})$ denote the symmetric subspace of $\mathcal{H}_M^{\otimes t_i}$. Let $\tilde{\rho}_{\Omega_i}$ be a quantum state that is a density operator that lives entirely within the symmetric subspace, i.e., $\tilde{\rho}_{\Omega_i}$ is an operator on $\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i})$. Let $U(d)$ be the group of $d \times d$ unitary matrices, and let $d\mu(U)$ be the unique, normalized Haar measure on $U(d)$. Then,*

$$\int_{U(d)} U^{\otimes t_i} \tilde{\rho}_{\Omega_i} (U^\dagger)^{\otimes t_i} d\mu(U) = \frac{\mathrm{Tr}(\tilde{\rho}_{\Omega_i}) \mathbb{I}_{\mathrm{Sym}(t_i)}}{\dim(\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i}))},$$

where $\mathbb{I}_{\mathrm{Sym}(t_i)}$ is the identity operator on the symmetric subspace $\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i})$.

Proof For any operator X on $\mathcal{H}_M^{\otimes t_i}$, the twirling channel is defined as

$$\mathcal{E}(X) = \int_{U(d)} U^{\otimes t_i} X (U^\dagger)^{\otimes t_i} d\mu(U).$$

First, we prove that the channel commutes with all operators of the form $V^{\otimes t_i}$ for any $V \in U(d)$.

$$\begin{aligned}
V^{\otimes t_i}\mathcal{E}(X)(V^\dagger)^{\otimes t_i} &= V^{\otimes t_i}\left(\int_{U(d)} U^{\otimes t_i} X (U^\dagger)^{\otimes t_i} d\mu(U)\right)(V^\dagger)^{\otimes t_i} \\
&= \int_{U(d)} (VU)^{\otimes t_i} X (U^\dagger V^\dagger)^{\otimes t_i} d\mu(U) \\
&= \int_{U(d)} (VU)^{\otimes t_i} X ((VU)^\dagger)^{\otimes t_i} d\mu(U)
\end{aligned}$$

As the Haar measure $d\mu(U)$ is left-invariant, $d\mu(U) = d\mu(VU)$ for any fixed $V \in U(d)$. Let $U' = VU$. As U ranges over the entire group $U(d)$, so does U'. Therefore, we can change the integration variable as follows

$$\int_{U(d)} (U')^{\otimes t_i} X ((U')^\dagger)^{\otimes t_i} d\mu(U') = \mathcal{E}(X)$$

Thus, for any $V \in U(d)$, we have,

$$V^{\otimes t_i}\mathcal{E}(X)(V^\dagger)^{\otimes t_i} = \mathcal{E}(X) \quad \Longrightarrow \quad V^{\otimes t_i}\mathcal{E}(X) = \mathcal{E}(X)V^{\otimes t_i}$$

This means that the operator $\mathcal{E}(X)$ lies in the commutant of the representation $U \mapsto U^{\otimes t_i}$.

The algebra of linear operators on $\mathcal{H}_M^{\otimes t_i}$ that commute with $U^{\otimes t_i}$ for all $U \in U(d)$ is precisely the algebra spanned by the permutation operators $\{P(\pi) \mid \pi \in \mathfrak{S}_{t_i}\}$, where $\mathfrak{S}_{t_i}$ is the symmetric group on t_i elements. The permutation operator $P(\pi)$ acts as

$$P(\pi)(|v_1\rangle \otimes \cdots \otimes |v_{t_i}\rangle) = \left|v_{\pi^{-1}(1)}\right\rangle \otimes \cdots \otimes \left|v_{\pi^{-1}(t_i)}\right\rangle$$

The Schur-Weyl duality implies that the output of our channel, $\mathcal{E}(X)$, must be a linear combination of these permutation operators

$$\mathcal{E}(X) = \sum_{\pi \in \mathfrak{S}_{t_i}} c_\pi(X) P(\pi)$$

where the coefficients $c_\pi(X)$ depend linearly on the input operator X. We compute the coefficient $c_\pi(X)$ for $X = \tilde{\rho}_{\Omega_i}$. The state $\tilde{\rho}_{\Omega_i}$ lies in the symmetric subspace. An operator ρ is in the symmetric subspace if and only if it is invariant under conjugation by any permutation operator, i.e.,

$$P(\sigma)\tilde{\rho}_{\Omega_i}P(\sigma)^\dagger = \tilde{\rho}_{\Omega_i} \quad \Longrightarrow \quad P(\sigma)\tilde{\rho}_{\Omega_i} = \tilde{\rho}_{\Omega_i}P(\sigma) \quad \text{for all } \sigma \in \mathfrak{S}_{t_i},$$

since $P(\sigma)$ is unitary.

Therefore

$$\begin{aligned}
P(\sigma)\mathcal{E}(\tilde{\rho}_{\Omega_i})P(\sigma)^\dagger &= P(\sigma)\left(\int_{U(d)} U^{\otimes t_i}\tilde{\rho}_{\Omega_i}(U^\dagger)^{\otimes t_i} d\mu(U)\right)P(\sigma)^\dagger \\
&= \int_{U(d)} (P(\sigma)U^{\otimes t_i})\tilde{\rho}_{\Omega_i}(U^\dagger)^{\otimes t_i}P(\sigma)^\dagger d\mu(U) \\
&= \int_{U(d)} (U^{\otimes t_i}P(\sigma))\tilde{\rho}_{\Omega_i}(P(\sigma)^\dagger(U^\dagger)^{\otimes t_i}) d\mu(U) \\
&= \int_{U(d)} U^{\otimes t_i}\left(P(\sigma)\tilde{\rho}_{\Omega_i}P(\sigma)^\dagger\right)(U^\dagger)^{\otimes t_i} d\mu(U) \\
&= \int_{U(d)} U^{\otimes t_i}\tilde{\rho}_{\Omega_i}(U^\dagger)^{\otimes t_i} d\mu(U) = \mathcal{E}(\tilde{\rho}_{\Omega_i})
\end{aligned}$$

Hence the operator $\mathcal{E}(\tilde{\rho}_{\Omega_i})$ commuted with all $V^{\otimes t_i}$ and with all $P(\sigma)$.

By Schur's Lemma, an operator that commutes with all operators in an irreducible representation must be a scalar multiple of the identity operator on the representation space. The combined action of the unitary group $U(d)$ and the symmetric group $\mathfrak{S}_{t_i}$ on $\mathcal{H}_M^{\otimes t_i}$ decomposes the space into irreducible subspaces. The symmetric subspace $\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i})$ is one such irreducible subspace.

Therefore, an operator that acts on $\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i})$ and commutes with both representations must be a scalar multiple of the identity operator on that subspace. This means

$$\mathcal{E}(\tilde{\rho}_{\Omega_i}) = c \cdot \mathbb{I}_{\mathrm{Sym}(t_i)}$$

for some constant $c \in \mathbb{C}$.

Computing the trace of the left-hand side, we get,

$$\begin{aligned}
\mathrm{Tr}\left(\mathcal{E}(\tilde{\rho}_{\Omega_i})\right) &= \mathrm{Tr}\left(\int_{U(d)} U^{\otimes t_i}\tilde{\rho}_{\Omega_i}(U^\dagger)^{\otimes t_i} d\mu(U)\right) \\
&= \int_{U(d)} \mathrm{Tr}\left(U^{\otimes t_i}\tilde{\rho}_{\Omega_i}(U^\dagger)^{\otimes t_i}\right) d\mu(U) \\
&= \int_{U(d)} \mathrm{Tr}\left((U^\dagger)^{\otimes t_i}U^{\otimes t_i}\tilde{\rho}_{\Omega_i}\right) d\mu(U) \\
&= \int_{U(d)} \mathrm{Tr}\left(\mathbb{I}\cdot\tilde{\rho}_{\Omega_i}\right) d\mu(U) \\
&= \int_{U(d)} \mathrm{Tr}(\tilde{\rho}_{\Omega_i}) d\mu(U) \\
&= \mathrm{Tr}(\tilde{\rho}_{\Omega_i})\int_{U(d)} d\mu(U) = \mathrm{Tr}(\tilde{\rho}_{\Omega_i}).
\end{aligned}$$

Computing trace of the right-hand side,

$$\mathrm{Tr}\Big(c\cdot \mathbb{I}_{\mathrm{Sym}(t_i)}\Big) = c\cdot \mathrm{Tr}\Big(\mathbb{I}_{\mathrm{Sym}(t_i)}\Big)$$

and $\mathrm{Tr}\Big(\mathbb{I}_{\mathrm{Sym}(t_i)}\Big) = \dim(\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i}))$.

Equating the traces of the left and right sides, we get,

$$c = \frac{\mathrm{Tr}(\tilde{\rho}_{\Omega_i})}{\dim(\mathrm{Sym}(\mathcal{H}_M^{\otimes t_i}))}.$$

□

Computation of normalization constant for general mixed states. The normalization constant for the case when the initial state $\rho_{\Omega_i} \in \mathcal{D}(\mathcal{H}_M^{\otimes t_i})$ is an arbitrary mixed density matrix is computed below.

An arbitrary mixed state ρ_{Ω_i} can be written using its spectral decomposition as a convex sum of orthogonal pure states:

$$\rho_{\Omega_i} = \sum_{j=1}^{D} \lambda_j \,|\phi_j\rangle\langle\phi_j|$$

where $D = d^{t_i} = (\dim \mathcal{H}_M)^{t_i}$, $\{|\phi_j\rangle\}_{j=1}^{D}$ is an orthonormal basis of eigenvectors for ρ_{Ω_i}, and $\{\lambda_j\}_{j=1}^{D}$ are the corresponding non-negative eigenvalues with $\sum_{j=1}^{D}\lambda_j = 1$.

$$
\begin{aligned}
\tilde{\rho}_{\Omega_i} &= \frac{\Pi_{\mathrm{Sym}(t_i)} \left(\sum_{j=1}^{D} \lambda_j \left|\phi_j\middle\rangle\!\middle\langle\phi_j\right|\right) \Pi_{\mathrm{Sym}(t_i)}}{\mathrm{Tr}\left[\Pi_{\mathrm{Sym}(t_i)} \rho_{\Omega_i} \Pi_{\mathrm{Sym}(t_i)}\right]} \\
&= \frac{\sum_{j=1}^{D} \lambda_j \left(\Pi_{\mathrm{Sym}(t_i)} \left|\phi_j\middle\rangle\!\middle\langle\phi_j\right| \Pi_{\mathrm{Sym}(t_i)}\right)}{\mathrm{Tr}\left[\Pi_{\mathrm{Sym}(t_i)} \rho_{\Omega_i} \Pi_{\mathrm{Sym}(t_i)}\right]} \\
&= \frac{\sum_{j=1}^{D} \lambda_j \left(\frac{1}{t_i!} \sum_{\pi \in \mathfrak{S}_{t_i}} P_d(\pi)\right) \left|\phi_j\middle\rangle\!\middle\langle\phi_j\right| \left(\frac{1}{t_i!} \sum_{\sigma \in \mathfrak{S}_{t_i}} P_d(\sigma)^\dagger\right)}{\sum_{j=1}^{D} \lambda_j \,\mathrm{Tr}\left[\Pi_{\mathrm{Sym}(t_i)} \left|\phi_j\middle\rangle\!\middle\langle\phi_j\right| \Pi_{\mathrm{Sym}(t_i)}\right]} \\
&= \frac{\frac{1}{(t_i!)^2} \sum_{j=1}^{D} \lambda_j \sum_{\pi,\sigma \in \mathfrak{S}_{t_i}} P_d(\pi) \left|\phi_j\middle\rangle\!\middle\langle\phi_j\right| P_d(\sigma)^\dagger}{\sum_{j=1}^{D} \lambda_j \,\mathrm{Tr}\left[\Pi_{\mathrm{Sym}(t_i)} \left|\phi_j\middle\rangle\!\middle\langle\phi_j\right| \Pi_{\mathrm{Sym}(t_i)}\right]} \\
&= \frac{\frac{1}{(t_i!)^2} \sum_{j=1}^{D} \lambda_j \sum_{\pi,\sigma \in \mathfrak{S}_{t_i}} \left|P_d(\pi)\phi_j\middle\rangle\!\middle\langle P_d(\sigma)\phi_j\right|}{\sum_{j=1}^{D} \lambda_j \,\mathrm{Tr}\left[\Pi_{\mathrm{Sym}(t_i)} \left|\phi_j\middle\rangle\!\middle\langle\phi_j\right| \Pi_{\mathrm{Sym}(t_i)}\right]}
\end{aligned}
$$

Let

$$
\mathcal{N} = \mathrm{Tr}\left[\sum_{j=1}^{D} \lambda_j \left(\Pi_{\mathrm{Sym}(t_i)} \left|\phi_j\middle\rangle\!\middle\langle\phi_j\right| \Pi_{\mathrm{Sym}(t_i)}\right)\right]
$$

By linearity of the trace, we get,

$$
\mathcal{N} = \sum_{j=1}^{D} \lambda_j \,\mathrm{Tr}\left[\Pi_{\mathrm{Sym}(t_i)} \left|\phi_j\middle\rangle\!\middle\langle\phi_j\right| \Pi_{\mathrm{Sym}(t_i)}\right]
$$

Then

$$
\begin{aligned}
&\mathrm{Tr}\left[\Pi_{\mathrm{Sym}(t_i)} \left|\phi_j\middle\rangle\!\middle\langle\phi_j\right| \Pi_{\mathrm{Sym}(t_i)}\right] \\
&= \mathrm{Tr}\left[\left|\phi_j\middle\rangle\!\middle\langle\phi_j\right| \Pi_{\mathrm{Sym}(t_i)}\right] \\
&= \left\langle\phi_j\right| \Pi_{\mathrm{Sym}(t_i)} \left|\phi_j\right\rangle \\
&= \left\langle \phi_j \middle| \left(\frac{1}{t_i!} \sum_{\pi \in \mathfrak{S}_{t_i}} P_d(\pi)\right) \middle| \phi_j \right\rangle \\
&= \frac{1}{t_i!} \sum_{\pi \in \mathfrak{S}_{t_i}} \left\langle\phi_j\right| P_d(\pi) \left|\phi_j\right\rangle .
\end{aligned}
$$

Therefore, the total normalization factor becomes

$$
\mathcal{N} = \sum_{j=1}^{D} \frac{\lambda_j}{t_i!} \sum_{\pi \in \mathfrak{S}_{t_i}} \left\langle\phi_j\right| P_d(\pi) \left|\phi_j\right\rangle .
$$

Finally, we get,

$$\tilde{\rho}_{\Omega_i} = \frac{\sum_{j=1}^{D} \lambda_j \left(\Pi_{\mathrm{Sym}(t_i)} |\phi_j\rangle\langle\phi_j| \Pi_{\mathrm{Sym}(t_i)}\right)}{\sum_{k=1}^{D} \lambda_k \langle\phi_k| \Pi_{\mathrm{Sym}(t_i)} |\phi_k\rangle}.$$

8.1 Example: Two-Policy Quantum Broadcast

We illustrate our MP-QPBE construction with a concrete example.

8.1.1 Setup

Consider an example with:

- **Message dimension:** $d = 2$ (qubit messages)
- **Number of policy blocks:** $m = 2$
- **Block 1:** $\Omega_1 = \{\mathsf{Policy}_1, \mathsf{Policy}_2\}$ with $t_1 = 2$ messages
- **Block 2:** $\Omega_2 = \{\mathsf{Policy}_3\}$ with $t_2 = 1$ message
- **Attribute universe:** $\Omega = \{A, B, C, D\}$
- **Policies:**
  - Policy_1: $(A \wedge B)$ — requires attributes A AND B
  - Policy_2: (C) — requires attribute C
  - Policy_3: $(A \vee D)$ — requires attribute A OR D
- **Unitary 3-design:** Single-qubit Clifford group Cliff_1 which is an exact 3-design with 24 elements.

8.1.2 Step 1: Construction of Access Trees and LSSS Matrices

Block 1: Composite Tree $\mathbb{T}_1 = \mathsf{AND}(\mathsf{T}^{(1)}, \mathsf{T}^{(2)})$

The composite tree requires both Policy_1 and Policy_2:

- Root: AND gate with threshold $(3, 3)$
- Children: $\mathsf{T}^{(1)}$ (AND gate for $A \wedge B$), $\mathsf{T}^{(2)}$ (leaf C)

Converting to LSSS matrix M_1 over $\mathbb{F}_{29}$ (prime > 24):

$$M_1 = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad \pi_1 = \begin{pmatrix} A \\ B \\ C \end{pmatrix}. \tag{105}$$

The target vector is $\vec{e}_1 = (1, 0, 0)^T$. A user with attributes $\{A, B, C\}$ can reconstruct using rows 1, 2, 3:

$$\omega_A(1, 1, 0) + \omega_B(1, 2, 0) + \omega_C(1, 0, 1) = (1, 0, 0). \tag{106}$$

Solving: $\omega_A = 2, \omega_B = -1 \equiv 28 \pmod{29}, \omega_C = 0$.

Block 2: Tree T_2 ***for*** $(A \vee D)$

LSSS matrix for OR gate:

$$M_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \pi_2 = \begin{pmatrix} A \\ D \end{pmatrix}. \tag{107}$$

Either row alone spans the target vector.

Block-Diagonal MS-LSSS

$$\mathcal{M} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \varrho = \begin{pmatrix} (A,1) \\ (B,1) \\ (C,1) \\ (A,2) \\ (D,2) \end{pmatrix}. \tag{108}$$

8.1.3 Step 2: Secret Generation and Sharing

Choose encryption unitary indices:

- Block 1: $k_1^* = 7$ (7th Clifford unitary)
- Block 2: $k_2^* = 15$ (15th Clifford unitary)

Random padding vectors:

$$\vec{v}_1 = (7, 13, 21)^T, \quad \vec{v}_2 = (15)^T \tag{109}$$

Compute shares $\mathbf{\Lambda} = \mathcal{M}\mathbf{R}$:

$$\mathbf{\Lambda} = \begin{pmatrix} 1 \cdot 7 + 1 \cdot 13 + 0 \cdot 21 \\ 1 \cdot 7 + 2 \cdot 13 + 0 \cdot 21 \\ 1 \cdot 7 + 0 \cdot 13 + 1 \cdot 21 \\ 1 \cdot 15 \\ 1 \cdot 15 \end{pmatrix} = \begin{pmatrix} 20 \\ 4 \\ 28 \\ 15 \\ 15 \end{pmatrix} \pmod{29}. \tag{110}$$

Key Distribution

- User with $S = \{A, B, C\}$: receives SK $= \{(\lambda_1, A, 1), (\lambda_2, B, 1), (\lambda_3, C, 1), (\lambda_4, A, 2)\}$.
- User with $S = \{A, D\}$: receives SK $= \{(\lambda_1, A, 1), (\lambda_4, A, 2), (\lambda_5, D, 2)\}$.

8.1.4 Step 3: Quantum Encryption

Block 1: Two-Qubit Message

Messages: $|\psi_1\rangle = |0\rangle$, $|\psi_2\rangle = |1\rangle$ (mutually orthogonal).

Tensor product: $\rho_{\Omega_1} = |0\rangle\langle 0| \otimes |1\rangle\langle 1| = |01\rangle\langle 01|$.

Symmetrization:

$$\Pi_{\mathrm{Sym}(2)} = \frac{1}{2}(I + \mathrm{SWAP}) = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{111}$$

The state $|01\rangle$ projects to:

$$\Pi_{\mathrm{Sym}(2)}|01\rangle = \frac{1}{2}(|01\rangle + |10\rangle) = \frac{1}{\sqrt{2}}|\psi^+\rangle \text{ (unnormalized)}. \tag{112}$$

Normalized symmetric state:

$$|\tilde{\psi}_{\Omega_1}\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \tag{113}$$

We choose the encryption unitary $U_7 = H$ as the 7-th Clifford operator:

$$\sigma_{\Omega_1} = (H \otimes H)|\tilde{\psi}_{\Omega_1}\rangle\langle\tilde{\psi}_{\Omega_1}|(H^\dagger \otimes H^\dagger). \tag{114}$$

Block 2: Single-Qubit Message

Message: $|\psi_3\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.
No symmetrization needed ($t_2 = 1$).
We choose the encryption unitary $U_{15} = S$ as the 15-th Clifford operator:

$$\sigma_{\Omega_2} = S|+\rangle\langle+|S^\dagger = |{+i}\rangle\langle{+i}|, \tag{115}$$

where $|{+i}\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$.

Broadcast State.

$$\rho_{\mathrm{Broadcast}} = \frac{1}{2}\left(|1\rangle\langle 1|_{\mathrm{index}} \otimes \sigma_{\Omega_1} + |2\rangle\langle 2|_{\mathrm{index}} \otimes \sigma_{\Omega_2}\right). \tag{116}$$

8.1.5 Step 4: Resource Summary

Table 9 Resource requirements for the example.

Resource	**Block 1**	**Block 2**
Message qubits	2	1
Symmetrization gates	1 SWAP + 1 H	0
Encryption gates	2 Clifford	1 Clifford
LSSS rows	3	2
Field operations (sharing)	9	2
Total qubits	1 (index) + 2 (max message) = 3	
Total Clifford gates	≤ 5	

8.1.6 Step 5: Authorized Decryption

A user with $S = \{A, B, C\}$:

1. Measures index register, obtains outcome "1" (Block 1).
2. Reconstructs k_1^*: $k_1^* = 2 \cdot 20 + 28 \cdot 4 + 0 \cdot 28 = 40 + 112 = 152 \equiv 7 \pmod{29}$.
3. Applies $(H^\dagger)^{\otimes 2}$ to recover $|\tilde{\psi}_{\Omega_1}\rangle$.
4. Performs POVM to identify message set $\{|0\rangle, |1\rangle\}$.

8.1.7 Step 6: Security Verification

An unauthorized user with $S = \{A\}$ for Block 1:

- Has shares: $\lambda_1 = 20$ (from attribute A).
- Cannot span target vector $(1, 0, 0)$ with row $(1, 1, 0)$ alone.
- From their perspective, k_1^* is uniformly distributed over $\{0, 1, \ldots, 23\}$.
- Perceived ciphertext:

$$\rho_{\text{unauth}}^{(1)} = \frac{1}{24} \sum_{k=0}^{23} U_k^{\otimes 2} |\tilde{\psi}_{\Omega_1}\rangle\langle\tilde{\psi}_{\Omega_1}| (U_k^\dagger)^{\otimes 2} = \frac{\mathbb{I}_{\text{Sym}(2)}}{3}, \tag{117}$$

which is the maximally mixed state on the 3-dimensional symmetric subspace.

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